

In Safe Hands: The Financial and Real Impact of Investor Composition Over the Credit Cycle

Antonio Coppola*

June 2024

Abstract

I show that investor composition impacts bond price dynamics and capital allocation during crises. Using large-scale holdings data and within-firm ownership variation across near-identical bonds, I causally identify bond returns' investor composition elasticities. Corporate bonds held predominantly by insurers rather than mutual funds suffer milder losses in downturns: increasing insurer holdings by half a bond's size causes 20% shallower drawdowns. A shift-share instrument isolates variation from large insurers' idiosyncratic primary-market allocations. Differences in intermediaries' liability structures drive these results, which hold across countries. During crises, firms with more stable bondholders maintain higher borrowing at lower cost and invest more.

*Stanford University. Email: acoppola@stanford.edu. I thank John Campbell, Chris Clayton, Ric Colacito, Olivier Darmouni, Wenxin Du, Xavier Gabaix, Zhengyang Jiang, Ye Li, Matteo Maggiori, Adrien Matray, Atif Mian, Lira Mota, Diego Perez, Kenneth Rogoff, Alexi Savov, Kerry Siani, Adrien Verdelhan, Yao Zeng, and seminar participants at the Boston Fed, Chicago Booth, Columbia Business School, Darden, Dartmouth Tuck, the European Central Bank, Harvard, LSE, MIT Sloan, NYU Stern, Olin, Princeton, the San Francisco Fed, Stanford GSB, Texas A&M, UNC Kenan-Flagler, the University of Maryland, the University of Washington, Wharton, and Yale SOM for helpful comments. This analysis makes use of data that are proprietary to Morningstar and/or its content providers. Neither Morningstar nor its content providers are responsible for any of the views expressed in this article. Financial support from the John Hall Jones Research Fund is gratefully acknowledged.

Large institutional investors dominate the bond markets and vary widely in their propensities to participate in fire sales of their assets. Accordingly, market practitioners and policymakers often refer to certain patient, long-term investors as *safe hands*, in contrast to *weak hand* investors who are prone to quickly liquidate their investments in times of market-wide distress. This market environment points to the consequential interplay of two distinct ideas in macro-finance. On the one hand, models that are now canonical stress the key role of credit market disruptions and asset fire sales (Williamson 1988) in firm dynamics during macroeconomic crises (Kiyotaki and Moore 1997, Bernanke et al. 1999). On the other hand, a class of asset pricing models emphasizing portfolio balance (Modigliani and Sutch 1966, Brainard and Tobin 1968, Kouri 1976, Kojen and Yogo 2019) highlights the importance of diverse institutional demand in determining equilibrium in financial markets. Bridging these concepts, in this paper I provide quantitative causal evidence that the distribution of ownership of financial assets across heterogeneous institutional investors is an important determinant of the magnitude of fire sales over the credit cycle, ultimately impacting capital allocation and real investment outcomes.

These findings contrast with the traditional frictionless view of bond markets, in which prices and allocative outcomes are entirely determined by a common discount factor and fundamentals such as borrower financials, by instead highlighting a role for the institutional structure of ownership in the market. They also have a parallel in the policy discourse. Institutional behavior in the bond market, for instance, shapes the domestic response to quantitative easing policies. In international macroeconomics, similarly, sovereign and corporate reliance on bond-market credit from flighty foreign investors has traditionally given rise to concerns about investor fickleness and sudden capital flow reversals (Calvo 1998). Hence one might hear reassurances that Argentina’s sovereign bonds are in *safe hands* and unlikely to soar in yield if they are instead held by investors who fully understand Argentina’s risks and will remain steady during volatile market conditions.¹

To measure the quantitative impact of this form of investor heterogeneity on fire sale dynamics in a causally identified manner, I turn to the corporate bond market as a laboratory. I focus on two investor sectors, insurance companies and mutual funds, which are the most

¹For an example of the *safe hands* and *weak hands* terminology, see The Economist (2006). Differences in propensities to engage in fire sales across institutional sectors are of course not primitive demand parameters, but rather—as I emphasize throughout the paper—emerge in equilibrium as the result of differences in the liability structures and contractual provisions involved in financial intermediation.

important participants in the market: together, they own two thirds of all U.S. investment-grade corporate bonds and therefore constitute a major source of debt financing for firms. Financial intermediation in these two sectors is starkly different. Mutual fund liabilities predominantly adopt an open-end contractual structure, which allows clients to redeem their capital easily at any time, but at the same time exposes funds to sudden, household-driven capital withdrawals. In contrast, insurers enter into long-term contracts with policyholders that typically stipulate constraints and penalties on early capital withdrawals. As a result of these differences in the contractual structure of liabilities across sectors, in equilibrium mutual funds have a higher propensity to engage in fire sales than insurers.

I show that during crises bonds held predominantly by insurance companies fall in value less than bonds held largely by mutual funds. All else equal, increasing insurance holdings in a bond by 50 percentage points leads to price declines during downturns that are 20 percent shallower. Firms whose bonds are held mainly by insurers therefore experience less severe slumps in the value of their existing debt, with significant real consequences. These firms are more likely to raise new bond financing in crises and face a lower cost of capital when doing so, which enables them to invest relatively more. Partly because of regulatory reform following the Great Recession, the share of corporate bonds held by funds has been steadily rising over the past two decades, which makes it all the more relevant to examine this source of financial fragility and understand its spillovers on the real economy.²

I carry out my analysis using comprehensive security-level portfolio holdings data for these institutional sectors. As outlined in Section 1, I combine regulatory data on the security holdings of U.S. insurance firms with an extensive dataset covering the holdings of mutual funds and exchange-traded funds (ETFs) around the world.³ By 2020, the security-level positions in this combined dataset represent nearly \$60 trillion in wealth held globally. I begin in Section 2 by investigating *investor base effects* in the context of the U.S. domestic corporate bond market. In this setting, I show that bonds held *ex ante* by domestic insurers experience lower losses in value in market downturns including the financial crisis of 2008-09, the COVID crisis of March 2020, and other periods featuring spikes in credit spreads.

²The Dodd-Frank Act of 2010, for example, sharply curtailed the corporate bond holdings of banks, shifting intermediation away from this sector and towards less leveraged institutions such as mutual funds.

³Corporate bond holdings by ETFs are small relative to those of mutual funds, although they have been growing in importance: they are less than 1 percent of the fund sector’s total holdings in 2005 and grow to 12 percent by 2020. For ease of exposition, I often omit the mention of the word “ETFs” and simply refer to the entire sector as “mutual funds” or “funds”. Appendix Section L discusses the role of ETFs more extensively.

The key empirical challenge is that the assignment of bonds to insurers or mutual funds might not be random, but rather endogenous to the bonds’ inherent safety. Insurers might, for instance, disproportionately hold bonds that are issued by firms whose revenues are more resilient and that will therefore do better in a crisis. I address this challenge with a two-pronged econometric approach. I start by introducing issuer-level fixed effects interacted with fixed effects for bond characteristics (such as credit rating, duration, issue size, debt seniority, and so forth). This approach fully absorbs any confounding variation that is fixed at the issuer level or explained by observable bond characteristics. Intuitively, the resulting estimates only compare bonds issued by the same firms, with the same credit rating, of very similar duration and size, and so on, one of which is held relatively more by insurers and the other by funds.

In Section 3, I show that insurers hold certain bonds more than others, even conditionally on being issued by the same firms and having the same characteristics, because of three institutional features of the market. First, large insurers mostly buy bonds when they are initially issued (on what is known as the *primary bond market*) and hold them for a long time—unlike mutual funds, which generally trade more actively.⁴ Second, even the very largest insurers are each only allocated a small subset of bonds at issuance through the primary-market distribution process: each insurer therefore ends up holding relatively few of the bonds available at any given time. These allocations at issuance in one particular bond versus a nearly identical one are driven by each insurer’s idiosyncratic liquidity needs, relationships with underwriters, and purchase timing, so that they are plausibly exogenous. Third, these idiosyncratic allocations do not cancel out in the aggregate—but rather they affect the overall holdings of the insurance sector—because of the heavily fat-tailed nature of the size distribution of insurers.

These institutional features motivate a shift-share instrument that zeroes in on this exogenous component of the identifying variation—the second prong of my empirical strategy. I remove all potentially endogenous variation due to trading that happens after a bond is first issued, as well as variation stemming from fluctuations over time in the total amount of capital available for new bond purchases by the insurance sector. Instead, I focus purely on the *extensive margin* of primary market allocations: that is, on whether each insurer is allocated a particular bond at issuance. Formally, the first stage regresses the insurance sector’s

⁴Primary market distributions are often the only occasions when these large insurers can accumulate their desired substantial positions. In the primary market the entire bond is available for sale, while on the secondary market only small shares of the bond turn over. Smaller insurers and funds are more active participants in the secondary market.

aggregate holdings of a bond on binary indicators for whether each insurer purchased that bond at issuance, combined linearly according to insurer sizes. The instrument thus captures the overall size of the set of insurers that buy a bond at issuance: intuitively, it takes a high value if a bond is allocated to large rather than small insurers. The identifying assumption is that this matching between *individual insurers* and *particular bonds* on the primary market is uncorrelated with unobserved determinants of price dynamics in crises that are unexplained by issuer effects and observed bond characteristics. Correspondingly, identification would be threatened if larger insurers were to systematically sort into acquiring bonds on the primary market that do better in subsequent crises for reasons not explained by firm effects or bond observables. Section 3 provides further supportive evidence for this identifying assumption.

The effects that I highlight imply relative mispricing of related securities. In times of distress, the prices of very similar corporate bonds can in fact diverge significantly: for instance, at the height of the Great Recession the market yields of corporate bonds issued by the same firms and sharing the same characteristics differed by as much as 90 basis points or more. While the most extreme price dislocations only occur at the worst of the crisis and rapidly abate afterwards, more muted yield differentials tend to persist at horizons of 4 to 6 quarters following severe downturns. It is not unusual to observe persistent mispricing of similar securities during crises (Duffie 2010, Du, Tepper and Verdelhan 2018), even for extremely liquid assets such as U.S. Treasury bonds.⁵

In Section 4, I show that these effects are not specific to the U.S. context, but rather occur pervasively across countries. I study the price dynamics of bonds issued by firms in three other advanced economies: the European Monetary Union (EMU), the United Kingdom, and Canada. Since outside of the United States I do not observe the holdings of domestic investors other than funds, I focus on the positions of funds. The complement of the funds' positions proxies for the holdings of domestic insurers as well as other investor sectors that one might expect to have a low propensity to engage in fire sales, such as defined-benefit pension funds.⁶ This analysis shows that the same pattern of investor base effects is pervasive across markets, with magnitudes similar to the ones that I estimate in the baseline.

In Section 5, I document how these fire sales ultimately associate with real firm outcomes. I

⁵See for example Hu et al. (2013), who show that yield differentials for nearly identical U.S. Treasuries reach the order of about 20 to 30 basis points at the peak of the financial crisis.

⁶In Section 4, I verify that this indirect proxy would have also worked well in the U.S. context.

focus on an event study of the dynamics of firms’ real investment, new bond issuance, and cost of capital in the Great Recession for U.S. firms. Since this analysis is at the firm level, I cannot use the identification strategy employed so far, which relies on variation across securities issued by the same firm. Instead, I couple a difference-in-differences design with an aggregated version of the instrument, which continues to be relevant at the firm level since the cross-section of bonds outstanding per firm is on average sufficiently small. Naturally, as common in most macro-finance settings (Khwaja and Mian 2008), this firm-level aggregation requires stronger assumptions than the within-firm analysis. Ranking firms by the share of their bonds held by insurers prior to the crisis, firms at the 90th percentile of the distribution—as compared to firms at the 10th percentile—have capital expenditure rates higher by 1.5 percentage points, are 25 percentage points more likely to issue new bonds in each of the crisis years, and pay offering yields lower by 120 basis points when doing so. The association between fire sales and investment dynamics is especially pronounced for firms that have a high share of their existing bond debt due during the crisis and hence experience rollover issues. These magnitudes are broadly consistent with those found in the bank lending literature: for example, Amiti and Weinstein (2018) find that a 20 percent contraction in bank lending supply leads to a decline of about 2 percentage points in firm investment rates.

Related Literature. By documenting the causal role of investor heterogeneity in the dynamics of fire sales, I contribute to the research agenda studying the determinants and outcomes of fire sales in financial markets (Williamson 1988, Shleifer and Vishny 1992), including theoretical work that links credit market disruptions to aggregate macroeconomic fluctuations (Bernanke and Gertler 1986, Kiyotaki and Moore 1997, Bernanke et al. 1999) and a class of more recent macro-finance models (He and Krishnamurthy 2013, Brunnermeier and Sannikov 2014, Di Tella 2017). I connect this literature to asset pricing models emphasizing portfolio balance and institutional investor heterogeneity (Modigliani and Sutch 1966, Brainard and Tobin 1968, Tobin 1969, Gromb and Vayanos 2002, Koijen and Yogo 2019, Gabaix and Koijen 2020, Vayanos and Vila 2021), as well as to empirical work studying market segmentation in fixed income (Mitchell et al. 2007, Hu et al. 2013, Greenwood and Vayanos 2010, 2014, Siriwardane 2019) and equity markets (Greenwood and Thesmar 2011, Anton and Polk 2014).⁷

⁷A key advantage of studying investor base effects in the bond market is that I can compare similar bonds issued by the same firms, since—unlike equity securities—multiple bonds are normally issued by the same

In the context of the bond market, papers have highlighted that investor characteristics contribute to equilibrium prices in particular settings. [Siani \(2021\)](#) estimates a model of differentiated investors in segmented primary bond markets. [Ellul et al. \(2011\)](#) highlight the role of regulatory constraints in shaping insurers' bond sales when bonds are downgraded to speculative grade.⁸ [Barbosa and Ozdagli \(2021\)](#) stress the importance of relationship lending by insurers in insulating firm financing during the COVID episode. [Bretscher et al. \(2020\)](#) apply the methodology of [Koijen and Yogo \(2019\)](#) to study U.S. corporate bond holdings and find heterogeneity in estimated demand elasticities across sectors. [Jiang et al. \(2021\)](#) examine the financial stability implications of liquidity transformation by fixed income mutual funds, while [Falato et al. \(2021a\)](#) document a fire-sale spillover channel within the bond fund sector, and [Morelli et al. \(2022\)](#) study the impact of global banks' financial health on emerging market sovereign bond price dynamics.

Understanding the consequences of the heterogeneous behavior of different classes of investors is similarly important in international finance, where it matters in modeling the dynamics of sudden stops ([Calvo 1998](#), [Caballero and Krishnamurthy 2001](#), [Mendoza 2010](#), [Forbes and Warnock 2012](#), [Caballero and Simsek 2020](#)) and is key in the exchange rate portfolio balance literature ([Kouri 1976](#), [Gabaix and Maggiori 2015](#), [Itskhoki and Mukhin 2021](#)).

In studying the real effects of bond market dynamics on firm financing and investment, I relate to work including [Gilchrist and Zakrajšek \(2007\)](#), [Philippon \(2009\)](#), [Almeida et al. \(2011, 2017\)](#), [Chernenko and Sunderam \(2012\)](#), [Harford and Uysal \(2014\)](#), [Benmelech et al. \(2019\)](#), [Darmouni et al. \(2020\)](#), [Mota \(2020\)](#), and [Irani et al. \(2021\)](#). [Kundu \(2021\)](#) studies the real impact of the related market for collateralized loan obligations, while [Kubitza \(2022\)](#) analyzes corporate financing in relation to variation in insurers' asset demand. Further, I contribute to the literature on the financial intermediation carried in the mutual fund and insurance sectors, which has examined the importance of the liability structure of these institutions ([Coval and Stafford 2007](#), [Frazzini and Lamont 2008](#), [Lou 2012](#), [Goldstein et al. 2017](#), [Wardlaw 2020](#),

firm. This allows me to isolate the causal effect of the investor base on pricing. For earlier work related to the equity literature, see also [Barberis and Shleifer \(2003\)](#) and [Barberis et al. \(2005\)](#). Work on asset pricing with heterogeneous investors in complete markets includes [Dumas \(1989\)](#), [Alvarez and Jermann \(2000\)](#), and [Alvarez et al. \(2002\)](#). In addition, for models that integrate segmented fixed income markets with the portfolio balance view of exchange rates, see [Greenwood et al. \(2020\)](#) and [Gourinchas et al. \(2020\)](#).

⁸In [Ellul et al. \(2011\)](#), regulation-driven insurer behavior *exacerbates* fire sales in this localized sense. Within the universe of investment-grade bonds, which comprises the majority of bond financing, I show that insurer behavior *mitigates* fire sales. Relatedly, see also [Becker et al. \(2021\)](#), who show that regulatory incentives shape insurers' holdings of mortgage-backed securities.

Chodorow-Reich et al. 2021) and of their behavior (Becker and Ivashina 2015, Ellul et al. 2015, Koijen and Yogo 2015, Koijen and Yogo 2016, Sen 2020, Sen and Sharma 2020).

My shift-share approach employs recent advances in the applied econometrics literature on Bartik instruments (Bartik 1991, Blanchard and Katz 1992, Goldsmith-Pinkham et al. 2020, Borusyak et al. 2021a), while the difference-in-differences strategy that I use to quantify firm outcomes relates to the literature on event study designs (Autor 2003, Borusyak et al. 2021b, Callaway and Sant’Anna 2021, Wooldridge 2021).

This paper contributes to the above strands of the literature by offering causal evidence that quantifies how the liability structure of two major institutional investor sectors impacts bond price dynamics over the credit cycle. I do so across a number of countries, stressing the commonality of these patterns rather than the idiosyncrasy of each market, and trace out the linkage between these mechanisms and real economic outcomes.

1 Institutional Setting and Security-Level Holdings

The insurance sector and the mutual fund sector (including ETFs) are some of the largest holders of corporate debt securities. I use two sets of micro-data on security-level holdings covering the positions of these two institutional sectors. The data on U.S. insurers’ holdings come from the Schedule D regulatory filings that insurance firms file quarterly with the National Association of Insurance Commissioners (NAIC).⁹ The data cover the complete universe of security holdings of U.S. insurers and have been previously used in several papers (Ellul et al. 2011, Chodorow-Reich et al. 2021, Coppola et al. 2021). As shown in Table 1, in 2015 the sample contains 1,225 insurers with an average of \$3.5 billion invested in bonds and equities each, although with considerable dispersion. The insurance sector is very concentrated: the largest 10 insurers hold 34 percent of the sector’s assets, while the largest 50 own 71 percent.

Data on the holdings of mutual funds and ETFs come from Morningstar, as used in Maggiori et al. (2020) and Coppola et al. (2021).¹⁰ As reported in Table 1, in 2015 the sample includes 8,193 funds domiciled in the United States and 22,069 funds domiciled abroad, with an average of \$1.9 billion and \$0.9 billion in assets each. The fund industry holds a wider

⁹I use the version of the data provided by S&P Global via its division formerly known as SNL Financial.

¹⁰Chen et al. (2021) confirm the accuracy of the Morningstar holdings data, while pointing out that the summary descriptions of fund portfolios reported to Morningstar can be inaccurate. I do not use these latter summary descriptions.

range of securities than the insurance sector, which is instead predominantly invested in bonds. Hence, in comparing the two sectors it is helpful to focus on funds that are specialists in fixed income. These specialist funds hold more than 80 percent of all corporate bond positions in the Morningstar data.¹¹ In 2015, there are 1,957 U.S. specialist funds and 5,090 non-U.S. ones, with an average of \$1.7 billion and \$0.7 billion in assets each. The fund sector is less concentrated than the insurance sector. In the United States, the largest 10 specialist funds only hold 21 percent of all securities, while the largest 50 own 44 percent. Outside of the United States, these numbers are 12 and 26 percent.¹²

Data on transacted bond prices in my baseline analysis are from the TRACE database, which includes nearly all corporate bond transactions in the United States.¹³ Table 2 shows the characteristics of the corporate bond holdings of insurance firms and fixed income specialist funds. The two sectors are fairly well-aligned, for instance in terms of the size of the average bond held and the shares of bonds held that are senior or have callable features. However, there are also differences: for example, insurers tend to hold bonds of slightly longer duration, with an average duration of 7 years as opposed to 5.5 years for U.S. funds. These differences highlight the necessity of imposing fixed effects for bond characteristics, as they could otherwise introduce confounding variation. Importantly, insurers are very averse to holding high-yield bonds (that is, bonds with a speculative-grade rating). While 37 percent of U.S. funds' corporate bond holdings are in high-yield debt, only 5 percent of insurers' positions are. This pattern results from the significantly higher risk-based regulatory capital charges on high-yield bonds faced by U.S. insurers (Ellul et al. 2011, Becker and Ivashina 2015). Since high-yield bonds are rarely held by insurers, my analysis throughout the paper excludes them.

Figure 1 plots the aggregate value of the positions in U.S. investment-grade corporate bonds observed in the insurance and fund holdings micro-data over time, alongside the total size of the market and the corresponding sectoral shares. An advantage of this empirical setting is the high degree of coverage of the entire market that I can achieve with security-

¹¹I classify funds as fixed income specialists using Morningstar's industry categorization. I select funds where Morningstar's *broad category group* is fixed income.

¹²My analysis also makes use of additional data on bond characteristics, sourced from Mergent's Fixed Income Securities Database (FISD), Factset, S&P Global, Moody's Default and Debt Recovery (DRD) Database, and Dealogic. Appendix Section A provides more details on data construction and further descriptive statistics.

¹³TRACE is distributed by FINRA. I apply to it the data cleaning steps in Dick-Nielsen (2014), which are common in the literature. Throughout the paper, I define U.S. corporate bonds as bonds that are reportable in TRACE. These include virtually all U.S. dollar-denominated bonds issued by U.S. firms and by U.S. affiliates of foreign corporate groups. Bonds denominated in non-USD currencies are not included in TRACE.

level holdings. By 2020, insurers' positions cover 35 percent of the market, with a further 22 and 9 percent accounted for by U.S. and foreign mutual funds and ETFs. Altogether, the positions in the micro-data account for two thirds of the entire size of the market. The share of these bonds held by insurers has been declining, starting at 52 percent in 2007, while the share held by funds has been increasing, from a baseline level of 14 percent for U.S. funds and 4 percent for foreign funds in 2007.

2 Investor Base Effects for U.S. Corporate Bonds

This section of the paper introduces the baseline empirical results in the setting of the U.S. domestic corporate bond market. I discuss their robustness to varying empirical specifications, and finally document the sharply different propensities of insurers and funds to fire-sell corporate bonds in a crisis, which constitutes the portfolio mechanism underlying my estimates.

2.1 Empirical Definitions

My baseline empirical analysis focuses on the universe of U.S. investment-grade corporate bonds that are actively traded, as reported in TRACE.¹⁴ The number of outstanding bonds in the sample grows from 7,951 in 2007 to 11,293 in 2020, while the average issue size increases from \$407 million to \$698 million.¹⁵ For each bond in TRACE, I construct a time series of returns in excess of Treasury bonds of matched duration. I let $P_{i,t}$ be the traded price of bond i in week t : this is constructed as the median price in the last day of week t in which bond i trades, plus accrued interest. The *raw* weekly net return on bond i is then $R_{i,t} = \frac{P_{i,t} + C_{i,t}}{P_{i,t-1}} - 1$, where $C_{i,t}$ is the coupon payment in the relevant time period, if any.¹⁶ I also let $D_{i,t}$ be the duration of bond i at time t , and $R_t^f(D)$ be the weekly return on a basket of Treasury bonds of duration D . The excess return on bond i is then defined relative to a duration-matched

¹⁴I define a bond as actively traded if—after implementing the cleaning steps in [Dick-Nielsen \(2014\)](#), which remove erroneous records—there is at least one completed trade observation recorded in TRACE in the full sample period (2005 to 2020). Hence my analysis excludes the set of bonds which, although potentially TRACE-eligible, do not enter the dataset since they are never traded in the relevant sample period.

¹⁵Appendix Table [A.5](#) provides summary statistics on the sample of U.S. investment-grade corporate bonds.

¹⁶The baseline definition of the return $R_{i,t}$ uses prices inclusive of accrued interest: equivalently, if working with prices $P_{i,t}^c$ defined to be exclusive of accrued interest, the return can be written as $R_{i,t} = \frac{P_{i,t}^c + AI_{i,t} + C_{i,t}}{P_{i,t-1}^c + AI_{i,t-1}} - 1$.

Treasury benchmark:¹⁷

$$R_{i,t}^e = R_{i,t} - R_t^f(D_{i,t}). \quad (1)$$

Subtracting the returns on a matched basket of Treasuries lets me remove the component of returns that is purely due to risk-free interest rate variation, and instead focus on that which is due to credit spread variation. I refer to the cumulative excess return since the first trading date t_i^0 of bond i as:

$$\text{CR}_{i,t} = \prod_{\tau=t_i^0}^t (1 + R_{i,\tau}^e). \quad (2)$$

The *insurer share* in bond i at time t is

$$\phi_{i,t} = \frac{X_{i,t}^I}{V_{i,t}}, \quad (3)$$

where $V_{i,t}$ is the outstanding market value of bond i and $X_{i,t}^I$ is the market value of all insurers' holdings in bond i . Given the frequency of the holdings data, these positions are measured on a quarterly basis, so in practice $X_{i,t}^I$ corresponds to the latest recorded position.

The baseline analysis hones in on the return properties of corporate bonds during aggregate market downturns. In order to do this, I compute *drawdowns* for each bond in each of the major market downturns in the U.S. corporate bond market: these are summary measures that capture a bond's total loss in value during a given downturn event. With t_e^S denoting the start date of event e and t_e^E its end date, the drawdown of corporate bond i in the event window $\mathcal{T}_e = (t_e^S, \dots, t_e^E)$ is defined as

$$\zeta_{i,e} = \max_{\tau \in \mathcal{T}_e} \left(1 - \frac{\text{CR}_{i,\tau}}{\text{CR}_{i,t_e^S}} \right). \quad (4)$$

The drawdown is therefore a number between 0 and 1, with a higher value corresponding to a more severe loss in value of a particular bond in a given market downturn episode. So for example, a value $\zeta_{i,e} = .4$ for the Great Recession episode would correspond to a bond whose cumulative return index $\text{CR}_{i,t}$ —at its trough during the Great Recession of 2008-09—reached

¹⁷If a bond does not trade in a given week, the returns $R_{i,t}$ and $R_t^f(D_{i,t})$ are computed relative to the latest trading week. I show in Section 2.3 that excluding these cases does not meaningfully affect my estimates. The sovereign returns at varying durations are obtained from CRSP whenever available, including only regular non-callable Treasury issues, and from Factset's zero-coupon sovereign benchmarks otherwise.

a value equal to 60 percent of its starting one in June 2007.¹⁸

2.2 Insurer Ownership and Bond Return Properties in Crises

I investigate how bond drawdowns vary with investor base composition in the four largest bond market downturns in the sample period. The shaded areas of Figure 2 highlight the event periods, whose start and end dates correspond to the prior trough and subsequent peak values in the Gilchrist and Zakrajšek (2012, henceforth GZ) index, which provides an aggregate measure of credit spreads on U.S. corporate bonds. The first downturn event is the Great Recession of 2008-09, during which the GZ index rose more than four-fold from a prior trough of 1.7 percent in June 2007 to a peak of 7.9 percent in December 2008. The second event window, during which the U.S. sovereign debt was downgraded and global markets tumbled, starts in April 2011 and ends in October 2011. The third window runs from August 2014 to February 2016, a period which featured large collapses in commodity prices. Finally, the fourth event window, starting in January 2020 and ending in March 2020, corresponds to the global COVID crisis and the resulting market crash, which involved large price dislocations for corporate bonds (Haddad et al. 2021, Ma et al. 2021).¹⁹

The baseline specification pools these events and regresses the log drawdown on bond i in event e on *ex ante* insurer shares ϕ_{i,t_e^S-12} and a series of interacted fixed effects that are progressively saturated:

$$\log \zeta_{i,e} = \alpha + \beta \cdot \phi_{i,t_e^S-12} + \text{Interacted Fixed Effects} + \varepsilon_{i,e}. \quad (5)$$

The coefficient of interest in these regressions is β , which (in the absence of confounders) approximates the semi-elasticity of drawdowns to *ex ante* insurer shares. Importantly, the insurer shares ϕ_{i,t_e^S-12} are measured in the quarter prior to the start of each event window, to avoid concerns of reverse causation.

¹⁸As detailed in Section 2.2, the start and end dates for each episode are chosen mechanically by identifying peaks and the corresponding prior troughs in the Gilchrist and Zakrajšek (2012) corporate credit spread index. The event window corresponding to the Great Recession starts in June 2007 and ends in December 2008. My results are not dependent on the use of the maximum operator in equation (4): in fact, computing drawdowns on a fixed window corresponding to the full event range yields analogous results, as I show in Section 2.3.

¹⁹The unusually rapid recovery of the corporate bond market in 2020 might be linked to the Federal Reserve’s announcement of its willingness to intervene in the market. See also Becker and Benmelech (2021) and Falato et al. (2021b) for discussions of the resilience of the U.S. corporate bond market in the COVID crisis. Relatedly, Kargar et al. (2021) and O’Hara and Zhou (2021) study corporate bond liquidity in the COVID episode.

The inclusion of fixed effects in equation (5) is necessary in order to tackle the concern that the estimated relationship between insurer shares and subsequent drawdowns may be driven by omitted factors that influence both the inherent riskiness of a given security and the likelihood of insurers to hold it. At their most saturated level, these fixed effects interact event dummies, issuer dummies, and security characteristics dummies.²⁰ The security characteristics in the fixed effects set include all of the following: (i) bond credit rating, measured using S&P and Moody’s ratings and ranging from AAA to BBB-,²¹ which is the threshold between high-yield and investment-grade debt; (ii) bond duration, which is non-parametrically clustered into categories;²² (iii) bond size, as captured by dummies for deciles of value outstanding; (iv) bond seniority, as captured by a binary dummy variable; (v) bond coupon type, either floating-rate or fixed-rate; and (vi) bond callability, also captured via a binary dummy variable. I do not include currency fixed effects since TRACE only includes U.S. dollar-denominated bonds.

When including the most saturated set of interacted fixed effects, therefore, the estimates only rely on variation in insurer ownership and drawdowns across nearly identical bonds. Intuitively, the regression only compares bonds issued by the same legal entity, of similar duration and size, with the same credit rating, and so forth. As an example of the identifying variation, consider two bonds issued by American Express Credit Corporation, the credit card issuing subsidiary of American Express, with respective CUSIP codes 0258M0BX6 and 0258M0CN7. These two bonds were both senior, five-year notes issued six months apart, in December 2005 and June 2006, and had analogous characteristics: besides being issued by the same firm and having comparable duration, they were both rated A+ at issuance and in the years following, were both \$500 million bonds, and matched on all other characteristics included in the fixed effect set. In the quarter prior to the Great Recession, however, these two bonds had different investor bases, with insurer shares of 22 percent and 4 percent. Despite

²⁰Issuers are defined via Mergent FISD’s issuer IDs, which capture legal entities. In the few cases in which Mergent FISD issuer IDs are not available, issuers are defined by pooling six-digit CUSIP issuer codes at the legal entity level via the CUSIP Global Services (CGS) associated issuers file. These issuer definitions are tighter than at the ultimate parent level since they correspond to individual firm subsidiaries. This provides a more rigorous degree of control for issuer fixed effects.

²¹I use the S&P conventions in listing credit ratings, although the analysis uses ratings from both S&P and Moody’s. The Moody’s ratings are converted to the equivalent on the S&P scale. Therefore, the analysis includes dummies for the following rating categories: AAA, AA+, AA, AA-, A+, A, A-, BBB+, BBB, BBB-.

²²The duration categories used in the baseline exercise are the following, which correspond to commonly used industry classifications: 0-1 years, 1-3 years, 3-5 years, 5-10 years, 10-15 years, 15-20 years, and greater than 20 years. Section 2.3 shows that these results are robust to using stricter categories in which duration is rounded to the nearest year. It also demonstrates robustness to other varying specification choices.

their similarities, these two bonds experienced different return dynamics during the Great Recession, as plotted in Appendix Figure A.1, with drawdowns of 32 percent and 51 percent respectively. This exemplifies the type of sample variation that the estimates are based on.

Table 3 reports the estimates from equation (5) as I progressively saturate the set of included fixed effects. These estimates are obtained via weighted least squares (WLS) using bond sizes as weights. Column 1 reports results with event fixed effects only, with an estimate of $\hat{\beta} = -.43$. Moving to the right, as more interactions are added to the fixed effects set, the estimated coefficient remains overall stable. Column 2 interacts event dummies with size and duration dummies, while columns 3 and 4 further introduce rating and issuer dummies. As intuitive, the inclusion of duration dummies increases the estimated magnitude—given that insurers tend to hold longer-duration bonds which are more volatile, biasing the estimates towards zero in the absence of a duration covariate—while the inclusion of rating and issuer dummies reduces it. Finally, the estimates in column 5 include the fully-saturated set of fixed effects described above, which encompasses all of event, size, duration, rating, issuer, bond callability, seniority, and coupon type. The estimated regression coefficient from the fully-saturated specification in column 5 is $\hat{\beta} = -.41$. This semi-elasticity estimate has the interpretation that increasing *ex ante* insurer holdings in each bond by 50 percentage points (half of the issue size) reduces the drawdown in a subsequent crisis by about 20 percent.

These effects display heterogeneity across the various events. Appendix Figure A.2 demonstrates this by estimating equation (5) on different subsamples of the data, split by event.²³ The estimates are higher in magnitude for the Great Recession and the COVID crisis than for the other two events in the sample. The market downturns in the Great Recession and in the COVID crisis were faster and more severe than those in 2011 and 2014-16, suggesting a certain degree of non-linearity, with investor base effects mattering more in harsher crises. I explore this non-linearity further in Appendix Section B, while Appendix Section F investigates heterogeneity among insurers, including across more and less financially constrained ones (Kojien and Yogo 2015, Becker et al. 2021, Ge and Weisbach 2021).

To quantify the degree of relative bond mispricing that accompanies these effects, it is helpful to consider the average standard deviation of yield spreads to Treasury within groups of

²³Appendix Figure A.2 also repeats the estimation on two mutually exclusive and collectively exhaustive subsamples, one for bonds rated AAA through A- and one for bonds rated BBB+ through BBB-. The estimates are quite homogeneous across these two credit rating groups.

matching bonds—that is, bonds issued by the same firms and having the same characteristics. This metric provides a simple measure of the level of mispricing in the corporate bond market at a particular point in time.²⁴ Prior to the Great Recession, the standard deviation of yield spreads for matching bonds averaged less than 20 basis points. However, the metric increases gradually over the course of the Great Recession event window, reaching a peak of nearly 100 basis points between September and December of 2008. This metric similarly spikes during other crisis periods, particularly in the COVID episode, peaking at nearly 60 basis points in March 2020, consistent with the evidence of price dislocations in this period by [Haddad et al. \(2021\)](#). These magnitudes align with other instances of relative mispricing of related securities in crises: for example, [Hu et al. \(2013\)](#) document yield differences with a standard deviation of about 20 basis point for nearly identical U.S. Treasuries during the 2008-09 financial crisis. Given the vastly lower liquidity and higher degree of segmentation of the corporate bond market relative to the Treasury market, it is natural to expect yield discrepancies to be more pronounced in the former.²⁵

A natural prediction of an asset fire sales mechanism is that these price dislocations for similar bonds should tend to revert over time. To examine this, in Appendix Section J, I perform a reversal analysis in the spirit of [Coval and Stafford \(2007\)](#): I construct a long-short portfolio which takes positions in investment-grade bonds that are outstanding as of 2007Q1 and have at least one counterpart bond that is issued by the same firm and has the same characteristics. Within each group of matching bonds, the portfolio takes long positions in bonds that have a pre-crisis insurer share below the group median, and short positions in bonds above the group median. The cumulative abnormal returns on this portfolio trace out the dynamics of the within-firm pricing pattern, revealing that the within-firm mispricing during the Great Recession fully reverted on a 4 to 6 quarters horizon, consistent with theoretical models of fire sales.

An elevated degree of relative mispricing for nearly identical bonds that lasts at such horizons indicates the presence of some persistent limits to arbitrage in the corporate bond market and of slow arbitrage capital adjustment ([Duffie 2010](#)). While a full investigation

²⁴The time series for this measure is shown in Appendix Figure A.15. Since this measure also depends partly on the overall level of credit spreads, it is not best suited to assess the dynamics of price reversal, which are instead studied formally in the analysis of Appendix Section J that is introduced later in the present section.

²⁵Indeed, the magnitude of the price discrepancies I document more closely aligns with spreads such as the CDS-bond basis, which persistently amounted to hundreds of basis points for investment-grade debt after the financial crisis ([Bai and Collin-Dufresne 2019](#)). See [Musto et al. \(2018\)](#) for further evidence from Treasuries.

of the reasons for these limits to arbitrage and capital adjustment is beyond the scope of this paper, several institutional features make these limits likely. Limits to arbitrage may be supported, for example, by search and matching frictions, given the over-the-counter (OTC), decentralized nature of the corporate bond market (Duffie, Gârleanu and Pedersen 2005, 2007). Additionally, regulatory frictions may prevent insurers from shifting their portfolios to underpriced bonds: life insurers, for instance, have flexibility to hold assets at historical cost towards regulatory capital requirements (Ellul et al. 2015), which generates incentives not to trade bonds in downturn periods since this would entail having to recognize capital losses. Perhaps most saliently, in large crises arbitrage capital is unavailable because all investor sectors with a large presence in the corporate bond market are, on net, attempting to sell. Appendix Figure A.3, for example, uses transactions data from the U.S. Flow of Funds to demonstrate that between 2008Q1 and 2009Q1 banks, broker-dealers, government-sponsored enterprises, insurers, mutual funds and ETFs, pensions funds, and foreign investors were all net sellers of corporate bonds.²⁶

2.3 Ruling Out Other Potential Explanations

I now consider the robustness of these baseline estimates to further variations in the empirical specification. The bars labeled 2 through 6 in Figure 3 demonstrate the robustness of the estimates to the introduction of further interactions to the full set of fixed effects. Bar 2 introduces an additional interaction with stricter duration categories, with duration rounded to the nearest year, in order to provide an even more stringent matching of similar bonds. Bar 3 adds an interaction with bond age, rounded to the nearest year, to rule out alternative explanations based on the differential properties of on-the-run and off-the-run bonds. Bar 4 includes an additional interaction with a dummy for whether a bond is issued under SEC rule 144A, which establishes special trading rules for certain restricted or controlled securities. Bar 5 adds an interaction with dummies that capture the set of a bond’s lead underwriters, to rule out confounding driven by factors fixed at the firm-underwriter level. Lastly, bar 6 introduces an interaction with dummies for the set of a bond’s covenants, hence removing bonds with

²⁶The only non-corporate sector that was a net buyer of corporate bonds in the financial crisis was the household sector. In the Flow of Funds, the household sector is a residual category, including various institutions such as hedge funds and nonprofits. A further limitation of these statistics is that the Flow of Funds does not report asset holdings separately for U.S. corporate bonds and foreign bonds held by U.S. investors.

non-matching covenants from the estimation sample. The details of the covenants analysis are described in Appendix Section H, which also provides statistics on the prevalence of each type of covenant in the sample: as discussed there, a key reason for why the results are largely unaffected by accounting for covenants is that most variation in covenants occurs across firms rather than within-firm, so that the within-firm exercise already absorbs a majority of it.

Bars 7 through 12 in Figure 3 additionally consider alternative sample treatments. Bar 7 excludes bonds with a zero insurer share. Bar 8 excludes the holdings of AIG from the computation of the insurer positions $X_{i,t}^I$, to rule out that the results may be influenced by AIG’s special financial situation following its bailout under the Troubled Asset Relief Program (TARP) in 2008-09. Bar 9 repeats the estimation using drawdowns computed by cumulating the raw returns $R_{i,t}$ rather than the excess returns $R_{i,t}^e$. Bar 10 uses an alternative measure of drawdowns, $\tilde{\zeta}_{i,e} = \max\{1 - CR_{i,t_e^E} / CR_{i,t_e^S}, 0\}$, which applies the same window to all bonds for a given event. Bar 11 drops any bonds that trade less frequently than once a week in a given event period. Bar 12 excludes callable bonds. Bar 13 excludes firms in the financial and utilities sectors, as detailed in Appendix Section K: this is helpful to strengthen identification, since bonds issued by firms in these sectors tend to be more complex than average (for instance, due to special collateralization or covenants), and hence excluding them removes the possibility of investors sorting on these complexities. In all cases, these alternative empirical specifications result in what are substantively the same estimates as the baseline.

Further, to show that the dependency of price dynamics on investor composition is primarily operative during market downturns, I estimate an additional version of the baseline specification in which I add non-crisis bond betas as a control variable. I estimate betas for each bond-event pair using return observations in the two-year window prior to the start of the given downturn event (which therefore all fall outside of the downturn windows in Figure 2). The betas are computed with respect to the return on a value-weighted index of all actively traded US investment-grade bonds. Bar 14 shows the estimates including this control variable, which remain quantitatively similar to the baseline.

Lastly, to rule out that the results might be driven by the choice of the drawdown measure, I estimate a version of the model in equation (5) in which the left-hand side variable follows the simplest possible specification, being defined simply as the net cumulative excess return on each bond during each downturn event, given by $\bar{R}_{i,e} = \prod_{\tau=t_e^S}^{t_e^E} (1 + R_{i,\tau}^e) - 1$. The estimates from

this specifications are not directly comparable in magnitude to the others in this section, and they are also expected to have opposite sign (since a reduction in the drawdown corresponds to an increase in the return). For this reason, the estimates are shown separately in Appendix Table A.6. The resulting point estimate of .08 is fully consistent with the baseline analysis: since the average drawdown in the sample, across events, is 19 percentage points, the baseline estimate $\hat{\beta} = -.41$ implies that an increase in the insurer share from 0 to 1 should lead on average to a roughly 8 percentage points higher excess return, which is precisely what the alternative specification finds, confirming the findings.

2.4 Inspecting the Mechanism

The portfolio mechanism resulting in the presence of investor base effects is best understood by comparing the trading behavior of insurers and funds in market downturns. Figure 4 does this in the context of the Great Recession. The solid green line plots the cumulative change in corporate bond holdings for the asset-weighted average insurer in the sample over the course of the Great Recession, relative to their level in 2008Q1. The changes in corporate bond holdings are measured using par values, so that they reflect pure quantity changes rather than any valuation effects owing to variation in market prices. By 2009Q1, the (asset-weighted) average insurer had reduced its holdings of corporate bonds by less than 5 percent. The solid blue line shows the equivalent portfolio dynamics for the asset-weighted average mutual fund or ETF in the sample: over the same time period, funds had reduced their corporate bond holdings by 15 percent, more than three times as much as insurers. The much more aggressive portfolio response of funds relative to insurers is consistent with the safe hands hypothesis laid out in the introduction and exemplifies the heterogeneity in quantity dynamics that accompanies the price dynamics documented in Section 2.2.

The reason for such a dramatic portfolio response of investment funds in the Great Recession does not primarily trace back to discretionary trading decisions made by fund portfolio managers, but rather to redemptions coming from their clients. The dashed red line in Figure 4 plots the component of the cumulative net corporate bond sales by funds that is *household-driven*. The household-driven sales measure is defined by constructing counterfactual portfolio holdings for each fund assuming the fund were to maintain the same portfolio allocation weights as in 2008Q1, and varying the overall size of its portfolio according to its

net redemptions. The household-driven net sales shown in the dashed red line therefore purely reflect household decisions to redeem assets from the fund sector, and abstract from active portfolio management choices of allocation across asset classes.²⁷ Comparing the dashed red and solid blue lines, more than two thirds of the net sales of corporate bonds by the fund sector in the Great Recession were household-driven.

In contrast, insurers do not face such aggressive capital redemptions from their clients. Figure 5 compares the time series properties of the net inflows and outflows of assets for funds and insurance firms. The dashed red line plots the quarterly value of the aggregate net client inflows for fixed income specialist funds in the sample, scaled by the lagged value of their total assets under management. The solid blue line plots the analogous concept for U.S. insurance firms, which is the quarterly net change in the sector’s aggregate policy reserves, scaled by lagged total policy reserves.²⁸ Funds face volatile and oftentimes negative net client inflows, which are strongly procyclical (Feroli et al. 2014, Goldstein et al. 2017, Timmer 2018). Clients withdraw their capital from funds precisely at the time when the market is doing worst, as evidenced by a quarterly time-series correlation of -44 percent between net fund inflows and the GZ index. In contrast, the net growth in insurers’ policy reserves is almost never negative: while funds can experience severe household-driven capital liquidations, insurers are by and large insulated. Moreover, net inflows into the insurance sector have diametrically opposite dynamic properties: they are countercyclical, being largest when the market is doing poorly, with a quarterly correlation with the GZ index of 32 percent.²⁹

This sharp contrast between the properties of net capital inflows into the two sectors is explained by differences in the contractual structure of their liabilities. Mutual funds predominantly adopt an open-end contractual structure, which allows clients to redeem capital continuously and with no limitations or charges. Capital intermediated by insurance companies, on the other hand, is not as easily redeemable. Most insurance products, for example annuity policies (a popular product offered by life insurers consisting of fixed income asset management

²⁷For work on mutual funds’ liquidity management following capital withdrawals, see Chernenko and Sunderam (2016), Choi et al. (2020), and Ma et al. (2021). For ease of exposition, here I abstract from differences in the redemption process for ETFs, which are instead detailed in Appendix Section L.

²⁸Policy reserves are the primary liability item on insurers’ balance sheets and correspond to the actuarial value of future expected payments to policyholders: they therefore increase when clients subscribe to insurance products. As I discuss in Appendix Section E, the same-quarter net change in insurers’ policy reserves is the chief determinant of the net amount of new bond purchases by the insurer sector.

²⁹The counter-cyclicity of flows into the insurance sector may reflect rising household risk aversion and increased demand for defined-benefit investment products such as annuities following severe market downturns.

with embedded longevity insurance), impose contractual restrictions and surrender charges for early capital withdrawals (Securities and Exchange Commission 2020). Contractual provisions of this kind shield insurers from sudden, household-driven capital withdrawals.

3 Shift-Share Approach and Discussion of Identification

In this section I investigate why nearly identical bonds have different investor bases. I show that this results from the combination of three institutional features of the market. First, large insurers mostly buy bonds on the primary market and hold them for a long time. Second, each insurer only purchases a small subset of all the bonds available on the primary market at any given time. Third, these random idiosyncratic purchase decisions matter in the aggregate because of a deviation from the law of large numbers, as the size distribution of insurers is unusually kurtotic (that is, fat-tailed). I exploit these institutional features to construct a shift-share instrument and outline how the instrument relates to the identifying variation exploited by the baseline ordinary least squares (OLS) regression. I discuss causal identification, both via the shift-share IV approach and via a complementary partial identification approach.

3.1 Understanding Variation in Insurer Ownership

The results in Section 2 demonstrate that similar bonds with different investor base composition experience different drawdowns during crises. Figure 6 shows that these results rely on the presence of sizable residual variation in insurer ownership for bonds issued by the same firms and with the same characteristics. The blue histograms plot the sample distribution of drawdowns and insurer shares, both pooled across events. The red histograms show the distribution of the same quantities after residualizing them against the fixed effects included in the baseline.³⁰ It is evident that, even after accounting for all these observables, there is still considerable dispersion in both drawdowns and ownership.

³⁰Formally, the red histograms show the residuals $\nu_{i,e}^{\zeta}$ and $\nu_{i,e}^{\phi}$ from the following regressions of drawdowns $\zeta_{i,e}$ and insurer shares ϕ_{i,t_e^S-12} on the fully-saturated set of interacted fixed effects:

$$\zeta_{i,e} = \alpha + \text{Interacted Fixed Effects} + \nu_{i,e}^{\zeta}, \quad (6)$$

$$\phi_{i,t_e^S-12} = \alpha + \text{Interacted Fixed Effects} + \nu_{i,e}^{\phi}. \quad (7)$$

This residual variation in insurer ownership across similar securities results from the idiosyncratic portfolio allocations of a few large insurers, made predominantly at the time of bond issuance. Corporate bond markets have a much lower degree of liquidity and higher transaction costs than equity markets, so that it is difficult for market participants to hold all outstanding, actively traded securities. This low degree of liquidity and ease of trading leads to idiosyncrasy in holdings at the security level. To show this, I compare insurers’ holdings in the largest 500 actively traded investment-grade bonds to their holdings of equities in the S&P 500, an equally-sized benchmark in a different market.³¹ Figure 7 ranks insurers by the overall size of their portfolios, and shows the share of the 500 largest investment-grade bonds that they hold an open position in (*blue dots*) alongside the share of S&P 500 equities that they hold an open position in (*red dots*), all measured cross-sectionally as of 2017Q4. While there is dispersion in the share of S&P 500 equities that each insurer holds, a large number of insurers cluster at 100 percent, which reflects the feasibility of holding all securities in the market portfolio in equity markets. In contrast, none of the 300 largest insurers hold an open position in more than 61 percent of the 500 largest bonds, and most have positions in a much lower share of these bonds, often less than 20 percent.³² Appendix Figure A.8 repeats this exercise for mutual funds and ETFs and shows similar patterns, while Appendix Section E gives an overview of the primary-market bond distribution process—which is led by *underwriters* who collect institutional investors’ bids—and discusses further the incentives that insurers have to acquire large positions in a small number of bonds.

Importantly, most of the corporate bond trading of large insurers is on the primary market. The absence of sizable follow-up portfolio rebalancing means that the ownership initially determined at issuance persists strongly over time. To quantify this persistence, I let $\tau(i)$ be the date of issuance of bond i and consider the following regressions of insurer shares over

³¹Since some insurers hold equities through mutual funds and ETFs rather than directly, these estimates are based on a version of the holdings data that unwinds positions in funds and attributes ownership of the underlying positions to the insurers. Indirect ownership of corporate bonds by insurers through investment funds, in contrast, is very rare. While insurers specialize in fixed income investing in order to support their policy liabilities, equity holdings are small relative to their bond positions.

³²A t -test for differences in means across the bond and equity data in Figure 7 rejects the hypothesis of equal means with $p < .01$. Further, these results are not driven by the particular choice of bonds. To show this, Appendix Figure A.7 repeats the analysis in Figure 7 with smaller reference sets of bonds—either the 100 largest or the 250 largest actively traded U.S. investment corporate bonds. The results within these smaller sets of securities are analogous.

time against insurer shares at issuance, $\phi_{i,\tau(i)}$, at varying quarterly horizons s :

$$\phi_{i,\tau(i)+12s} = \alpha_s + \lambda_s \phi_{i,\tau(i)} + \varepsilon_{i,s}. \quad (8)$$

In the specification above, the quarterly horizon index s is multiplied by a factor of 12 since—for consistency with the rest of the paper—the time step frequency is weekly: therefore, incrementing s by one corresponds to a one-quarter shift in the series. The solid blue line in Figure 9a plots the estimated passthrough coefficients λ_s for progressively longer horizons s , which remain stably close to 1 up to a 36-quarter horizon. The solid red line in Figure 9b plots the R-squared values from these regressions, which plateau at a remarkably high level: nearly 10 years after the initial issuance of an investment-grade corporate bond, the insurer shares first determined at bond issuance still explain 40 percent of the variation in contemporaneous ownership. For comparison, Figure 8 also plots the same regression coefficients (*dashed blue line*) and R-squared values (*dashed red line*) using mutual fund and ETF ownership shares in place of insurer shares in equation (8), both of which decay much faster than the equivalent metrics for insurers: by quarter 16, the passthrough coefficient λ_s declines to less than .5 for funds while remaining close to .9 for insurers, while the R-squared value is nearly three times higher for the latter over the same horizon. The faster decay for funds reflects the more active trading of this investor sector over time. In contrast, the absence of meaningful portfolio rebalancing by insurers adds to the idiosyncratic nature of their bond investments.

Large insurers do so much of their trading on the primary market because of the drastically higher degree of liquidity on it vis-à-vis the secondary market. To characterize the decay in liquidity as bond trading moves to the secondary market, I consider the following specification for varying quarterly horizons s after bond issuance, which takes the same form as (8):

$$\mathcal{T}_{i,\tau(i)+12s} = \alpha_s + \lambda_s \mathcal{T}_{i,\tau(i)} + \varepsilon_{i,s}, \quad (9)$$

where $\mathcal{T}_{i,t}$ is a measure of quarterly trading activity in bond i , either the quarterly number of trades or quarterly dollar trading volume (both from TRACE). Figure 9 plots the estimated coefficients λ_s for U.S. investment-grade corporate bonds. Quarterly trading volume drops off sharply after issuance, on average being just 10 percent of the issuance-quarter level a single quarter after issuance. The number of trades decays as well starting in the first quarter after

issuance, although less markedly, as the average size of a trade also diminishes. Appendix Figure A.9 compares the size distribution of trades on the primary versus the secondary market: it shows that indeed the average trade on the primary market is more than five times as large as on the secondary market. Consistent with the view that large insurers buy on the primary market because of its enhanced liquidity, the preference for buying at issuance increases with insurer portfolio size. Appendix Figure A.10 plots the share of corporate bond holdings purchased on the primary market by insurer size decile and shows that this metric increases monotonically with insurer size: larger insurers value the primary market more.

Lastly, one might expect the idiosyncratic portfolio allocations that I have highlighted to cancel out in the aggregate if the law of large numbers applies in this particular empirical context. As already seen in Table 1, however, asset holdings in the insurance sector are very concentrated, with the largest 50 insurers owning more than 70 percent of the sector’s total invested assets. The idiosyncratic portfolio decisions of individual insurers matter in determining the sector’s aggregate ownership because of a large deviation from the law of large numbers, resulting from the heavily fat-tailed size distribution of insurer portfolios. Appendix Section C provides a formal statistical test and quantifies these deviations from the law of large numbers, thanks to which idiosyncratic portfolio decisions—made mostly at the time of bond issuance—generate the sizable residual variation in aggregate insurer ownership shown in Figure 6. As a concrete example of this data-generating process, suppose that MetLife—one of the very largest insurance firms—bids for an allocation in IBM’s 10-year bond issued in a given year but, having obtained this allocation, does not then go to the trouble of participating in the following year’s 10-year IBM offering, which it might view as largely substitutable. Since MetLife accounts for a large share of the insurance sector’s total assets, its lack of participation in the latter bond’s offering will tend to lower the resulting aggregate insurer share—even if underwriters might have otherwise allocated the two bonds so as to have a similar institutional ownership profile.

3.2 An Instrument Based on Random Purchases at Issuance

I now further narrow down the identifying variation in order to purely exploit the idiosyncratic allocations to large insurers on the primary market. I begin with an accounting decomposition that highlights the various sources of identifying variation in the OLS regressions of Section

2. The insurer share $\phi_{i,t}$ is the sum of its value at issuance $\phi_{i,\tau(i)}$ and deviations from it due to trading on the secondary market, $\hat{\phi}_{i,t} = \phi_{i,t} - \phi_{i,\tau(i)}$. One can further decompose the insurer share at issuance $\phi_{i,\tau(i)}$ by summing over potential outcomes for individual insurers:

$$\phi_{i,t} = \sum_{j \in \mathcal{J}} \xi_{i,j} \cdot \tilde{\phi}_{i,j,\tau(i)} + \hat{\phi}_{i,t}, \quad (10)$$

where $\mathcal{J}$ is the set of insurers in the sample, $\xi_{i,j}$ is a dummy variable indicating whether insurer j acquired bond i at issuance, and $\tilde{\phi}_{i,j,\tau(i)}$ is a potential outcome—the share in bond i acquired by insurer j on the primary market conditionally on purchasing the bond.³³

This expression has three sets of terms, which correspond to three distinct sources of variation. The variation due to secondary-market trading, which one might worry could be endogenous,³⁴ is captured by $\hat{\phi}_{i,t}$. A second source of variation comes from the *intensive margin* of primary-market trading and is captured by the potential outcomes $\tilde{\phi}_{i,j,\tau(i)}$: when insurers have more capital available to invest in new bonds, they will acquire bigger positions. As shown in Section 2.4, however, the capital available to insurers for new investments is not orthogonal to credit market conditions: rather, insurers have more capital on hand when aggregate credit spreads are high. Since the credit quality of new corporate bond issues is known to vary systematically over the credit cycle (Greenwood and Hanson 2013), it might be desirable to strip away this variation as well. This leaves us with the third and last source of variation in ownership: the *extensive margin* of primary market trading—meaning the binary allocations of new bonds to each insurer, captured by the purchase indicators $\xi_{i,j}$.

I construct a shift-share instrument that isolates the variation coming from this last channel. The instrument removes the secondary-market component $\hat{\phi}_{i,t}$ and replaces the intensive margin terms $\tilde{\phi}_{i,j,\tau(i)}$ with measures of insurer size that are time- and bond-invariant. Insurer size, $\bar{\phi}_j$, is measured as the average size of insurer j 's primary market bond purchases relative to the aggregate insurance sector (so that $\sum_{j \in \mathcal{J}} \bar{\phi}_j = 1$). The instrument therefore captures

³³Naturally, the potential outcome $\tilde{\phi}_{i,j,\tau(i)}$ is only observable for insurers who actually acquire bond i at issuance, so that $\xi_{i,j} = 1$. This potential outcomes interpretation corresponds to an assumed data generating process in which purchases are strictly positive conditionally on entering the market, so that $\tilde{\phi}_{i,j,\tau(i)} > 0$. This assumption is innocuous and only matters for exposition.

³⁴The analysis in Section 2.2 mitigates this concern by using insurer shares measured one quarter before each event, which are close to those at issuance, as shown in Section 3.1.

the collective size of the set of insurers that acquire bond i at issuance:

$$Z_i = \sum_{j \in \mathcal{J}} \xi_{i,j} \bar{\phi}_j. \quad (11)$$

This instrument follows the shift-share structure studied in the applied econometrics literature (Goldsmith-Pinkham et al. 2020, Borusyak et al. 2021a), with the purchase indicators $\xi_{i,j}$ functioning as the *shares* and the size coefficients $\bar{\phi}_j$ functioning as the *shifters*. Intuitively, the instrument takes a high value if the insurers who acquire a bond at issuance are particularly large ones rather than small ones.

The exclusion restriction requires that $\mathbb{E}[Z_i \varepsilon_{i,e} \mid \mathcal{F}_{i,e}] = 0$, where $\mathcal{F}_{i,e}$ contains all the observables included in the regressions and $\varepsilon_{i,e}$ is the error term from the linear model specified in equation (5). A sufficient condition for this restriction is that the extensive margin purchase indicators $\xi_{i,j}$ are exogenous conditionally on the set of interacted fixed effects included in the model, while there is no orthogonality assumption on the size coefficients $\bar{\phi}_j$.³⁵

$$\mathbb{E}[\xi_{i,j} \varepsilon_{i,e} \mid \mathcal{F}_{i,e}] = 0 \text{ for all } i, j. \quad (13)$$

Intuitively, the identifying assumption corresponding to condition (13) is that the matching between *individual insurers* and *particular bonds* on the primary market in a given quarter is uncorrelated with unobserved determinants of future drawdowns that are not explained by issuer effects interacted with observed bond characteristics. This identification approach, which relies on the exogeneity of the shares in the shift-share instrument, is similar to the one outlined and studied by Goldsmith-Pinkham et al. (2020).³⁶

The identifying assumption therefore relies on the idea that insurers are not able to (or do not attempt to) select bonds *within issuer* that will draw down less in a market downturn

³⁵The proof is simple. Using the linearity of expectations:

$$0 = \mathbb{E}[Z_i \varepsilon_{i,e} \mid \mathcal{F}_{i,e}] = \mathbb{E}\left[\sum_j \xi_{i,j} \bar{\phi}_j \varepsilon_{i,e} \mid \mathcal{F}_{i,e}\right] = \sum_j \bar{\phi}_j \mathbb{E}[\xi_{i,j} \varepsilon_{i,e} \mid \mathcal{F}_{i,e}]. \quad (12)$$

³⁶In Appendix Section G, I discuss the interpretation of the shift-share IV estimates if insurers are heterogeneous in their impact on bond drawdowns. Following Goldsmith-Pinkham et al. (2020), I decompose the IV estimator in terms of Rotemberg (1983) weights. In the presence of heterogeneity, the IV approach continues to provide a valid causal estimator—although it should then be interpreted as estimating the weighted average of heterogeneous treatment effects.

for reasons not explained by any of the included observable bond characteristics. If larger insurers were to systematically acquire bonds that do better in crises for unobservable reasons, for example, that would generate problematic sorting, as it would introduce confounding correlation between the purchase indicators $\xi_{i,j}$ and the error terms $\varepsilon_{i,e}$. This kind of sorting constitutes the primary threat to identification within the IV framework, but there is *a priori* reason to believe that it should be absent in practice. Insurers are benchmark investors who do not generally try to outperform the broad investment-grade bond market—and therefore do not engage in strategic allocations across similar bonds within firm. Moreover, while the identifying assumption is not directly testable, one can test a related condition: whether some insurers systematically acquire bonds with a high *residual alpha*—that is, a component of returns that is not explained by the performance of the overall bond market and by observable bond characteristics or issuer effects. The finance literature has developed approaches to tackle this kind of question and differentiate investment ability from noise in the cross-section of investors. Appendix Section D introduces a bootstrap approach following Fama and French (2010). The bootstrap-based test rejects the presence of residual alpha in the cross-section of insurers, providing further support for the identifying assumption.³⁷

Table 4 reports the results from estimating the baseline specification (5) via two-stage least squares (2SLS), using the shift-share variable Z_i to instrument for the aggregate insurer shares ϕ_{i,t_e^S-12} . The 2SLS estimates are shown in columns 3 and 4, with the corresponding OLS estimates shown in columns 1 and 2. The instrument is highly relevant, with first-stage F statistics ranging from 71.8 to 73.3, and the point estimates are similar to the ones from the OLS regressions, although with larger standard errors. Once I impose the fully saturated set of interacted fixed effects in column 8, the point estimate is $\hat{\beta} = -.45$, which aligns with the corresponding OLS estimate $\hat{\beta} = -.41$.³⁸

The relevance of the instrument results precisely from the same institutional features dis-

³⁷The reverse concern—that issuers might be able to forecast insurer demand and time their issuance accordingly—does not pose an issue. This is because the instrument only needs to be exogenous *conditionally* on issuer fixed effects, which would absorb this confounding firm-level behavior.

³⁸Since the insurance holdings data start in 2005, in constructing the instrument for bonds issued prior to this year I use insurers’ holdings in 2005 as a proxy for their holdings at issuance. As shown in Appendix Table A.7, the results are quantitatively consistent when restricting the sample to bonds issued in 2005 or later. Further, Appendix Figure A.11 visualizes the first-stage relationship using a binned scatterplot, while Appendix Figure A.12 demonstrates the robustness of the 2SLS estimates under a range of different specification variations similar to those introduced in the OLS case. The shift-share IV estimates remain significant at the 95 percent confidence level when adopting the *AKM* and *AKM0* inference procedures of Adão et al. (2019).

cussed in Section 3.1. The fact that insurers mostly trade on the primary market makes the primary-market purchase indicators $\xi_{i,j}$ a relevant predictor of the subsequent positions of each insurer j . Moreover, the high degree of idiosyncrasy in each insurer’s purchases generates dispersion in the distribution of the indicators $\xi_{i,j}$, and the deviations from the law of large numbers (reflected in the size terms $\bar{\phi}_j$) imply that this dispersion does not cancel out once I aggregate over insurers. On the whole, the shift-share IV approach discussed in this section provides a sharp characterization of the identifying variation underlying my estimates and confirms causal point-identification.

3.3 Partial Identification Approach

Beyond the shift-share IV framework laid out above, I also consider a complementary partial identification approach in the context of the baseline OLS regressions of Section 2. Suppose that the true data-generating model follows the specification in equation (5), but that it however also includes relevant unobservable omitted variables. In Appendix Section N, I follow Oster (2019) and define a ratio δ that relays the strength of selection on the unobservables relative to selection on the observables included in the regression, which correspond the fully-saturated set of interacted fixed effects for issuer and bond characteristics.

I use the partial identification econometric approach of Oster (2019) to ask how large the ratio δ would have to be in order for the baseline estimates of equation (5) to be consistent with a zero treatment effect of insurer ownership on bond drawdowns. This methodology uses variation in the estimated regression coefficients and R-squared values as I include the observed regressors in order to estimate this threshold value of δ . In this sense, it formalizes common informal arguments that appeal to coefficient stability in the face of selection on observables to argue for robustness to selection on unobservables.

The estimated threshold value of δ for a zero treatment effect is $\bar{\delta} = 3.2$. That is, one would require selection on unobservables to be more than three times as strong as selection on the included observables in order to generate a zero treatment effect. Both Altonji et al. (2005) and Oster (2019) argue that results with a threshold value $\bar{\delta}$ greater than or equal to 1 should be viewed as robust to the potential presence of selection on unobservables. This partial identification approach provides strong evidence that the baseline OLS results obtained by estimating equation (5) are robust to the potential presence of residual omitted variable

bias resulting from selection on unobservables. While point-identification of β in equation (5) through OLS requires the absence of residual omitted variable bias once we include the fully-saturated fixed effects (that is, $\delta = 0$), the partial identification of a negative treatment effect $\beta < 0$ only requires the much weaker assumption that selection on unobservables is less than roughly three times as strong as selection on the included observables ($\delta < 3.2$).

4 Investor Base Effects in Non-U.S. Markets

Having documented the presence of sizable investor base effects in the U.S. corporate bond market, I now turn to investigating this phenomenon across countries, and show that the same empirical patterns hold pervasively. I focus on corporate bonds issued by firms in three large advanced economies with well-developed securities markets: the European Monetary Union (EMU), the United Kingdom, and Canada. More specifically, I include in the analysis all firms resident in these countries as well as their foreign subsidiaries.³⁹ The sample includes the universe of investment-grade bonds issued by these firms except for those that are TRACE-reportable, which are instead included in the analysis of Section 2 and are predominantly U.S. dollar-denominated bonds issued by U.S. subsidiaries.

4.1 Empirical Setup

In the non-U.S. setting, I focus on the holdings of mutual funds and ETFs, since I do not directly observe the portfolios of other domestic institutional sectors. I define the proxy *safe hands share* in bond i at time t by looking at the complement of fund holdings:

$$\tilde{\phi}_{i,t} = 1 - \frac{X_{i,t}^F}{V_{i,t}}, \quad (14)$$

where $X_{i,t}^F$ is the market value of total mutual fund and ETF holdings in bond i . The measure $\tilde{\phi}_{i,t}$ is intended to proxy for the holdings of investors such as domestic insurers and pension funds that have a lower propensity to fire-sell their assets in market downturns. The positions $X_{i,t}^F$ include the holdings both of funds domiciled in the same country as the issuer of bond i and of foreign funds, although the former are usually dominant given the presence of home

³⁹To establish links between affiliates and their corporate parents, I use the ultimate parent aggregation algorithm of Coppola et al. (2021).

bias in portfolio investment. To confirm the validity of this proxy approach, I repeat my baseline estimates for U.S. corporate bonds, as in equation (5), using the proxy safe hands shares $\tilde{\phi}_{i,t}$ instead of the insurer shares $\phi_{i,t}$. Column 1 of Table 5 reports the resulting estimate ($\hat{\beta} = -.39$), which is very similar to the baseline ($\hat{\beta} = -.41$). The proxy shares and the insurer shares accordingly have a high correlation of 51 percent in the U.S. sample.

A second difference between the U.S. and non-U.S. settings is that traded prices (as in TRACE) are generally not available for the universe of corporate bonds outside the United States. I therefore use prices provided by S&P Global, which are traded when possible and otherwise estimated. Estimated bond prices are widely used in fixed income portfolio management and index construction (Bloomberg L.P. 2017) and have also been commonly used in academic research (He et al. 2017, Nozawa 2017, Kelly et al. 2021). In computing excess returns per equation (1), I use the risk-free returns applicable to each economy: these are returns on duration-matched baskets of German bunds for EMU corporate bonds, British gilts for U.K. corporate bonds, and Canadian sovereign bonds for Canadian corporate bonds.

4.2 Investor Composition and Bond Returns Across Countries

I consider the following specification for non-U.S. corporate bonds, which is analogous to the baseline one in equation (5). Letting $c(i)$ be the currency of denomination of bond i , the regression specification is:

$$\log \zeta_{i,e} = \alpha + \beta \cdot \tilde{\phi}_{i,t_e^S-12} + \rho_{c(i)} + \text{Interacted Fixed Effects} + \varepsilon_{i,e}. \quad (15)$$

This specification additionally includes dummies for the currency of denomination of each bond i . In the U.S. exercise, this is unnecessary since all bonds in the sample are denominated in U.S. dollars. In the non-U.S. sample, however, there is currency variation, with significant shares of the bonds in the sample denominated in Euros, British pounds, or Canadian dollars. While all bond prices are measured in the currency of denomination of the bond, so that nominal exchange rate movements do not directly contribute to variation in the drawdown measures $\zeta_{i,e}$, the currency dummies absorb any systematic variation in drawdowns that might indirectly be determined by a bond's currency of denomination. For parsimony and consistency with the U.S. exercise, I use the same set of events, with the same start and end dates, as in Section 2.

Column 2 in Table 5 shows the estimate from equation (15) in a pooled sample that includes all of the EMU, the United Kingdom, and Canada. Columns 3 through 5 show the estimates individually for each of the three countries. In all cases, the estimates show a negative and significant relationship between *ex ante* safe-hand shares $\tilde{\phi}_{i,t_e^s-12}$ and subsequent drawdowns. Moreover, the estimates are quantitatively consistent with the ones obtained from the U.S. exercise. For instance, the pooled non-U.S. semi-elasticity estimate $\hat{\beta} = -.51$ in column 2 aligns with the $\hat{\beta} = -.41$ estimate obtained in the U.S. setting. Altogether, the non-U.S. estimates shown in Table 5 demonstrate that the presence of investor base effects in corporate bond markets is pervasive across countries, in a way that is consistent with the mechanism laid out in this paper. These further results rule out potential alternative explanations that rely on the idiosyncratic characteristics of the U.S. corporate bond market and insurance or mutual fund sector, and underscore the generality of the effects that I document.

5 How Fire Sales Transmit to Real Firm Outcomes

An important question is whether the differential price impact for bonds in the secondary market that I documented in the preceding sections transmits to real firm outcomes. In this section, I focus on an event study of U.S. firms in the Great Recession and show that secondary market fire sales significantly influence funding conditions in the primary bond market for the affected firms, ultimately with consequences for their real investment dynamics. In documenting that variation in a firm’s cost of capital in the bond market impacts real investment, I provide evidence for a channel that, although theoretically central in macro-finance (Gilchrist and Zakrajšek 2007, Philippon 2009, Almeida et al. 2017), has been difficult to establish empirically. Since this analysis is at the firm level, I cannot use the identification strategy employed so far, which focuses on variation across similar bonds issued by the same firms. My approach instead couples a difference-in-differences empirical strategy with a version of the shift-share instrument of Section 3 that is aggregated to the firm level. This firm-level empirical strategy allows me to remove potential confounders, although under stronger identifying assumptions than required in the security-level exercise.

5.1 Aggregating the Instrument to the Firm Level

I begin by defining the relevant quantities at the firm level. Letting $\mathcal{I}_k$ be the set of bonds issued by firm k and outstanding in 2007Q1, I construct the firm-level insurer share Φ_k by aggregating insurer ownership across securities, with all positions also measured in 2007Q1, before the start of the financial crisis:

$$\Phi_k = \frac{\sum_{i \in \mathcal{I}_k} X_{i,2007Q1}^I}{\sum_{i \in \mathcal{I}_k} V_{i,2007Q1}}. \quad (16)$$

The firm-level insurer share Φ_k is plausibly endogenous to firm outcomes during the Great Recession, since for example insurers might sort into holding bonds issued by firms that are inherently more resilient in crises. To remove this type of confounding variation, I construct a firm-level instrument by aggregating the security-level instrument Z_i defined in equation (11):

$$\mathcal{Z}_k = \sum_{i \in \mathcal{I}_k} \rho_{i,k} \cdot Z_i, \quad (17)$$

where the terms $\rho_{i,k} = \frac{V_{i,2007Q1}}{\sum_{i' \in \mathcal{I}_k} V_{i',2007Q1}}$ are weights proportional to bond size. The firm-level instrument $\mathcal{Z}_k$ captures whether the bonds issued by a given firm are *on average* allocated to large rather than smaller insurers at issuance.

Aggregating to the firm level still retains the majority of the variation present at the security level: the sample standard deviation for the firm-level instrument $\mathcal{Z}_k$, for instance, only shrinks to 76 percent of that for the unaggregated firm-level instrument Z_i . This variation persists at the firm level because the cross-section of bonds outstanding per firm is generally small, although it has long right tails: within the sample, the median and mean number of bonds outstanding per firm in 2007 are 3 and 17, respectively.⁴⁰ The idiosyncratic portfolio allocations of the few largest insurers therefore matter even once I aggregate to the borrower level, so that the instrument continues to be relevant.

The identifying assumption required for the firm-level instrument to be valid is stronger than the one required for the security-level regressions of Section 3. Namely, the firm-level identifying assumption is that the matching between individual insurers and particular bonds

⁴⁰Naturally, these numbers are higher within the set of firms that contribute identifying variation in the highly-saturated, baseline security-level analysis. For example, among the issuers that contribute variation in the most saturated specification (column 5) reported in Table 3, the median and mean number of bonds outstanding per firm at a given time are 8 and 30.

on the primary market is uncorrelated with determinants of firm outcomes during the crisis that are not explained by *observable firm characteristics*, including average firm-level bond attributes. In contrast, identification at the security level only requires the much weaker condition that the primary-market matching between insurers and bonds is exogenous conditionally on *issuer fixed effects* interacted with *observed bond characteristics*. In requiring stronger assumptions for identification at progressively higher levels of aggregation, this setting parallels themes that are common in the macro-finance literature (Khwaja and Mian 2008, Wolf 2021).

5.2 Firm Financing and Investment Outcomes

To assess the association between bond ownership and firm investment outcomes, I merge the data on bond ownership with yearly firm financials from Compustat.⁴¹ I estimate the following dynamic difference-in-differences specification with continuous treatment intensity:

$$\text{Investment}_{k,t} = \alpha_k + \gamma_t + \sum_{\tau} \beta_{\tau} \Phi_k \mathbb{1}_{\tau=t} + \text{Controls} + \varepsilon_{k,t}, \quad (18)$$

where $\text{Investment}_{k,t}$ is the investment rate of firm k in year t , measured either as yearly capital expenditures (CAPX) over lagged total assets or as yearly CAPX plus acquisitions over lagged total assets. The firm fixed effects α_k remove confounding variation that is fixed at the firm level and is state- and time-invariant, while the time fixed effects γ_t absorb time trends in investment that are common to all firms. In turn, the coefficients β_{τ} trace out the dynamic equilibrium association between firm investment and the pre-crisis share of the firm's bonds that is owned by insurers.

The controls consist of firm-level covariates, all measured prior to the crisis, interacted with time dummies: these include fixed effects for issuer rating, average duration of the firm's outstanding bonds (using the same categories as in the baseline regressions), two-digit industry, and deciles of firm size and firm leverage. I estimate this specification via just-identified two-stage least squares, by also interacting the firm-level instrument $\mathcal{Z}_k$ with time dummies (Wooldridge 2010). The sample excludes speculative-grade issuers. While a series of recent papers has shown that two-way fixed effects specifications such as the one in equation (18) can suffer from bias in settings with staggered treatment adoption (Callaway and Sant'Anna

⁴¹In order to match the level at which the Compustat data is reported, I aggregate the security-level data to the ultimate parent level.

2021, Goodman-Bacon 2021), this type of design does not face the same issues when treatment timing is common across observations (Wooldridge 2021), as is the case here.⁴²

Figure 10 displays the estimates from this instrumented dynamic difference-in-differences specification. The red line plots the estimated coefficients using the CAPX measure, while the blue line plots the estimates for the CAPX plus acquisitions measure. The pre-crisis coefficients serve as an effective placebo test and are all close to zero. Following the crisis, there is a large and significant impact of *ex ante* insurer shares on firm investment, which gradually fades toward zero in 2010 and 2011. Increasing firm-level insurer ownership by 50 percentage points—which is equivalent to a movement from the 10th to the 90th percentile in the distribution of the insurer shares Φ_k —is associated with an increase in CAPX equal to roughly 1.5 percent of assets, and with an increase in CAPX and acquisitions ranging from 2 to 3 percent of assets. These estimated magnitudes are broadly similar to those found in the literature studying bank lending. Amiti and Weinstein (2018), for example, show that a 20 percent contraction in bank lending supply leads to a decline of about 2 percentage points in annualized fixed physical investment rates in a sample of Japanese firms.⁴³ Further, the results remain robust to excluding firms in the financial and utilities sectors, as discussed in Appendix Section K.

Longer-lived effects on real outcomes are consistent with the within-firm pricing dynamics documented in Section 2. This is because of the presence of different dynamics within firm and across firms: the within-firm mispricing dynamics—while helpful to achieve identification in saturated empirical specifications—are not the same channel as for real firm outcomes. Rather, fire sales at the level of the firm itself have a larger impact on firm outcomes, and these naturally have slower reversal dynamics. Indeed, faster within-firm reversal is consistent with theories that postulate different degrees and dynamics of demand-driven price impact for assets depending on the availability of close substitutes—as emphasized, for instance, in the models of Chaudhary et al. (2022), Gabaix and Koijen (2022), and Davis et al. (2023).

These investment effects are particularly strong for firms that have a high share of their existing debt coming due during the crisis. To see this, I estimate the following static version

⁴²The regressions include observations up to the year 2011. I end the sample in 2011 since later observations overlap with the U.S. sovereign downgrade event. The excluded year dummy is for 2007, so that all coefficients estimate effects relative to 2007. All financial ratios are winsorized at the 1st and 99th percentiles.

⁴³See column 5 of Table 2 in Amiti and Weinstein (2018). In non-crisis years, the average investment rates for firms in the sample are 6 percent (CAPX over assets) and 9.3 percent (CAPX and acquisitions over assets).

of the difference-in-differences design in equation (18):

$$\text{Investment}_{k,t} = \alpha_k + \gamma_t + \beta \cdot \Phi_k \cdot \text{Post}_t + \eta \cdot \Phi_k \cdot \text{Post}_t \cdot \mathcal{M}_k + \text{Controls} + \varepsilon_{k,t}, \quad (19)$$

where Post_t is a dummy that takes the value of one for post-crisis observations (i.e. observations for the years starting in 2008) and zero otherwise, and $\mathcal{M}_k$ is the share (by value) of a firm’s bonds outstanding as of 2007Q1 that are maturing in 2008 or 2009. The coefficient η captures the incremental impact of insurer ownership on firm investment that is due to maturing debt. Table 6 shows the estimates from this regression. Since the instrumented regressions are under-identified when including the additional parameter η , I show these results using ordinary least squares. Increasing the share of maturing bonds by 50 percentage points has a further impact of about 1 percent of assets on CAPX and of 1.5 percent of assets on CAPX plus acquisitions: fire sales matter more for firms with more debt coming due, which face challenges in rolling over existing bond financing.⁴⁴

The reason fire sales ultimately matter for these real investment outcomes is that they impact the allocation of capital to firms via new bond issues: while secondary market prices are not directly allocative, they set the reference point for what investors are willing to pay for new issues on the primary market. To investigate this channel, I examine the impact of investor composition on the dynamics of new bond issuance. I estimate the following instrumented dynamic specification, which is equivalent to the one in equation (18):

$$Y_{k,t} = \alpha_k + \gamma_t + \sum_{\tau} \beta_{\tau} \Phi_k \mathbb{1}_{\tau=t} + \text{Controls} + \varepsilon_{k,t}, \quad (20)$$

where $Y_{k,t}$ is one of two outcomes—either an indicator for whether firm k issued a *new* bond in year t , or the average offering yield on new bonds offered by firm k in year t , conditionally on issuing a new bond. I continue to use the same set of controls as in specification (18), and I estimate these regressions via two-stage least squares. Figure 12a shows the estimated dynamic coefficients when using the new issuance indicators as the outcome. These coefficients capture the dynamic response of the probability that a firm will issue a new bond in each of

⁴⁴For related discussions of the role of corporate debt maturity on real firm outcomes in crises, see also Almeida et al. (2011) and Benmelech et al. (2019, 2021). Additionally, He and Xiong (2012) introduce a model where rollover problems impact priced credit risk. Massa et al. (2013) study the relationship between bondholders’ investment horizon and firm capital structure.

the crisis years. Figure 12b plots the coefficients when using offering yields as the outcome: these coefficients capture the impact of insurer ownership on firms’ cost of new borrowing in the corporate bond market.

Firms with a higher *ex ante* insurer share have a substantially higher likelihood of issuing new debt following the crisis and face a lower user cost of capital when doing so: increasing the firm-level insurer share by 50 percentage points is associated with an increase in the likelihood of new bond issuances of between 25 and 30 percentage points in 2008 and 2009, and with a reduction in offering yields on new debt of the order of 120 basis points. In accordance with the idea that secondary market dynamics directly impact the pricing of new issues on the primary market, estimating specification (20) with secondary-market rather than primary-market yields results in quantitatively analogous dynamics.⁴⁵

Summing up, these results are consistent with the notion that fire sales impact real outcomes because firms become less able to raise new bond financing at a low cost when their existing debt trades at a large discount. In this view, firms whose debt is owned predominantly by insurers suffer milder declines in the demand for their bonds during sharp market downturns, or equivalently do not face as high an increase in their cost of capital. Overall, the evidence supports the hypothesis that firms which are unable to access new financing through the corporate bond market might ultimately reduce their investment in response to the reduced availability of capital, particularly if they need to use cash on hand to pay for existing debt that is coming due. These outcomes highlight the value of a safe-handed investor base from an issuer’s perspective.⁴⁶

5.3 Existing Relationships and Primary Market Purchases

As mentioned above, secondary-market bond prices set the reference point for investors’ willingness to pay for new issues on the primary market. I now document an additional channel through which existing holdings by insurers also impact primary market dynamics: the presence of existing lending relationships in the bond market between issuers and insurers makes the latter more likely to participate in new issuances in bad times. Extending the notation of

⁴⁵The secondary-market yield estimates are shown in Appendix Figure A.13.

⁴⁶For related themes in equity markets, see Edmans et al. (2012), Derrien et al. (2013), Aghion et al. (2013), and Dessaint et al. (2019), who study the impact of the institutional composition of firm equity-holders and equity price dynamics on real outcomes. Appendix Section M provides a structural interpretation of my real-side results using a *q*-theoretic model with costly external financing.

Section 3, I let $\xi_{i,j,k,t}$ be an indicator for whether investor j purchased bond i , issued by firm k , at the time of its issuance t . Further, I let $\mathcal{R}_{i,j,k,t}$ be an indicator for whether investor j already held in its portfolio other bonds issued by firm k at the time of issuance of the new bond i . I estimate the following specification:

$$\xi_{i,j,k,t} = \eta_t + \alpha_k + \gamma \mathcal{R}_{i,j,k,t} + \sum_s \beta_s \mathcal{R}_{i,j,k,t} D_{s,k,t} + \sum_s \rho_s D_{s,k,t} + \varepsilon_{i,j,k,t}, \quad (21)$$

where $D_{s,k,t}$ is a dummy indicating whether the average secondary-market yield spread to Treasury of firm k at time t falls within the decile s of firm-level yield spreads. This specification quantifies how existing bond market lending relationships (captured by $\mathcal{R}_{i,j,k,t}$) impact the probability that an investor will purchase a firm's new bond issues (captured by $\xi_{i,j,k,t}$): this effect is expressed by the coefficient γ . Further, the specification investigates how the impact of relationships on purchase likelihoods varies as a firm's existing bonds trade at a discount: this *incremental* fire sale effect is captured by the sequence of coefficients β_s . The firm fixed effects α_k absorb firm-specific but time-invariant differences in purchase likelihoods, while the time fixed effects η_t absorb common time trends.

Figure 12 plots the estimated coefficients β_s from specification (21), as well as the baseline relationship effect γ , for the sample including all new investment-grade U.S. bond issuances between 2005 and 2020, and all U.S. insurance firms. The estimated value of γ shows the presence of a strong relationship effect to begin with: insurers are 5.5 percentage points more likely to buy a firm's new bonds if they have an existing lending relationship in place, which is a large effect when compared to the average value of the purchase indicator $\xi_{i,j,k,t}$ in the sample, equal to 4.8 percent. Strikingly, this relationship effect strongly increases as the secondary-market yields on a firm's existing bonds go up: going from the bottom decile to the top decile of yield spreads, the relationship effect increases by nearly 8 percentage points.

These estimates illustrate that investors not only strongly favor buying bonds from the firms they already hold in their portfolios, but also become more and more reluctant to lend via the bond market to new firms as their debt falls in value on the secondary market.⁴⁷ From

⁴⁷This behavior would be compatible, for example, with a model in which institutional investors carry out in-depth research on a new firm before adding its bonds to their portfolio, and in which secondary-market yields are a particularly salient signal of risk in these initial issuer assessments. These facts are also consistent with the results of [Barbosa and Ozdagli \(2021\)](#), who find that, on the intensive margin, life insurance companies purchase larger amounts of new bond issues if they have existing relationships with corporate borrowers, and that during the COVID crisis these relationships insulated bonds from market-wide shocks.

a firm’s perspective, these relationship effects compound the value of insurers’ participation in their bond investor base: beyond not contributing to fire sales, these insurers will be more likely to continue to provide new credit in bad times.

6 Pricing Implications Outside of Downturns

While the analysis above has focused on price dynamics during market downturn periods, the impact of investor composition is also relevant in normal times. This section examines equilibrium pricing outcomes outside of crises via the following specification, which has each bond’s yield spread to Treasury $s_{i,t}$, in basis points, on the left hand side:

$$s_{i,t} = \alpha + \bar{\Lambda} \phi_{i,t-12} + \tilde{\Lambda} \phi_{i,t-12} \text{GZ}_t + \text{Interacted Fixed Effects} + \varepsilon_{i,t}. \quad (22)$$

The interaction with the [Gilchrist and Zakrajšek \(2012\)](#) index captures variation in the relationship between insurer ownership and yield spreads over the credit cycle in a simple parametric manner, and as before the fixed effects interaction also includes time fixed effects. The red line in [Figure 13](#) plots the estimated total effect $\Lambda_t = \bar{\Lambda} + \tilde{\Lambda} \text{GZ}_t$ over time in the sample of U.S. investment-grade bonds, while the blue shaded area shows the 95 percent confidence band. These estimates show that outside of crises higher insurance ownership is associated with higher yield spreads, with a total effect of the order of 20 basis points as the insurer share varies from 0 to 1. During crises, the sign of the estimated effect inverts as bonds with higher insurer ownership experience lower value losses.

A literature in finance has documented the presence of a corporate bond liquidity premium, leading less liquid corporate bonds to have higher equilibrium yield spreads ([Chen et al. 2007](#), [Bao et al. 2011](#)). Higher yield spreads on insurer-held bonds reflect these bonds’ endogenous liquidity properties. This liquidity channel may interact with a potential additional channel owing to the endogenous differences in intertemporal risk generated by investor base heterogeneity. The return properties I have documented give bonds with a higher insurer share desirable state-contingent properties, in that they do better in bad states of the world: if these properties are priced, this channel would tend to counteract the liquidity mechanism, increasing the prices of bonds with a high insurer share (and therefore lowering their yield spreads). The price outcomes documented in [Figure 13](#) plausibly reflect the equilibrium in-

teraction of these two pricing channels, although I do not quantitatively disentangle them, as that would require imposing a specific asset pricing model.

Additionally, in Appendix Section I, I examine the relationship between investors composition and primary market prices outside of crisis periods, by estimating hedonic models of offering yields as a function of bond characteristics and investor holdings. The primary market results confirm and extended the secondary market equilibrium pricing patterns reported here. Higher insurance ownership is associated with slightly higher offering yields in the primary market: a 10 percentage points higher insurer share associates with a 12 to 13 basis points offering yield increase, consistent with the secondary market pricing. In addition to revealing the endogenous illiquidity of insurer-held bonds, in the primary market context the higher offering yields are evidence that firms pay a slightly higher cost of capital when their bonds are allocated in larger share to insurers. In this sense, the impact of insurer ownership on firms' cost of capital is state contingent: issuers effectively pay a few basis points in higher cost of capital in bad times, but this results in a lower cost of capital during crisis times—such that placing bonds with insurers has features of an insurance contract.

7 Conclusion

In this paper I provide quantitative causal evidence demonstrating that investor base composition is an important determinant of bond price dynamics and firm financing outcomes in response to aggregate credit cycle fluctuations. I establish causal identification by comparing nearly identical bonds issued by the same firms and via a complementary shift-share instrumental variable approach. In both the United States and in international markets, ownership by *safe-handed* institutional investors such as insurers shields securities from large value losses in crises, to a large extent thanks to the lower exposures of these financial intermediaries' liabilities to sizable household-driven capital withdrawals. Firms whose debt is owned by investors less prone to fire sales experience milder disruptions in their primary-market bond financing and real investment during crises: they issue more new bonds and pay a lower cost of capital when doing so, which ultimately allows them to invest relatively more. These facts demonstrate how the distribution of ownership of financial assets across heterogeneous institutional investors affects the magnitudes of fire sales and real firm outcomes over the credit cycle.

References

- Adão, Rodrigo, Michal Kolesár, and Eduardo Morales**, “Shift-share designs: Theory and inference,” *The Quarterly Journal of Economics*, 2019, *134* (4), 1949–2010.
- Aghion, Philippe, John Van Reenen, and Luigi Zingales**, “Innovation and institutional ownership,” *American Economic Review*, 2013, *103* (1), 277–304.
- Almeida, Heitor, Igor Cunha, Miguel A Ferreira, and Felipe Restrepo**, “The real effects of credit ratings: The sovereign ceiling channel,” *The Journal of Finance*, 2017, *72* (1), 249–290.
- Almeida, Heitor, Murillo Campello, Bruno Laranjeira, and Scott Weisbenner**, “Corporate debt maturity and the real effects of the 2007 credit crisis,” *Critical Finance Review*, 2011, *1* (1), 3–58.
- Altonji, Joseph G, Todd E Elder, and Christopher R Taber**, “An evaluation of instrumental variable strategies for estimating the effects of Catholic schooling,” *Journal of Human Resources*, 2005, *40* (4), 791–821.
- Alvarez, Fernando and Urban J Jermann**, “Efficiency, equilibrium, and asset pricing with risk of default,” *Econometrica*, 2000, *68* (4), 775–797.
- Alvarez, Fernando, Andrew Atkeson, and Patrick J Kehoe**, “Money, interest rates, and exchange rates with endogenously segmented markets,” *Journal of Political Economy*, 2002, *110* (1), 73–112.
- Amiti, Mary and David E Weinstein**, “How much do idiosyncratic bank shocks affect investment? Evidence from matched bank-firm loan data,” *Journal of Political Economy*, 2018, *126* (2), 525–587.
- Anton, Miguel and Christopher Polk**, “Connected stocks,” *The Journal of Finance*, 2014, *69* (3), 1099–1127.
- Autor, David H**, “Outsourcing at will: The contribution of unjust dismissal doctrine to the growth of employment outsourcing,” *Journal of Labor Economics*, 2003, *21* (1), 1–42.
- Bai, Jennie and Pierre Collin-Dufresne**, “The CDS-bond basis,” *Financial Management*, 2019, *48* (2), 417–439.
- Bao, Jack, Jun Pan, and Jiang Wang**, “The illiquidity of corporate bonds,” *The Journal of Finance*, 2011, *66* (3), 911–946.
- Barberis, Nicholas and Andrei Shleifer**, “Style investing,” *Journal of Financial Economics*, 2003, *68* (2), 161–199.
- Barberis, Nicholas, Andrei Shleifer, and Jeffrey Wurgler**, “Comovement,” *Journal of Financial Economics*, 2005, *75* (2), 283–317.
- Barbosa, Monica and Ali Ozdagli**, “Is public debt arm’s length? Evidence from corporate bond purchases of life insurance companies,” *Working Paper*, 2021.
- Bartik, Timothy J**, “Who benefits from state and local economic development policies?,” *WE Upjohn Institute for Employment Research Paper*, 1991.
- Becker, Bo and Efraim Benmelech**, “The resilience of the US corporate bond market during

- financial crises,” *Working Paper*, 2021.
- Becker, Bo and Victoria Ivashina**, “Reaching for yield in the bond market,” *The Journal of Finance*, 2015, 70 (5), 1863–1902.
- Becker, Bo, Marcus M Opp, and Farzad Saidi**, “Regulatory forbearance in the US insurance industry: The effects of removing capital requirements for an asset class,” *Forthcoming at the Review of Financial Studies*, 2021.
- Benmelech, Efraim, Carola Frydman, and Dimitris Papanikolaou**, “Financial frictions and employment during the great depression,” *Journal of Financial Economics*, 2019, 133 (3), 541–563.
- Benmelech, Efraim, Nittai Bergman, and Amit Seru**, “Financing labor,” *Review of Finance*, 2021, 25 (5), 1365–1393.
- Bernanke, Ben S and Mark Gertler**, “Agency costs, collateral, and business fluctuations,” *American Economic Review*, 1986, 79 (1), 14–31.
- Bernanke, Ben S, Mark Gertler, and Simon Gilchrist**, “The financial accelerator in a quantitative business cycle framework,” *Handbook of Macroeconomics*, 1999, 1, 1341–1393.
- Blanchard, Olivier J and Lawrence F Katz**, “Regional evolutions,” *Brookings Papers on Economic Activity*, 1992.
- Bloomberg L.P.**, “Fixed income cash pricing sources,” *Technical Report*, 2017.
- Borusyak, Kirill, Peter Hull, and Xavier Jaravel**, “Quasi-experimental shift-share research designs,” *Forthcoming at the Review of Economic Studies*, 2021.
- Borusyak, Kirill, Xavier Jaravel, and Jann Spiess**, “Revisiting event study designs: Robust and efficient estimation,” *Working Paper*, 2021.
- Brainard, William C and James Tobin**, “Pitfalls in financial model building,” *American Economic Review*, 1968, 58 (2), 99–122.
- Bretscher, Lorenzo, Lukas Schmid, Ishita Sen, and Varun Sharma**, “Institutional corporate bond demand,” *Working Paper*, 2020.
- Brunnermeier, Markus K and Yuliy Sannikov**, “A macroeconomic model with a financial sector,” *American Economic Review*, 2014, 104 (2), 379–421.
- Caballero, Ricardo J and Alp Simsek**, “A model of fickle capital flows and retrenchment,” *Journal of Political Economy*, 2020, 128 (6), 2288–2328.
- Caballero, Ricardo J and Arvind Krishnamurthy**, “International and domestic collateral constraints in a model of emerging market crises,” *Journal of Monetary Economics*, 2001, 48 (3).
- Callaway, Brantly and Pedro HC Sant’Anna**, “Difference-in-differences with multiple time periods,” *Forthcoming at the Journal of Econometrics*, 2021.
- Calvo, Guillermo A**, “Capital flows and capital-market crises: The simple economics of sudden stops,” *Journal of Applied Economics*, 1998, 1 (1), 35–54.
- Chaudhary, Manav, Zhiyu Fu, and Jian Li**, “Corporate bond multipliers: Substitutes matter,” *Working Paper*, 2022.

- Chen, Huaizhi, Lauren Cohen, and Umit G Gurun**, “Don’t take their word for it: The misclassification of bond mutual funds,” *The Journal of Finance*, 2021, 76 (4), 1699–1730.
- Chen, Long, David A Lesmond, and Jason Wei**, “Corporate yield spreads and bond liquidity,” *The Journal of Finance*, 2007, 62 (1), 119–149.
- Chernenko, Sergey and Adi Sunderam**, “The real consequences of market segmentation,” *The Review of Financial Studies*, 2012, 25 (7), 2041–2069.
- Chernenko, Sergey and Adi Sunderam**, “Liquidity transformation in asset management: Evidence from the cash holdings of mutual funds,” *NBER Working Paper No. 22391*, 2016.
- Chodorow-Reich, Gabriel, Andra Ghent, and Valentin Haddad**, “Asset insulators,” *The Review of Financial Studies*, 2021, 34 (3), 1509–1539.
- Choi, Jaewon, Saeid Hoseinzade, Sean Seunghun Shin, and Hassan Tehranian**, “Corporate bond mutual funds and asset fire sales,” *Journal of Financial Economics*, 2020, 138 (2), 432–457.
- Coppola, Antonio, Matteo Maggiori, Brent Neiman, and Jesse Schreger**, “Redrawing the map of global capital flows: The role of cross-border financing and tax havens,” *The Quarterly Journal of Economics*, 2021, 136 (3), 1499–1556.
- Coval, Joshua and Erik Stafford**, “Asset fire sales (and purchases) in equity markets,” *Journal of Financial Economics*, 2007, 86 (2), 479–512.
- Darmouni, Olivier, Oliver Giesecke, and Alexander Rodnyansky**, “The bond lending channel of monetary policy,” *Working Paper*, 2020.
- Davis, Carter, Mahyar Kargar, and Jiacui Li**, “Why is asset demand inelastic?,” *Working Paper*, 2023.
- Derrien, François, Ambrus Kecskés, and David Thesmar**, “Investor horizons and corporate policies,” *Journal of Financial and Quantitative Analysis*, 2013, 48 (6), 1755–1780.
- Dessaint, Olivier, Thierry Foucault, Laurent Frésard, and Adrien Matray**, “Noisy stock prices and corporate investment,” *The Review of Financial Studies*, 2019, 32 (7), 2625–2672.
- Di Tella, Sebastian**, “Uncertainty shocks and balance sheet recessions,” *Journal of Political Economy*, 2017, 125 (6), 2038–2081.
- Dick-Nielsen, Jens**, “How to clean enhanced TRACE data,” *Working Paper*, 2014.
- Du, Wenxin, Alexander Tepper, and Adrien Verdelhan**, “Deviations from covered interest rate parity,” *The Journal of Finance*, 2018, 73 (3), 915–957.
- Duffie, Darrell**, “Presidential address: Asset price dynamics with slow-moving capital,” *The Journal of finance*, 2010, 65 (4), 1237–1267.
- Duffie, Darrell, Nicolae Gârleanu, and Lasse H Pedersen**, “Over-the-counter markets,” *Econometrica*, 2005, 73 (6), 1815–1847.
- Duffie, Darrell, Nicolae Gârleanu, and Lasse H Pedersen**, “Valuation in over-the-counter markets,” *The Review of Financial Studies*, 2007, 20 (6), 1865–1900.
- Dumas, Bernard**, “Two-person dynamic equilibrium in the capital market,” *The Review of Financial Studies*, 1989, 2 (2), 157–188.

- Edmans, Alex, Itay Goldstein, and Wei Jiang**, “The real effects of financial markets: The impact of prices on takeovers,” *The Journal of Finance*, 2012, 67 (3), 933–971.
- Ellul, Andrew, Chotibhak Jotikasthira, and Christian T Lundblad**, “Regulatory pressure and fire sales in the corporate bond market,” *Journal of Financial Economics*, 2011, 101 (3), 596–620.
- Ellul, Andrew, Chotibhak Jotikasthira, Christian T Lundblad, and Yihui Wang**, “Is historical cost accounting a panacea? Market stress, incentive distortions, and gains trading,” *The Journal of Finance*, 2015, 70 (6), 2489–2538.
- Falato, Antonio, Ali Hortacsu, Dan Li, and Chaehee Shin**, “Fire-sale spillovers in debt markets,” *Journal of Finance*, 2021.
- Falato, Antonio, Itay Goldstein, and Ali Hortaçsu**, “Financial fragility in the COVID-19 crisis: The case of investment funds in corporate bond markets,” *Journal of Monetary Economics*, 2021.
- Fama, Eugene F and Kenneth R French**, “Luck versus skill in the cross-section of mutual fund returns,” *The Journal of Finance*, 2010, 65 (5), 1915–1947.
- Feroli, Michael, Anil K Kashyap, Kermit L Schoenholtz, and Hyun Song Shin**, “Market tantrums and monetary policy,” *Conference Paper, US Monetary Policy Forum*, 2014.
- Forbes, Kristin J and Francis E Warnock**, “Capital flow waves: Surges, stops, flight, and retrenchment,” *Journal of International Economics*, 2012, 88 (2), 235–251.
- Frazzini, Andrea and Owen A Lamont**, “Dumb money: Mutual fund flows and the cross-section of stock returns,” *Journal of Financial Economics*, 2008, 88 (2), 299–322.
- Gabaix, Xavier and Matteo Maggiori**, “International liquidity and exchange rate dynamics,” *The Quarterly Journal of Economics*, 2015, 130 (3), 1369–1420.
- Gabaix, Xavier and Ralph SJ Koijen**, “In search of the origins of financial fluctuations: The inelastic markets hypothesis,” *Working Paper*, 2020.
- Gabaix, Xavier and Ralph SJ Koijen**, “In search of the origins of financial fluctuations: The inelastic markets hypothesis,” *Working Paper*, 2022.
- Ge, Shan and Michael S Weisbach**, “The role of financial conditions in portfolio choices: The case of insurers,” *Journal of Financial Economics*, 2021.
- Gilchrist, Simon and Egon Zakrajšek**, “Investment and the cost of capital: New evidence from the corporate bond market,” *NBER Working Paper No. 13174*, 2007.
- Gilchrist, Simon and Egon Zakrajšek**, “Credit spreads and business cycle fluctuations,” *American Economic Review*, 2012, 102 (4), 1692–1720.
- Goldsmith-Pinkham, Paul, Isaac Sorkin, and Henry Swift**, “Bartik instruments: What, when, why, and how,” *American Economic Review*, 2020, 110 (8), 2586–2624.
- Goldstein, Itay, Hao Jiang, and David T Ng**, “Investor flows and fragility in corporate bond funds,” *Journal of Financial Economics*, 2017, 126 (3), 592–613.
- Goodman-Bacon, Andrew**, “Difference-in-differences with variation in treatment timing,” *Forthcoming at the Journal of Econometrics*, 2021.
- Gourinchas, Pierre-Olivier, Walker Ray, and Dimitri Vayanos**, “A preferred-habitat model

- of term premia and currency risk,” *Working Paper*, 2020.
- Greenwood, Robin and David Thesmar**, “Stock price fragility,” *Journal of Financial Economics*, 2011, *102* (3), 471–490.
- Greenwood, Robin and Dimitri Vayanos**, “Price pressure in the government bond market,” *American Economic Review*, 2010, *100* (2), 585–590.
- Greenwood, Robin and Dimitri Vayanos**, “Bond supply and excess bond returns,” *The Review of Financial Studies*, 2014, *27* (3), 663–713.
- Greenwood, Robin and Samuel G Hanson**, “Issuer quality and corporate bond returns,” *The Review of Financial Studies*, 2013, *26* (6), 1483–1525.
- Greenwood, Robin, Samuel G Hanson, Jeremy C Stein, and Adi Sunderam**, “A quantity-driven theory of term premia and exchange rates,” *Working Paper*, 2020.
- Gromb, Denis and Dimitri Vayanos**, “Equilibrium and welfare in markets with financially constrained arbitrageurs,” *Journal of Financial Economics*, 2002, *66* (2-3), 361–407.
- Haddad, Valentin, Alan Moreira, and Tyler Muir**, “When selling becomes viral: Disruptions in debt markets in the COVID-19 crisis and the Fed’s response,” *The Review of Financial Studies*, 2021, *34* (11), 5309–5351.
- Harford, Jarrad and Vahap B Uysal**, “Bond market access and investment,” *Journal of Financial Economics*, 2014, *112* (2), 147–163.
- He, Zhiguo and Arvind Krishnamurthy**, “Intermediary asset pricing,” *American Economic Review*, 2013, *103* (2), 732–70.
- He, Zhiguo and Wei Xiong**, “Rollover risk and credit risk,” *The Journal of Finance*, 2012, *67* (2), 391–430.
- He, Zhiguo, Bryan Kelly, and Asaf Manela**, “Intermediary asset pricing: New evidence from many asset classes,” *Journal of Financial Economics*, 2017, *126* (1), 1–35.
- Hu, Grace Xing, Jun Pan, and Jiang Wang**, “Noise as information for illiquidity,” *The Journal of Finance*, 2013, *68* (6), 2341–2382.
- Irani, Rustom M, Rajkamal Iyer, Ralf R Meisenzahl, and Jose-Luis Peydro**, “The rise of shadow banking: Evidence from capital regulation,” *The Review of Financial Studies*, 2021, *34* (5), 2181–2235.
- Itskhoki, Oleg and Dmitry Mukhin**, “Exchange rate disconnect in general equilibrium,” *Journal of Political Economy*, 2021, *129* (8), 2183–2232.
- Jiang, Hao, Yi Li, Zheng Sun, and Ashley Wang**, “Does mutual fund illiquidity introduce fragility into asset prices? Evidence from the corporate bond market,” *Journal of Financial Economics*, 2021.
- Kargar, Mahyar, Benjamin Lester, David Lindsay, Shuo Liu, Pierre-Olivier Weill, and Diego Zuniga**, “Corporate bond liquidity during the COVID-19 crisis,” *The Review of Financial Studies*, 2021, *34* (11), 5352–5401.
- Kelly, Bryan T, Diogo Palhares, and Seth Pruitt**, “Modeling corporate bond returns,” *Forth-*

- coming at the Journal of Finance*, 2021.
- Khwaja, Asim I and Atif Mian**, “Tracing the impact of bank liquidity shocks: Evidence from an emerging market,” *American Economic Review*, 2008, *98* (4), 1413–42.
- Kiyotaki, Nobuhiro and John Moore**, “Credit cycles,” *Journal of Political Economy*, 1997, *105* (2), 211–248.
- Koijen, Ralph SJ and Motohiro Yogo**, “The cost of financial frictions for life insurers,” *American Economic Review*, 2015, *105* (1), 445–75.
- Koijen, Ralph SJ and Motohiro Yogo**, “Shadow insurance,” *Econometrica*, 2016, *84* (3), 1265–1287.
- Koijen, Ralph SJ and Motohiro Yogo**, “A demand system approach to asset pricing,” *Journal of Political Economy*, 2019, *127* (4), 1475–1515.
- Kouri, Pentti JK**, “The exchange rate and the balance of payments in the short run and in the long run: A monetary approach,” *The Scandinavian Journal of Economics*, 1976, pp. 280–304.
- Kubitza, Christian**, “Investor-driven corporate finance: Evidence from insurance markets,” *Working Paper*, 2022.
- Kundu, Shohini**, “The externalities of fire sales: Evidence from collateralized loan obligations,” *Working Paper*, 2021.
- Lou, Dong**, “A flow-based explanation for return predictability,” *The Review of Financial Studies*, 2012, *25* (12), 3457–3489.
- Ma, Yiming, Kairong Xiao, and Yao Zeng**, “Mutual fund liquidity transformation and reverse flight to liquidity,” *Forthcoming at the Review of Financial Studies*, 2021.
- Maggiori, Matteo, Brent Neiman, and Jesse Schreger**, “International currencies and capital allocation,” *Journal of Political Economy*, 2020, *128* (6), 2019–2066.
- Massa, Massimo, Ayako Yasuda, and Lei Zhang**, “Supply uncertainty of the bond investor base and the leverage of the firm,” *Journal of Financial Economics*, 2013, *110* (1), 185–214.
- Mendoza, Enrique G**, “Sudden stops, financial crises, and leverage,” *American Economic Review*, 2010, *100* (5), 1941–66.
- Mitchell, Mark, Lasse Heje Pedersen, and Todd Pulvino**, “Slow moving capital,” *American Economic Review: Papers & Proceedings*, 2007, *97* (2), 215–220.
- Modigliani, Franco and Richard Sutch**, “Innovations in interest rate policy,” *American Economic Review*, 1966, *56* (1/2), 178–197.
- Morelli, Juan M, Pablo Ottonello, and Diego J Perez**, “Global banks and systemic debt crises,” *Econometrica*, 2022, *90* (2), 749–798.
- Mota, Lira**, “The corporate supply of (quasi) safe assets,” *Working Paper*, 2020.
- Musto, David, Greg Nini, and Krista Schwarz**, “Notes on bonds: Illiquidity feedback during the financial crisis,” *The Review of Financial Studies*, 2018, *31* (8), 2983–3018.
- Newey, Whitney K and Kenneth D West**, “A simple, positive semi-definite, heteroskedasticity

- and autocorrelation consistent covariance matrix,” *Econometrica*, 1987, 55 (3), 703–708.
- Newey, Whitney K and Kenneth D West**, “Automatic lag selection in covariance matrix estimation,” *The Review of Economic Studies*, 1994, 61 (4), 631–653.
- Nozawa, Yoshio**, “What drives the cross-section of credit spreads? A variance decomposition approach,” *The Journal of Finance*, 2017, 72 (5), 2045–2072.
- O’Hara, Maureen and Xing Alex Zhou**, “Anatomy of a liquidity crisis: Corporate bonds in the COVID-19 crisis,” *Journal of Financial Economics*, 2021.
- Oster, Emily**, “Unobservable selection and coefficient stability: Theory and evidence,” *Journal of Business & Economic Statistics*, 2019, 37 (2), 187–204.
- Philippon, Thomas**, “The bond market’s q ,” *The Quarterly Journal of Economics*, 2009, 124 (3).
- Rotemberg, Julio**, “Instrument variable estimation of misspecified models,” *MIT Sloan Working Paper*, 1983.
- Securities and Exchange Commission**, “Introduction to Investing: Annuities,” *Technical Report*, 2020.
- Sen, Ishita**, “Regulatory limits to risk management,” *Working Paper*, 2020.
- Sen, Ishita and Varun Sharma**, “Internal models, make believe prices, and bond market cornering,” *Working Paper*, 2020.
- Shleifer, Andrei and Robert W Vishny**, “Liquidation values and debt capacity: A market equilibrium approach,” *The Journal of Finance*, 1992, 47 (4), 1343–1366.
- Siani, Kerry**, “Raising bond capital in segmented markets,” *Working Paper*, 2021.
- Siriwardane, Emil N**, “Limited investment capital and credit spreads,” *The Journal of Finance*, 2019, 74 (5), 2303–2347.
- The Economist**, “The dark side of debt,” *Press Article*, 2006.
- Timmer, Yannick**, “Cyclical investment behavior across financial institutions,” *Journal of Financial Economics*, 2018, 129 (2), 268–286.
- Tobin, James**, “A general equilibrium approach to monetary theory,” *Journal of Money, Credit and Banking*, 1969, 1 (1), 15–29.
- Vayanos, Dimitri and Jean-Luc Vila**, “A preferred-habitat model of the term structure of interest rates,” *Econometrica*, 2021, 89 (1), 77–112.
- Wardlaw, Malcolm**, “Measuring mutual fund flow pressure as shock to stock returns,” *The Journal of Finance*, 2020, 75 (6), 3221–3243.
- Williamson, Oliver E**, “Corporate finance and corporate governance,” *The Journal of Finance*, 1988, 43 (3), 567–591.
- Wolf, Christian K**, “The missing intercept: A demand equivalence approach,” *Working Paper*, 2021.
- Wooldridge, Jeffrey M**, *Econometric analysis of cross section and panel data*, MIT Press, 2010.
- Wooldridge, Jeffrey M**, “Two-way fixed effects, the two-way Mundlak regression, and difference-in-differences estimators,” *Working Paper*, 2021.

		<i>N</i>	Total Invested Assets (\$M)				Concentration Ratios		
			Mean	Median	95 th Percentile	SD	CR ₁₀	CR ₅₀	CR ₁₀₀
A. US Insurers									
	2005	1,208	2,804	110	10,914	12,616	0.34	0.73	0.85
	2010	1,242	3,048	118	12,397	13,994	0.34	0.72	0.86
	2015	1,225	3,510	138	14,770	15,399	0.34	0.71	0.85
	2020	1,216	4,396	163	19,813	19,821	0.35	0.70	0.85
B. US Funds (All)									
	2005	5,597	1,080	219	3,816	4,948	0.14	0.32	0.42
	2010	6,675	1,570	267	6,000	6,587	0.11	0.29	0.39
	2015	8,193	1,948	291	7,167	8,546	0.11	0.27	0.37
	2020	8,418	3,715	404	12,768	23,616	0.16	0.34	0.44
C. Non-US Funds (All)									
	2005	11,135	681	97	1,989	5,303	0.18	0.38	0.46
	2010	20,135	580	81	1,676	9,360	0.22	0.33	0.39
	2015	22,069	878	118	2,449	8,563	0.18	0.33	0.40
	2020	25,152	1,136	151	3,441	10,209	0.13	0.25	0.33
D. US Funds (Specialists)									
	2005	1,565	794	196	2,839	4,968	0.28	0.46	0.58
	2010	1,529	1,503	307	6,160	7,243	0.26	0.48	0.61
	2015	1,957	1,667	324	6,236	6,206	0.21	0.44	0.56
	2020	2,097	2,881	464	11,197	11,666	0.21	0.45	0.58
E. Non-US Funds (Specialists)									
	2005	2,506	574	150	2,454	1,527	0.12	0.29	0.42
	2010	4,150	522	119	2,135	1,624	0.11	0.27	0.38
	2015	5,090	687	149	2,744	2,481	0.12	0.26	0.36
	2020	6,087	980	208	3,716	3,884	0.12	0.27	0.36

Table 1: **Holdings data: sample summary statistics.** This table shows the total number of insurers as well as mutual funds and exchange traded funds (ETFs) in the sample. I report statistics separately for all funds and for fixed income specialist funds (as determined via Morningstar’s categorization). I also show statistics on total invested assets (equities and bonds), as well as concentration ratios detailing the share of total invested assets held by largest 10, 50, and 100 institutions in each group. Insurers and funds with less than \$25 million in total invested assets are excluded. Data shown for the years 2005, 2010, 2015, and 2020.

	Average	Average	Share of Holdings in Category (%)			
	Duration (Years)	Issue Size (\$M)	High Yield	Junior	Floating	Callable
A. US Insurers						
2005	6.2	563	8	8	3	57
2010	6.4	678	6	5	2	70
2015	7.0	726	5	4	1	78
2020	8.1	785	4	4	4	77
B. US Funds*						
2005	4.9	556	55	4	9	66
2010	5.1	733	40	5	7	61
2015	5.5	761	37	6	7	71
2020	5.9	814	27	6	13	77
C. Non-US Funds*						
2005	5.7	615	18	9	19	24
2010	4.7	688	28	7	17	28
2015	5.3	772	36	10	15	32
2020	5.4	808	30	11	21	45
*Fixed income specialists only						

Table 2: **Overview of corporate bond holdings of insurers and fixed income specialist funds.** This table reports summary statistics on the characteristics of the corporate bond holdings of insurance companies and fixed income specialist funds. These include average bond duration, the share of holdings that is in high-yield bonds, the share of holdings that is in junior bonds, the share of holdings that is in floating-rate bonds, the share of holdings that is in bonds with callable features, and the average issue size for bonds held.

	Log Bond Drawdown				
	(1)	(2)	(3)	(4)	(5)
Insurer Share	-.43 (.06)	-.60 (.05)	-.46 (.05)	-.40 (.10)	-.41 (.10)
<i>Fixed Effects Interaction:</i>					
Event Dummy	Yes	Yes	Yes	Yes	Yes
Bond Size	—	Yes	Yes	Yes	Yes
Bond Duration	—	Yes	Yes	Yes	Yes
Bond Rating	—	—	Yes	Yes	Yes
Issuer Dummy	—	—	—	Yes	Yes
Callable Dummy	—	—	—	—	Yes
Senior Dummy	—	—	—	—	Yes
Coupon Type	—	—	—	—	Yes
Identifying Observations	15,012	14,996	14,649	5,328	4,945
Identifying Issuers	1,430	1,430	1,412	452	437
R^2	.29	.60	.68	.84	.85

Table 3: **Investor base effects for U.S. corporate bonds: baseline OLS estimates.** This table reports the estimates of the regression specification in equation (5) in the sample of actively traded investment-grade U.S. corporate bonds. The specification in column 1 includes event fixed effects. Column 2 includes interacted event by bond size decile by duration category fixed effects. Columns 3 and 4 add additional interactions with rating and issuer dummies. Column 5 includes the fully-saturated set of interacted fixed effects, which include all of the above variables plus dummies for bond seniority, bond callability, and coupon type. The estimates are from weighted least squares (WLS) regressions, with bond sizes as weights. Standard errors are clustered at the issuer by event level and are reported in parentheses.

	OLS		2SLS	
	(1)	(2)	(3)	(4)
Insurer Share	-.40 (.10)	-.41 (.10)	-.55 (.15)	-.45 (.15)
<i>Fixed Effects Interaction:</i>				
Event Dummy	Yes	Yes	Yes	Yes
Bond Size	Yes	Yes	Yes	Yes
Bond Duration	Yes	Yes	Yes	Yes
Bond Rating	Yes	Yes	Yes	Yes
Issuer Dummy	Yes	Yes	Yes	Yes
Callable Dummy	—	Yes	—	Yes
Senior Dummy	—	Yes	—	Yes
Coupon Type	—	Yes	—	Yes
Identifying Observations	5,328	4,945	5,328	4,945
Identifying Issuers	452	437	452	437
R^2	.84	.85	.84	.85
First Stage F	—	—	71.8	73.3

Table 4: **Shift-share IV estimates.** This table reports the estimates of the shift-share instrumental variable (IV) regression specification in equation (5) in the sample of actively traded investment-grade U.S. corporate bonds, alongside the baseline OLS estimates (for comparison). The OLS estimates are shown in columns 1 and 2, while the two-stage least squares (2SLS) IV estimates are shown in columns 3 and 4. The various columns correspond to differing sets of bond characteristics included in the interacted fixed effects. The estimates are from weighted least squares (WLS) regressions, with bond sizes as weights. Standard errors are clustered at the issuer by event level and are reported in parentheses.

	Log Bond Drawdown				
	(1)	(2)	(3)	(4)	(5)
Proxy Safe Share	-.39 (.14)	-.51 (.12)	-.40 (.15)	-.64 (.21)	-.35 (.20)
Sample	USA	Pooled Non-US	EMU	GBR	CAN
<i>Fixed Effects Interaction:</i>					
Event Dummy	Yes	Yes	Yes	Yes	Yes
Bond Size	Yes	Yes	Yes	Yes	Yes
Bond Duration	Yes	Yes	Yes	Yes	Yes
Bond Rating	Yes	Yes	Yes	Yes	Yes
Issuer Dummy	Yes	Yes	Yes	Yes	Yes
Callable Dummy	Yes	Yes	Yes	Yes	Yes
Senior Dummy	Yes	Yes	Yes	Yes	Yes
Coupon Type	Yes	Yes	Yes	Yes	Yes
Currency FE	—	Yes	Yes	Yes	Yes
Identifying Observations	4,945	6,890	4,090	1,484	1,316
Identifying Issuers	437	582	393	102	87
R^2	.84	.87	.84	.89	.88

Table 5: **Investor base effects for non-U.S. corporate bonds and proxy approach for U.S. corporate bonds.** Column 1 reports the estimates of the regression specification in equation (5), estimated by proxying safe-hand holdings via the complement of fund positions, in the sample of actively traded investment-grade U.S. corporate bonds. Columns 2 through 5 report the estimates of the regression specification in equation (15). The sample in column 2 pools all investment-grade corporate bonds issued by firms in the European Monetary Union (EMU), United Kingdom, and Canada that are not TRACE-reportable. Columns 3 through 5 show the results separately for each of these three economies. All specifications include fully-saturated interacted fixed effects. The estimates are from weighted least squares (WLS) regressions, with bond sizes as weights. Standard errors are clustered at the issuer by event level and are reported in parentheses.

	Investment Rate (%)					
	CAPX			CAPX and Acquisitions		
	(1)	(2)	(3)	(4)	(5)	(6)
$\hat{\beta}$	2.80 (1.11)	2.14 (.68)	1.80 (.69)	5.42 (1.73)	3.94 (1.04)	2.93 (1.08)
$\hat{\eta}$	— —	— —	1.95 (.88)	— —	— —	3.27 (1.31)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,994	7,994	7,994	7,994	7,994	7,994
Firms	1,028	1,028	1,028	1,028	1,028	1,028
R^2	.80	.80	.80	.57	.57	.57
Estimator	IV	OLS	OLS	IV	OLS	OLS
First Stage F	65.3	—	—	74.2	—	—

Table 6: **Insurer ownership and investment dynamics in the Great Recession.** This table reports estimates from the static difference-in-differences specification of equation (19). The dependent variables in these regressions are firm-level annual investment rates for U.S. firms. The coefficient $\hat{\beta}$ captures the effect of the interaction between the insurer share and the post-period dummy. The coefficient $\hat{\eta}$ captures the effect of the interaction of the insurer share, the post-period dummy, and the share of the firm's outstanding bonds that is due in 2008 or 2009. All regressions include two-way fixed effects for firm and year, as well as controls for firm-level covariates, measured in the pre-period and interacted with the post-period dummy: these are fixed effects for issuer rating, average duration of the firm's outstanding bonds, two-digit industry, and deciles of firm size and firm leverage. Columns 1 and 4 show estimates from two-stage least-squares, using the firm-level instrument in equation (17), interacted with the post-period dummy. The remaining columns use ordinary least squares. Standard errors are clustered at the firm level and reported in parentheses.

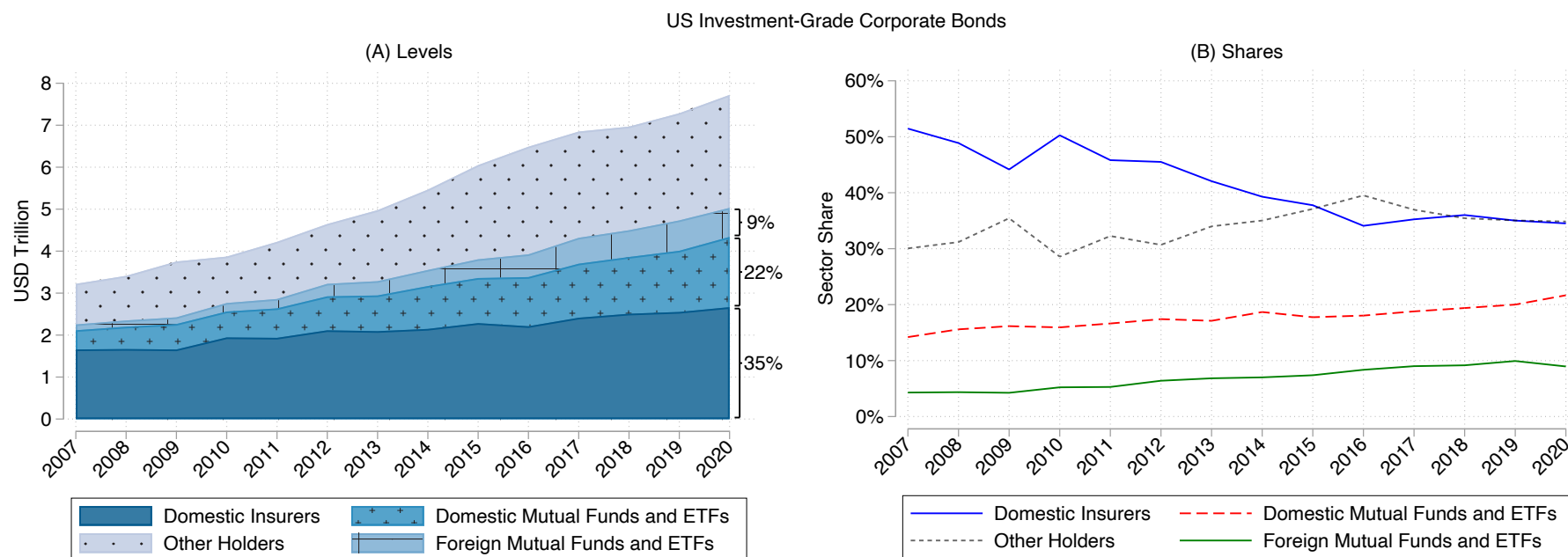


Figure 1: **Aggregate value of U.S. investment-grade corporate bonds and market coverage via security-level holdings micro-data.** The total shaded region in panel A shows the aggregate outstanding value of U.S. investment-grade corporate bonds in the sample, defined as bonds that are reportable in TRACE. The *dark blue area* shows the aggregate value of the positions in these bonds held by U.S. insurers, as reported in the NAIC data. The remaining shaded areas of panel A show the value of the positions in these bonds held by investment funds (mutual funds and ETFs) in the Morningstar data, split according to whether the funds are U.S. domiciled or not. Panel B shows the corresponding shares of the aggregate value that each sector accounts for, providing a counterpart to the picture in levels of panel A.

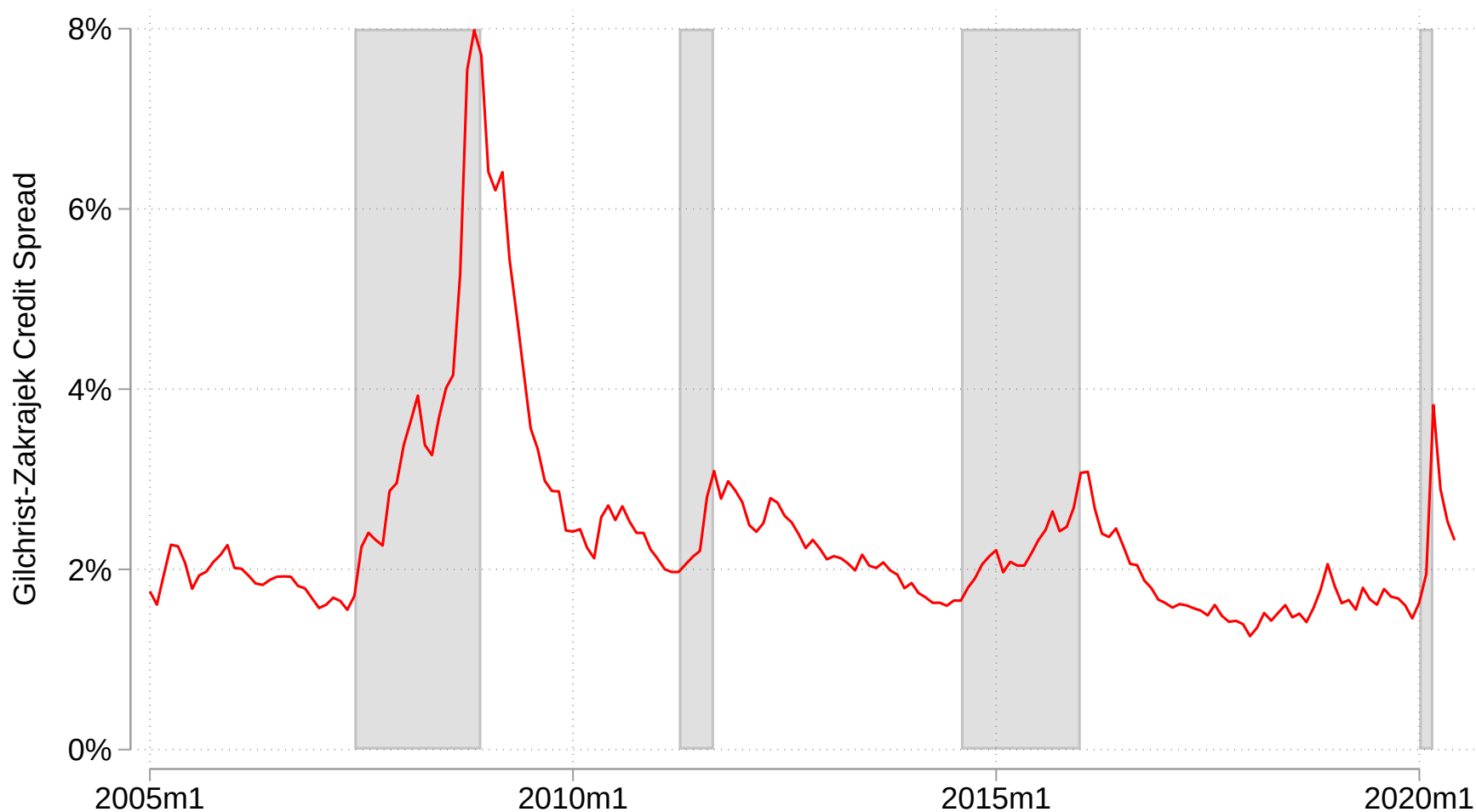


Figure 2: **Events of focus.** The *solid red line* plots the [Gilchrist and Zakrajšek \(2012\)](#) index, which provides an aggregate measure of credit spreads on U.S. corporate bonds. The *shaded gray areas* show the event windows used to estimate bond drawdowns in my analysis. The start and end dates for each event are defined mechanically so that they correspond to the prior trough and subsequent peak values in the GZ index. Section [2.2](#) discusses the drivers of aggregate credit spread variation in these episodes.

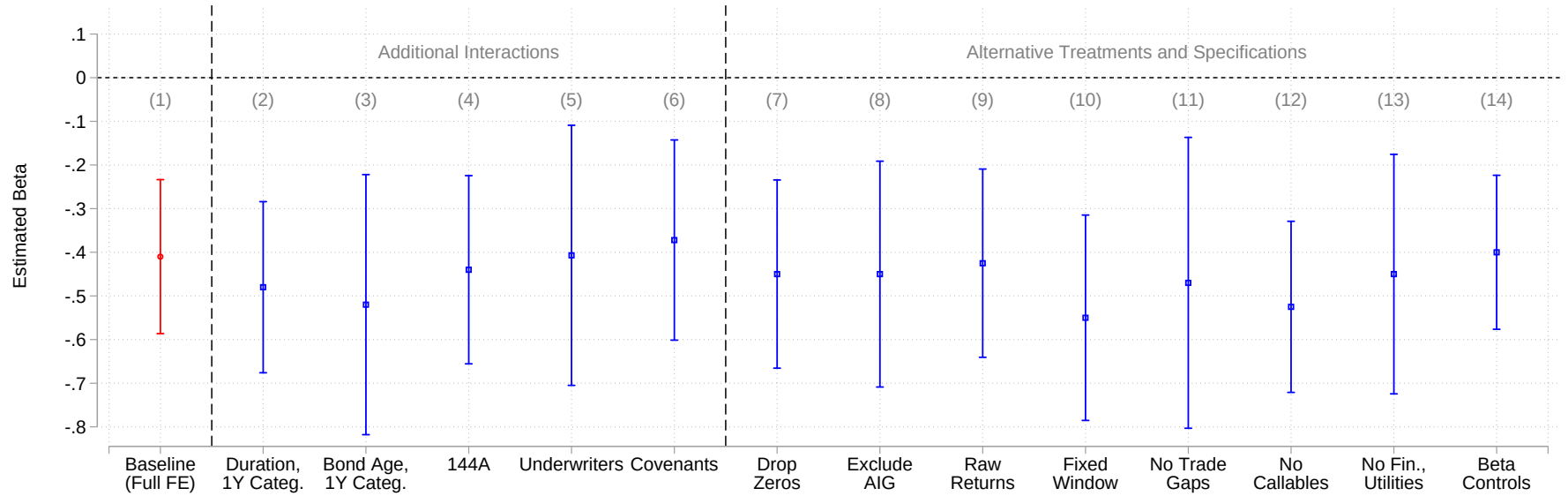


Figure 3: **Investor base effects for U.S. corporate bonds: robustness.** This figure demonstrates the robustness of the baseline estimate (*red bar*) from column 5 of Table 3 to further variations in the empirical specification. Bars 2 through 4 introduce further interactions to the fully-saturated set of fixed effects. Bar 2 introduces an additional interaction with stricter duration categories, with duration rounded to the nearest year. Bar 3 adds an interaction with bond age, rounded to the nearest year. Bar 4 includes an additional interaction with a dummy indicating whether a bond is issued under SEC rule 144A. Bar 5 adds an interaction with dummies for the set of a bond's lead underwriters. Bar 6 adds an interaction with dummies for the set of a bond's covenants, hence removing bonds with non-matching covenants from the estimation sample. Additionally, bars 7 through 14 consider alternative sample treatments and specifications. Bar 7 excludes bonds with a zero insurer share from the sample. Bar 8 excludes the positions of AIG from the computation of the insurer positions $X_{i,t}^I$. Bar 9 uses raw returns $R_{i,t}$ in place of the excess returns $R_{i,t}^e$. Bar 10 uses an alternative measure of drawdowns which applies the same window to all bonds for a given event. Bar 11 drops any bonds that trade less frequently than once a week in a given event period. Bar 12 excludes callable bonds. Bar 13 excludes financials and utilities. Bar 14 adds non-crisis betas as an additional control. The displayed confidence intervals are at the 95 percent confidence level and are clustered at the issuer by event level.

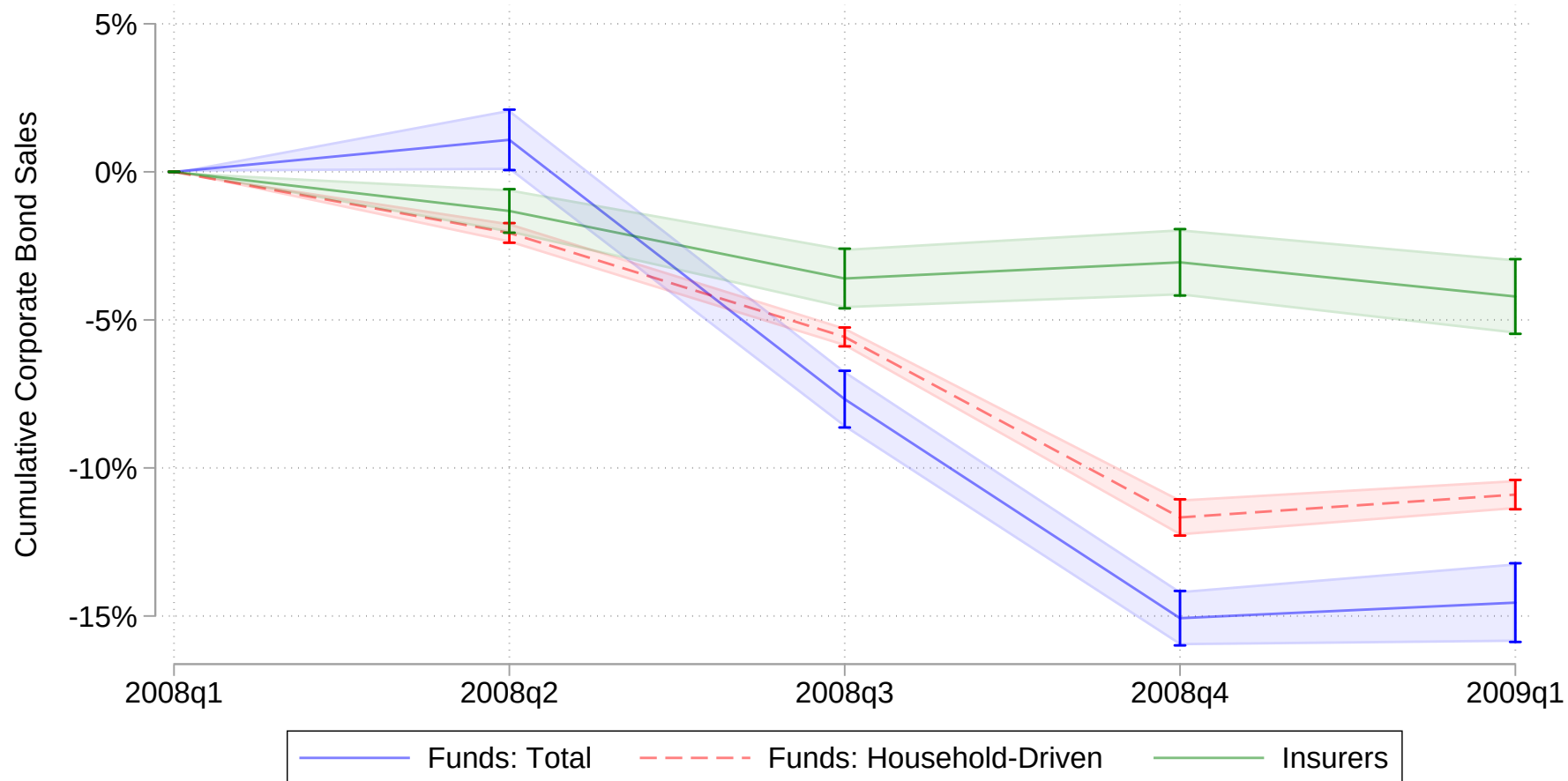


Figure 4: **Corporate bond sales dynamics in the Great Recession.** The *solid green line* plots the cumulative change in corporate bond holdings for the asset-weighted average insurer in the sample over the course of the Great Recession, relative to their level in 2008Q1. The changes in corporate bond holdings are measured using par values, so that they reflect pure quantity changes rather than any valuation effects owing to variation in market prices. The *solid blue line* shows the equivalent portfolio dynamics for the asset-weighted average mutual fund or ETF in the sample. The *dashed red line* plots the component of the cumulative net corporate bond sales by funds that is household-driven, as defined in Section 2.4. The vertical bars show 95 percent confidence intervals around these estimates.

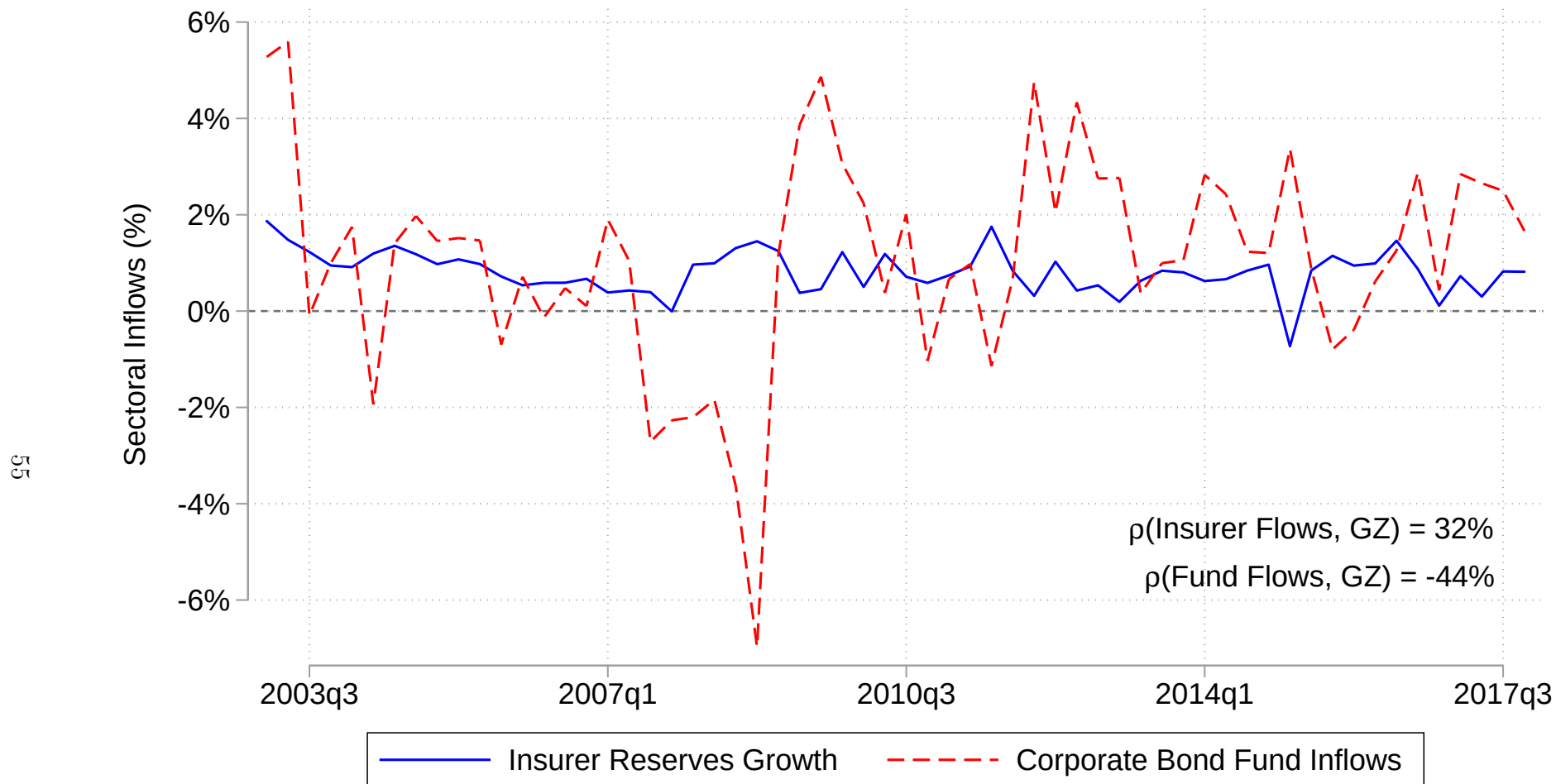


Figure 5: **Flows in and out of insurers and fixed income funds.** The *solid blue line* shows the quarterly net change in policy reserves for the aggregate insurance sector, scaled by lagged total policy reserves. The *dashed red line* shows the quarterly value of aggregate net client inflows and outflows for fixed income mutual funds and ETFs in the sample, scaled by lagged total assets under management. The *black annotations* report the correlation of these series with the quarterly GZ spread index.

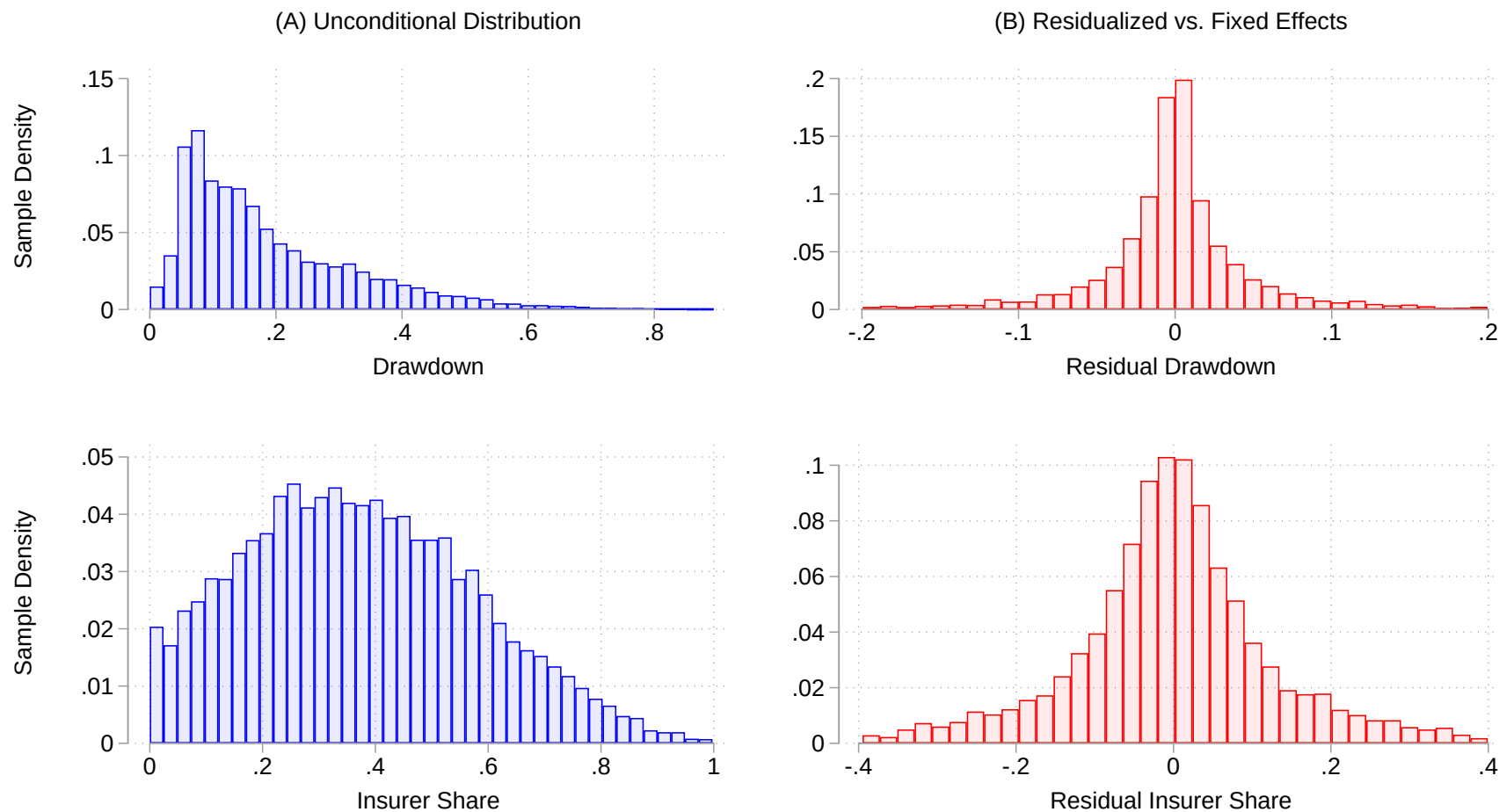


Figure 6: **Residual variation within the fixed effects.** The *blue histograms* in Panel A plot the sample distribution of drawdowns and insurer shares in the sample of actively traded U.S. investment-grade corporate bonds that underlies the regression estimates in Table 3. The *red histograms* in Panel B plot the sample distribution of the same quantities after residualizing them against the fully-saturated set of interacted fixed effects.

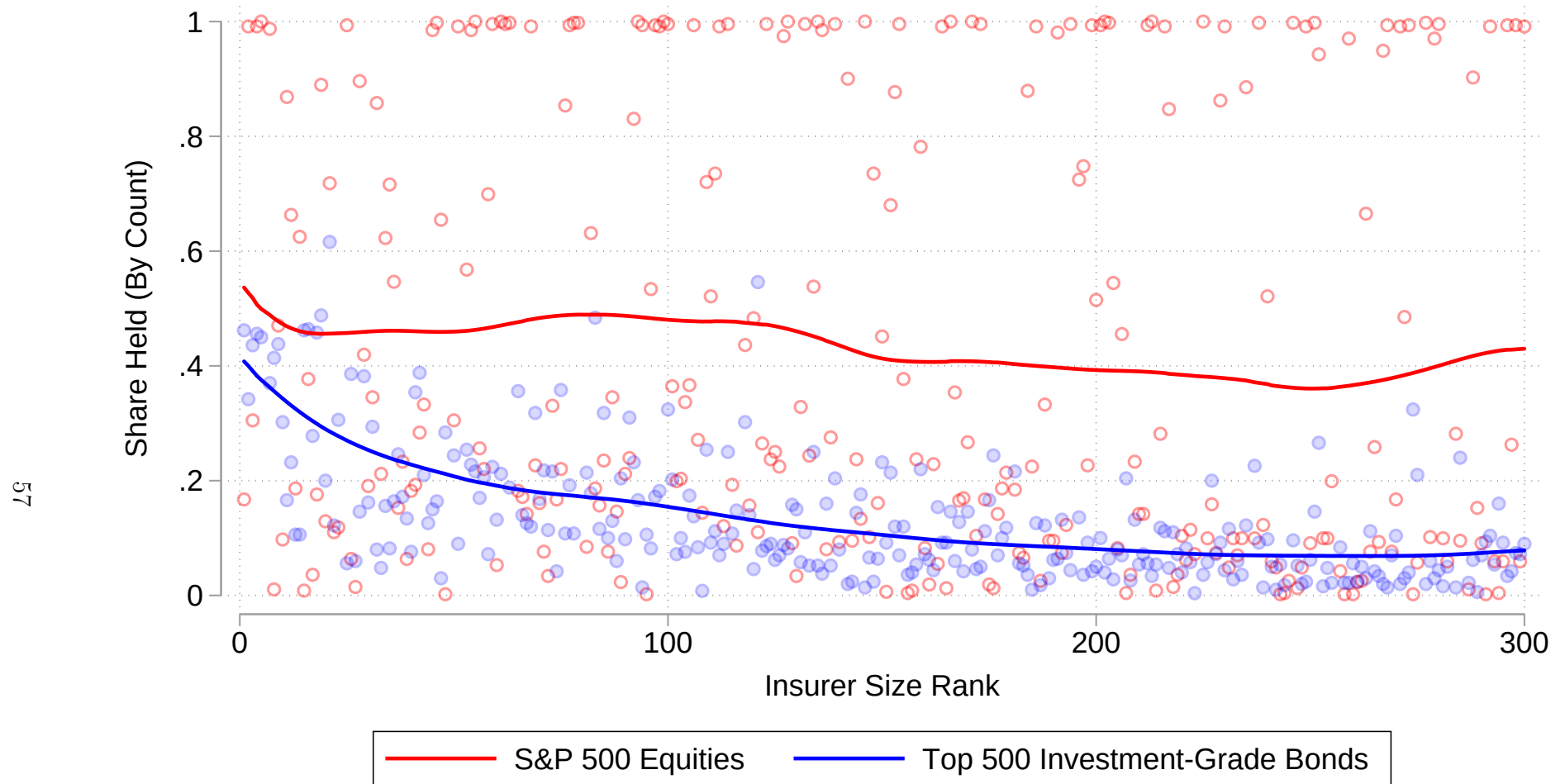


Figure 7: **Idiosyncrasy in insurers' corporate bond holdings at the security level.** This figure ranks insurers by the overall size of their portfolio investment, and shows the share of the 500 largest actively traded investment-grade U.S. corporate bonds that they hold an open position in (*blue dots*) as well as the share of S&P 500 equities that they hold an open position in (*red dots*), all measured cross-sectionally as of 2017Q4. Indirect equity positions held through mutual funds and ETFs are included. The thick lines display the fit from lowess regressions of the investment shares (shown by the dots) on the insurer rank.

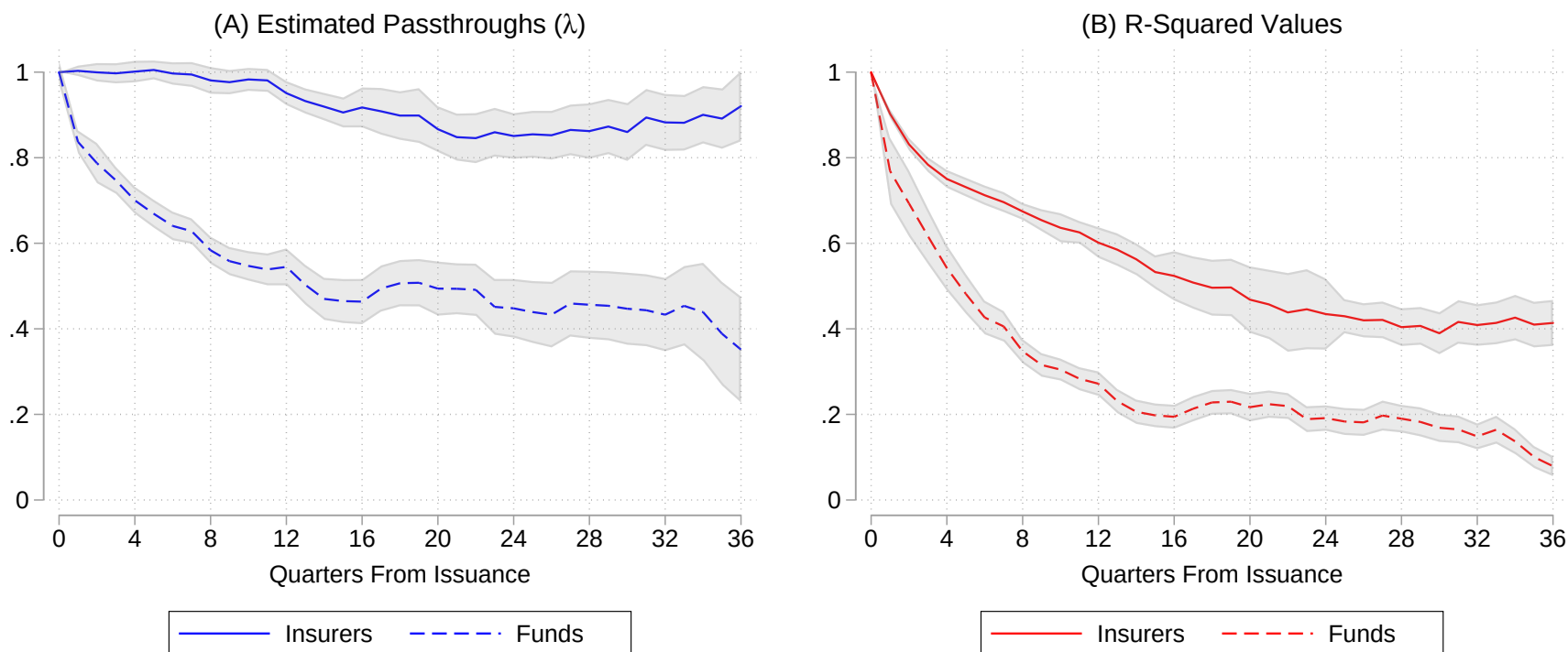


Figure 8: **Persistence of holdings decisions made at bond issuance.** This figure shows estimates from the regression specification in equation (8). The *solid blue line* in Panel A plots the estimated passthrough coefficients λ_s for insurer holdings against the quarterly horizon s . For comparison, the *dashed blue line* plots the same regression coefficients using mutual fund and ETF ownership shares in place of the insurer shares in equation (8). The *solid red line* and *dashed red line* in Panel B plot the corresponding R-squared values from these regressions. The *gray shaded areas* show 95 percent confidence bands, which use heteroskedasticity-robust standard errors in Panel A and are bootstrapped in Panel B.

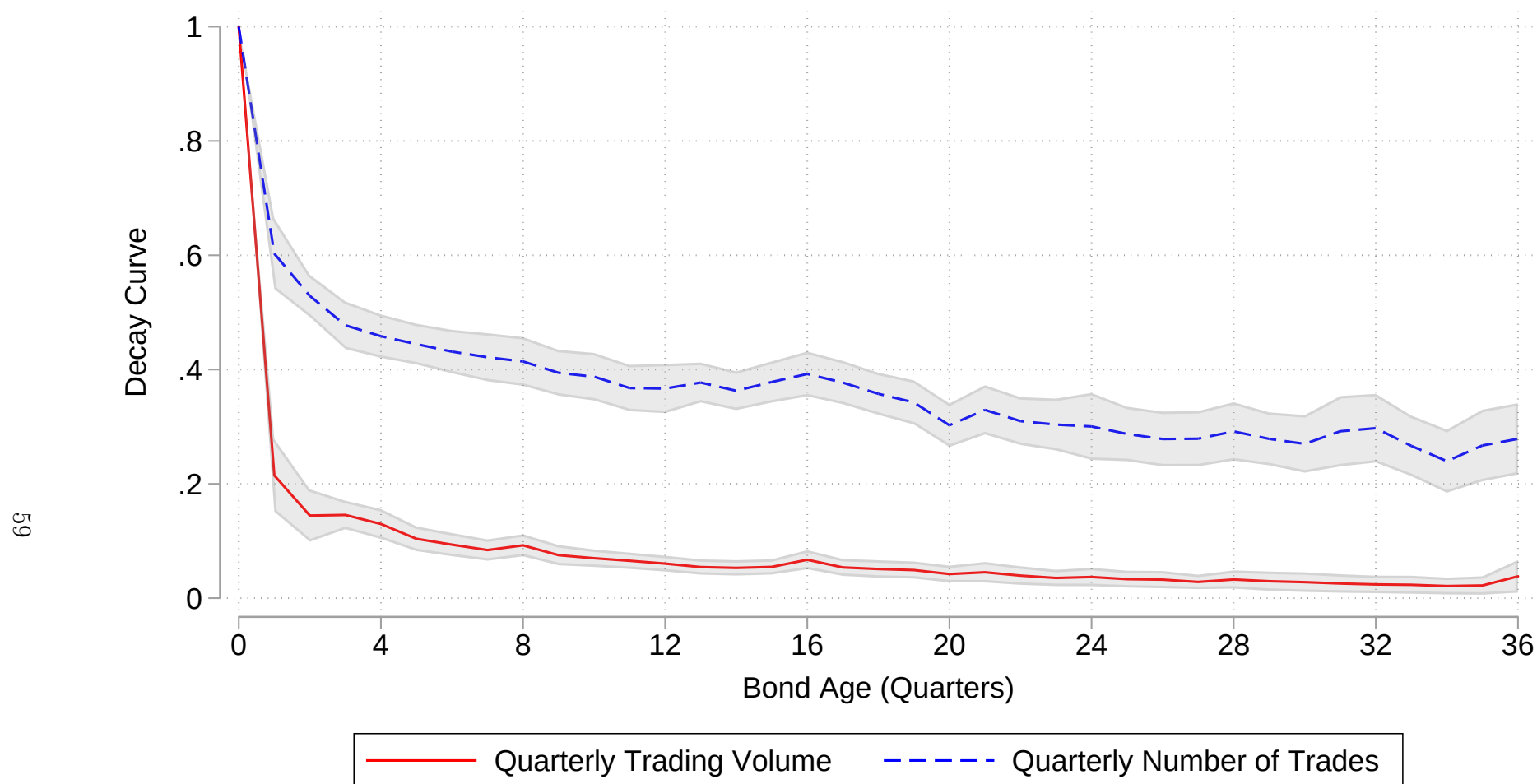


Figure 9: **Corporate bond liquidity in the primary market vs. the secondary market.** This figure shows the estimated coefficients from equation (9), at varying quarterly horizon. For each horizon, the specification regresses quarterly trading volume (*red solid line*) and quarterly number of trades (*dashed blue line*) for each U.S. investment-grade corporate bond against the same measures in the quarter of bond issuance. The estimated coefficients display a rapid decay toward zero as a consequence of the sharply reduced liquidity of the secondary corporate bond market as compared to the primary market. Standard errors are heteroskedasticity-robust. The *gray shared areas* correspond to 95 percent confidence bands.

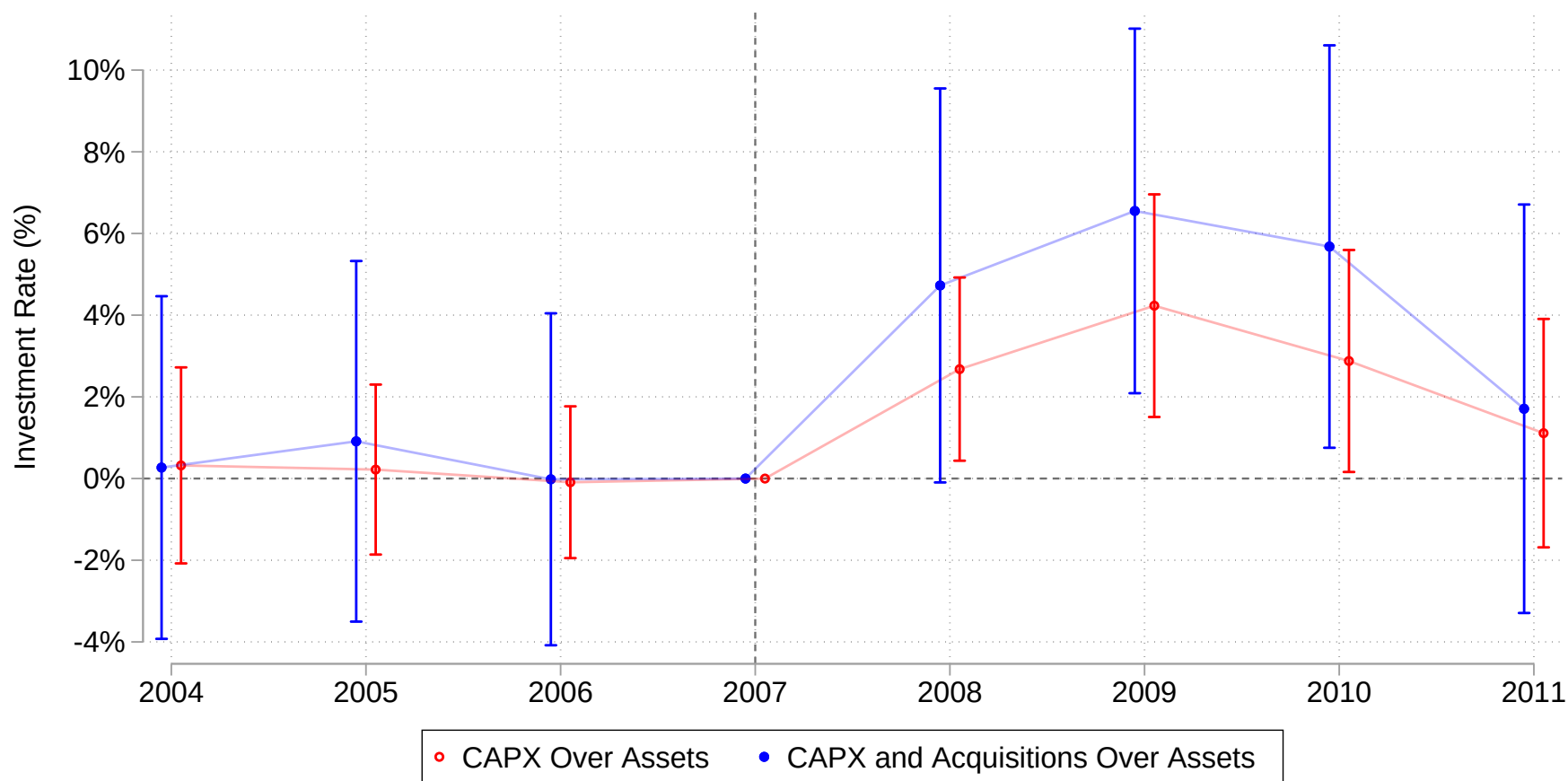


Figure 10: **Insurer ownership and firm investment dynamics in the Great Recession.** This figure shows the estimated coefficients from the difference-in-differences specification in equation (18), estimated on the sample of U.S. firms with outstanding bonds in 2007Q1, excluding speculative-grade issuers. The estimates are from just-identified two-stage least-squares, using the firm-level instrument in equation (17), interacted with year dummies. The coefficients quantify the dynamic association between firm investment rates and the share of a firm's bonds held by domestic insurers in 2007Q1. The controls consist of firm-level covariates, measured in the pre-period and interacted with time dummies: these include fixed effects for issuer rating, average duration of the firm's outstanding bonds, two-digit industry, and deciles of firm size and firm leverage. The *red line* shows the estimates for capital expenditures (CAPX) over lagged total assets, while the *blue line* shows the estimates for CAPX plus acquisitions over lagged total assets. Standard errors are clustered at the firm level. Bars correspond to 95 percent confidence intervals.

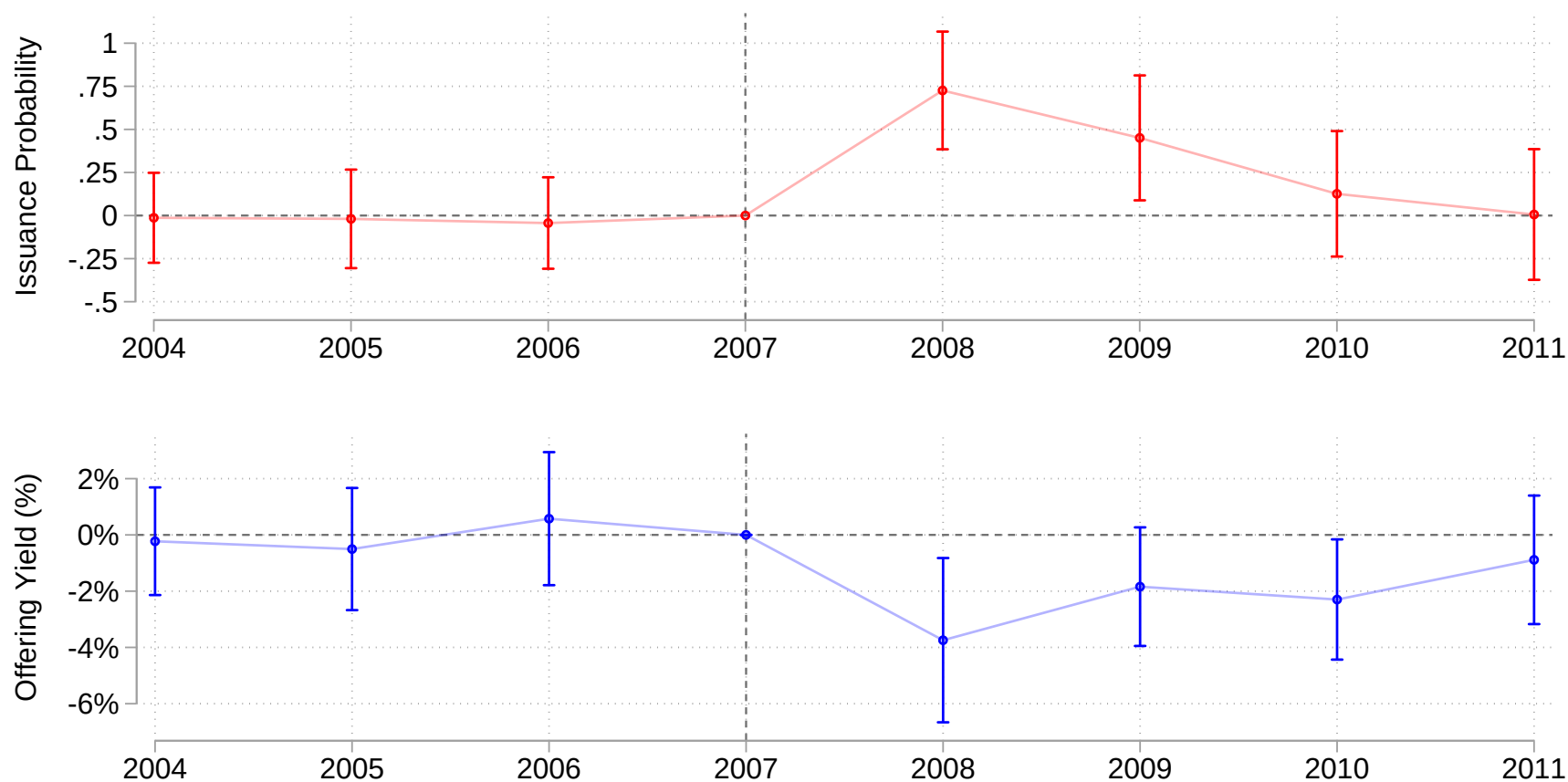


Figure 11: **Insurer ownership and firm credit outcomes in the Great Recession.** This figure shows the estimated coefficients from the difference-in-differences specification in equation (20), estimated on the sample of U.S. firms with outstanding bonds in 2007Q1, excluding speculative-grade issuers. The estimates are from just-identified two-stage least-squares, using the firm-level instrument in equation (17), interacted with year dummies. The coefficients quantify the association of yearly firm bond issuance probability (*red line*) and offering yields on new bonds (*blue line*) with the share of a firm's bonds held by domestic insurers in 2007Q1. The controls consist of firm-level covariates, measured in the pre-period and interacted with time dummies: these include fixed effects for issuer rating, average duration of the firm's outstanding bonds, two-digit industry, and deciles of firm size and firm leverage. Standard errors are clustered at the firm level. Bars correspond to 95 percent confidence intervals.

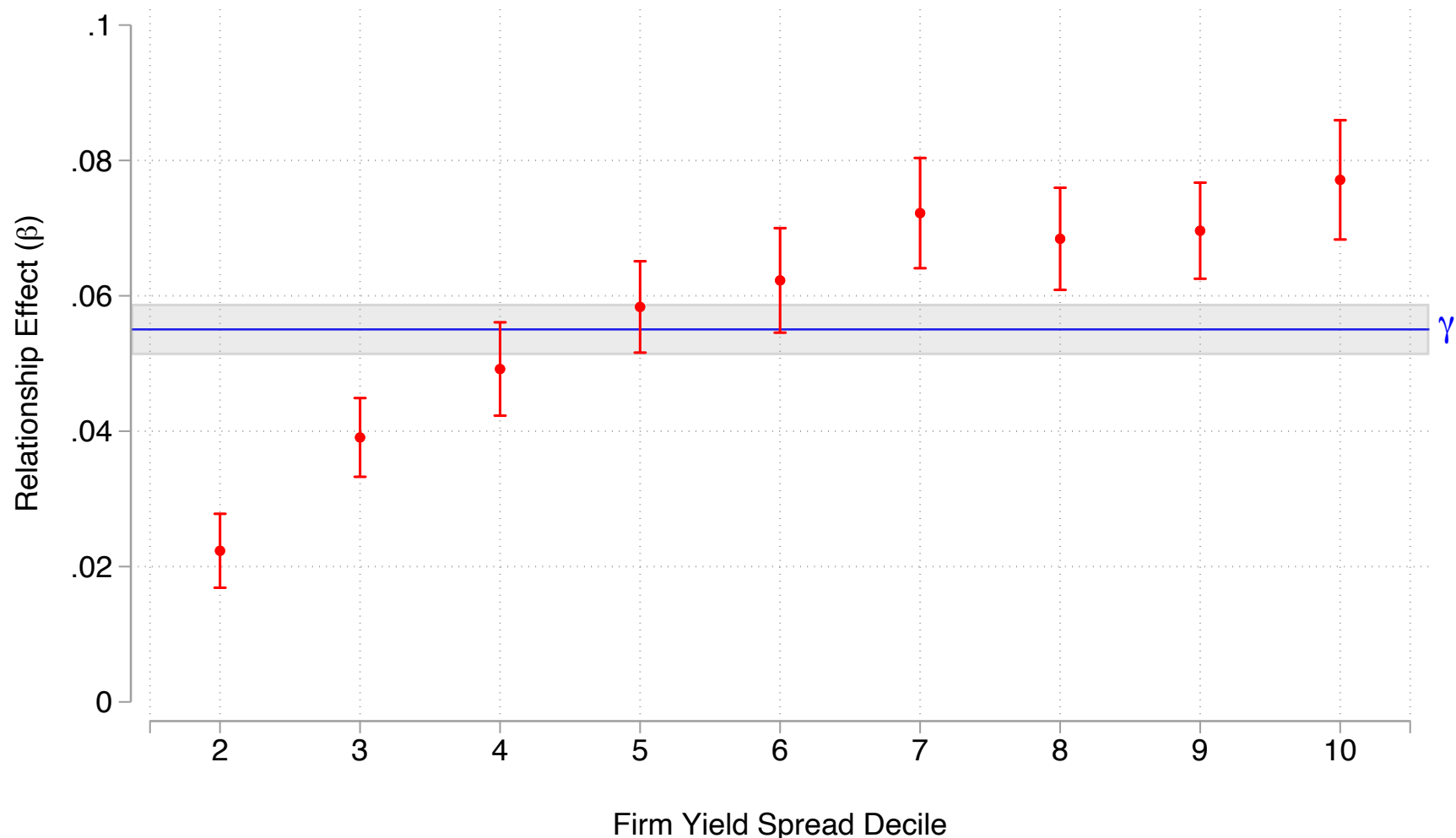


Figure 12: **The role of relationships in corporate bond market financing.** This figure displays the estimated coefficients from the regression specification in equation (21). The *solid blue line* displays the coefficient on the relationship indicators $\mathcal{R}_{i,j,k,t}$, which quantifies how much an existing lending relationship increases the probability that a given insurer will buy a firm's new bond offerings for firms in the lowest decile of secondary-market yield spreads. The *red dots* show the estimated coefficients on the interaction between the relationship indicators and firm spread decile dummies, quantifying how this relationship effect varies as firm yield spreads rise. The *red bars* and the *gray shaded area* plot 95 percent confidence intervals for these estimates. Standard errors are clustered at the issuer level.

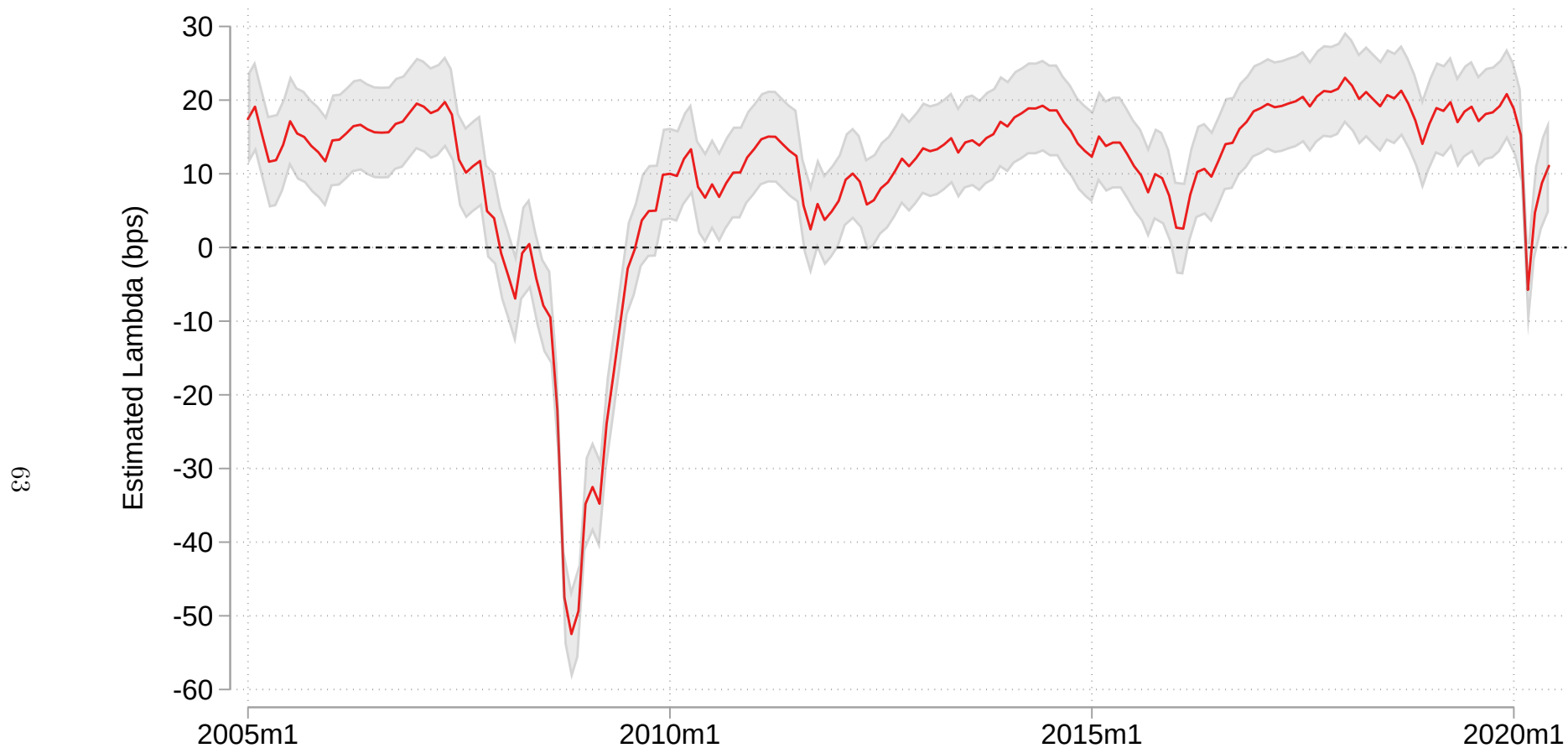


Figure 13: **Insurer ownership and yield spread levels.** This figure shows the estimates from the regression specification in equation (22). The *red line* plots the total estimated effect $\Lambda_t = \bar{\Lambda} + \tilde{\Lambda} GZ_t$ of security-level insurer shares on yield spreads to Treasury. The *blue shaded area* corresponds to a 95 percent confidence band estimated using Newey and West (1987) HAC standard errors, with lag selection as in Newey and West (1994).

ONLINE APPENDIX FOR
“IN SAFE HANDS: THE FINANCIAL AND REAL IMPACT OF INVESTOR
COMPOSITION OVER THE CREDIT CYCLE”

Not for Publication

This appendix contains the following sections:

1. Section [A](#) provides further details on the security-level holdings data used in the paper.
2. Section [B](#) introduces a full-panel specification that studies investor base effects over the full credit cycle.
3. Section [C](#) outlines further results concerning deviations from the law of large numbers (LLN) in the holdings of insurance companies.
4. Section [D](#) details the Fama-French bootstrap approach used to test for the presence of residual alpha in the cross-section of insurers.
5. Section [E](#) discusses the institutional features of the primary corporate bond market.
6. Section [F](#) provides further analysis on heterogeneity within the insurance sector.
7. Section [G](#) discusses the interpretation of the IV estimates in the presence of within-sector heterogeneity.
8. Section [H](#) provides further details concerning bond covenants.
9. Section [I](#) provides a hedonic analysis of primary market bond prices as a function of characteristics and investor composition.
10. Section [J](#) analyzes reversal in the pricing patterns and further relates the empirical findings to models of fire sales.
11. Section [K](#) provides results excluding financial and utility firms.
12. Section [L](#) explores the role of exchange traded funds (ETFs) in greater depth.
13. Section [M](#) outlines a structural interpretation of the real-side results under q -theory with costly external financing.
14. Section [N](#) provides further details on the Oster test of Section [3](#).
15. Section [O](#) studies how investor base composition impacts the volatility of bond turnover.

A Further Details on Holdings Data

In this section I provide further details on the two sets of security-level holdings data that I use throughout the paper: the Morningstar data on the holdings of worldwide mutual funds and ETFs, and the NAIC holdings data on the holdings of U.S. insurance firms.

Morningstar holdings data. The Morningstar data on the holdings of worldwide mutual funds and ETFs are self-reported by each fund on a monthly or sometimes quarterly basis. These data are collected by funds in all sub-sectors of the asset management industry, including funds that specialize in investing in equity, fixed income, real estate, commodities, and other asset classes. The funds report all their positions, including those in bonds, stocks, cash, and financial derivatives. Morningstar collects data for funds resident both in the United States and abroad. Throughout the paper, I use the same sample of nine developed economies as [Maggiori et al. \(2020\)](#) and [Coppola et al. \(2021\)](#), which these authors selected because of their high degree of data coverage and quality.¹ Altogether, the dataset includes observations on over 1.3 billion individual positions across asset classes. Positions in the Morningstar data are reported together with their nine-digit CUSIP codes. When CUSIP codes are missing, funds often report instead twelve-digit ISIN codes, which I convert to CUSIP codes using mapping data from CUSIP Global Services (CGS). In a few cases, funds will not report either a CUSIP or ISIN code, but rather a different type of security identifier, such as a SEDOL code, a FIGI identifier, or a Bloomberg ID. In these cases, I map the positions to CUSIP codes by performing a search for the codes via Bloomberg’s OpenFIGI API.²

The Morningstar data offer a high degree of coverage of the universe of mutual funds and ETFs. To show this, here I focus on fixed income specialist funds, which are the most relevant for my analysis. The dashed blue line in Figure [A.13a](#) shows the aggregate market value of the positions held by U.S. domiciled fixed income specialist investment funds in the Morningstar data over time, which has grown from about \$1 trillion in 2005 to nearly \$6 trillion by 2020. The red line in the same graph shows the aggregate value of the positions of U.S. fixed income mutual funds and ETFs estimated by the Investment Company Institute (ICI), a major industry association. The red line serves as a benchmark, and a comparison with the dashed blue line demonstrates a high degree of coverage of the Morningstar data in the United States. Figure [A.13b](#) repeats the same exercise for funds domiciled outside of the United States, plotting the total value of the positions of fixed income funds in the micro-data in the dashed blue line and the relevant ICI benchmark figures in the red line. Coverage outside of the United States is also high, although somewhat lower than in the U.S. setting.

¹The sample includes funds domiciled in the United States, the European Monetary Union (EMU), the United Kingdom, Canada, Switzerland, Australia, Sweden, Denmark, and Norway.

²These steps are analogous to the ones used in [Maggiori et al. \(2020\)](#) and [Coppola et al. \(2021\)](#).

NAIC holdings data. The NAIC data cover the full universe of security holdings of U.S. insurance firms. The primary security identifiers in the NAIC data are nine-digit CUSIP codes, which I use to perform merges with the rest of the data sources. Insurers report both the par value and the estimated market value of their bond holdings. However, since insurers have some discretion in determining the market value of their bond positions in these regulatory filings (Sen and Sharma 2020), I recompute market values by multiplying the par value of reported holdings by the latest available transacted prices from TRACE: this ensures that positions in the same bonds are marked to market consistently across insurers. For bonds that do not have recent traded prices in TRACE, I use price data from S&P Global where available, and otherwise default to the market value originally reported to NAIC. Since the empirical analysis focuses strictly on bonds that are in TRACE and are actively traded, this is immaterial to any of the results shown in the paper.

Further descriptive statistics. Table A.4 details the composition of the overall portfolios of insurers and fixed income specialist funds. Slightly more than half of insurers’ assets (57 percent in 2015) are invested in corporate bonds—nearly all U.S. corporate bonds. A further 23 percent of insurer portfolios is in sovereign bonds, with smaller shares in asset-backed securities (ABS) and equities. Specialist funds in the United States have a significant share of their assets (39 percent) in ABS, while ABS holdings are small for non-U.S. funds. Funds’ corporate bond holdings reflect a strong home bias: while U.S. funds hold 31 percent of their assets in U.S. corporate bonds and only 2 percent in non-U.S. corporate bonds, funds domiciled outside the United States hold 23 percent of their assets in U.S. corporate bonds and 37 percent in non-U.S. corporate bonds.

B Investor Base Effects Over the Full Credit Cycle

The baseline empirical analysis in Section 2 focuses on bond price dynamics in times of crisis. This particular focus is due to the asymmetry in the magnitude of investor base effects over the credit cycle. In order to understand this asymmetry more clearly, it is helpful to also consider a complementary full-panel approach that studies investor base effects over the entire credit cycle. In order to do this, I begin by computing the betas of each bond i with respect to the returns on a value-weighted index of all actively traded investment-grade bonds in the United States. These betas are estimated on a two-year rolling window, and I refer to them as $b_{i,t}$.³ I then consider the following regression specification:

$$b_{i,t} = \alpha + \bar{\Gamma} \phi_{i,t-12} + \tilde{\Gamma} \phi_{i,t-12} \text{GZ}_t + \text{Interacted Fixed Effects} + \varepsilon_{i,t}, \quad (\text{A.1})$$

³Appendix Figure A.5 repeats the estimates with betas computed on a shorter one-quarter rolling window. The results are analogous, albeit with larger standard errors.

where GZ_t is the value of the [Gilchrist and Zakrajšek \(2012\)](#) index at time t . The interaction between *ex ante* insurer shares and the GZ index serves to capture variation in investor base effects over the credit cycle in a simple parametric way. Consistent with the analysis carried out so far, the sample includes all actively traded U.S. investment-grade corporate bonds, and the fixed effects interact issuer dummies with all the security characteristics included in the baseline, plus time fixed effects. These regressions include one observation per security-month pair, with all variables measured at the end of the month.

The estimated total effect of insurer ownership on bond betas is $\Gamma_t = \bar{\Gamma} + \tilde{\Gamma} GZ_t$. The red line in [Figure A.4](#) plots the estimated value of Γ_t over time. This figure clearly demonstrates the non-linear nature of the relationship between insurer ownership and bond return properties. When aggregate credit spreads are low, as in 2018-19 or before 2007, investor base effects are negligible—as evidenced by the small estimated values of Γ_t , often statistically indistinguishable from zero. When credit spreads spike, however, higher insurer ownership leads to significantly lower betas with respect to market returns. For instance, at the height of the Great Recession the estimates reach a value of $\Gamma_t = -.54$, which is broadly consistent with the baseline results of [Section 2](#): as the *ex ante* insurer share rises from 0 to 1, the passthrough from aggregate market returns to individual bond returns roughly halves. In this sense, bonds with a higher insurer share are more insulated from market-wide fluctuations in crises.

C Further Results on LLN Deviations

In this section I elaborate on the deviations of the law of large numbers (LLN) in insurer holdings introduced in [Section 3](#). In order to study these LLN deviations, I examine the rate at which information is lost as we exclude the positions of the largest holders underlying each insurer share $\phi_{i,t}$. Letting N denote the number of insurers in the sample, I define $X_{i,t}^{I,k}$ to be the k -th largest position of an individual insurer in bond i at time t , so that $X_{i,t}^{I,1} \geq X_{i,t}^{I,2} \geq \dots \geq X_{i,t}^{I,N}$ and $X_{i,t}^I = \sum_{k=1}^N X_{i,t}^{I,k}$. I then construct the *truncated* insurer shares

$$\phi_{i,t}^{(K)} = \frac{\sum_{k=1+K}^N X_{i,t}^{I,k}}{V_{i,t}}, \quad (\text{A.2})$$

which exclude the largest K positions underlying each individual aggregate insurer share.

Consider now a regression of the actual insurer shares on the truncated insurer shares as we vary the number of excluded insurers K :

$$\phi_{i,t} = \alpha_K + \pi_K \phi_{i,t}^{(K)} + \alpha_{i,t,K}. \quad (\text{A.3})$$

A theoretical benchmark in which the LLN applies delivers sharp predictions for how the

R-squared values from this regression should decay with K . Namely, consider a benchmark model with equal-sized insurers whose holdings are subject to *i.i.d.* shocks, which I refer to as the *LLN benchmark*. In the theoretical LLN benchmark, the R-squared value from regression (A.3) declines with the number of excluded holders K as:⁴

$$R_{\text{LLN}}^2(K) \approx \sqrt{1 - \frac{K}{N}}. \quad (\text{A.4})$$

The solid red line in Figure A.17 plots this theoretical benchmark, while the solid blue line plots the empirical R-squared values for insurer shares obtained from estimating equation (A.3). The large discrepancy between the solid red line and the solid blue line demonstrates the presence of a large deviation from the law of large numbers: while in the LLN benchmark excluding the ten largest holders underlying each position leads to negligible information loss, in reality this removes most of the power of the truncated insurer shares in explaining the actual insurer shares. In this sense, the decisions of the few largest insurers matter disproportionately. For comparison, the dashed blue line in Figure A.17 also plots the same estimates for mutual funds and ETFs, which serve as a further empirical benchmark. While the fund sector also exhibits some degree of deviation from the LLN benchmark, this is less pronounced than for the insurance sector.

The empirical patterns documented here and in Section 3 also suggest a series of placebo exercises. A first placebo replaces the true insurer shares ϕ_{i,t_e^S-12} with the truncated shares $\phi_{i,t_e^S-12}^{(10)}$, which exclude the largest underlying 10 holders for each bond. A second placebo exercise regresses the true insurer shares ϕ_{i,t_e^S-12} against the insurer shares at issuance $\phi_{i,\tau(i)}$, and then uses the residuals from this regression as the right-hand side argument in equation (5). In both cases, we are removing some of the most informative components of the regressors, and we would expect to see the estimates attenuate. Figure A.18 shows the estimates from these placebo exercises, and in both cases the estimated coefficients are indeed driven to zero.

D Residual Alpha in the Cross-Section of Insurers

As discussed in Section 3, the identifying assumption in my shift-share IV approach requires that insurers are not able to (or do not attempt to) select bonds that will draw down less in a market downturn for reasons not explained by issuer effects or any of the included observable bond characteristics. While this condition is not directly testable, I test the related condition that certain insurers systematically acquire bonds with a high *residual alpha*—that is, alpha with respect to the overall bond market that is not explained by observable bond characteristics or issuer effects.

⁴The accuracy of this approximation was confirmed via Monte Carlo simulation.

Distinguishing between investment ability and noise in a cross-section of investors is a well-trodden question in finance. Here I use the methodology of [Fama and French \(2010\)](#), which is a non-parametric approach relying on the bootstrap. This method generates bootstrap distributions of measured alphas in the cross-section using simulations under which the null hypothesis of no true alphas is true by construction, and compares these bootstrap distributions to the actual empirical distribution of measured alphas.

I start by constructing, for each insurer and at a quarterly frequency, returns on their overall holdings of U.S. investment-grade corporate bonds. The portfolio return for insurer j in quarter t is $R_{j,t}$. I also estimate a quarterly benchmark market return R_t^m using a value-weighted basket of all outstanding U.S. investment-grade corporate bonds. I then consider, separately for each insurer j , the following time-series regressions:

$$R_{j,t} = \alpha_j + \beta_j R_t^m + \varepsilon_{j,t}. \quad (\text{A.5})$$

The coefficient α_j estimates the alpha of insurer j 's portfolio with respect to the market portfolio. The blue curves in [Figure A.16](#) plot kernel density estimates of the measured empirical alphas α_j and their corresponding t -statistics t_j^α in the cross-section of insurers.

Next, I construct bootstrap distributions in which the null of no residual alpha holds by construction. For each bootstrap simulation, I independently replace each insurer's bond positions with an equal-sized position in a randomly chosen matching bond: that is, a bond by the same issuer and with the same characteristics (using all characteristics included in the fully-saturated set of interacted fixed effects specified in [Section 2](#), and sampling with replacement).⁵ The red curves in [Figure A.16](#) plot kernel density estimates of the simulated measured alphas α_j and their corresponding t -statistics t_j^α , with each red curve averaging over 1000 independent bootstrap simulations.

If certain insurers were able to systematically select bonds with a high residual alpha, we would expect the empirical distributions of α_j and t_j^α (the blue curves) to have heavier right tails than the corresponding simulated bootstrap distributions (the red curves). Instead, the blue curves fall squarely within the range traced out by the red curves, so that this bootstrap-based test rejects the presence of residual alpha in the cross-section of insurers.

E Additional Institutional Details

In this section I provide an overview of the functioning of the primary corporate bond market, as surveyed for example in [The Bond Market Association \(2004\)](#) and [Nikolova et al. \(2020\)](#). Firms that wish to raise debt financing by issuing new corporate bonds normally hire a financial intermediary—such as an investment bank's capital markets desk—to act as the

⁵Naturally, if there is no bond that matches a given one, that position is simply left as-is in the simulation.

lead underwriter for the offering. The lead underwriter typically invites other institutions to be part of the underwriting syndicate and drafts the initial terms of the offering (such as issue size, target pricing, and so forth) after review with the issuer.

The terms of the offering are only finalized after the completion of a process known as *book-building*. The book-building process commences with the underwriting syndicates reaching out to prospective institutional investors (such as insurers and funds) with the preliminary terms of the issue. Institutional investors submit information about their demand to the underwriters in the form of indications of interest (IOIs), which specify how much of the issue the investor is willing buy at a given offering yield. The lead underwriter then adjusts the terms of the issue to clear the market. Finally, the lead underwriter and the underwriting syndicate establish and distribute allocations in the new bond to the investors who submitted IOIs.

New issues are typically oversubscribed, so that the process of receiving a primary-market allocation is competitive from the perspective of the investors. Allocations are valuable given the high liquidity of the primary market and because new issues are typically slightly under-priced relative to current secondary-market yields. [Nikolova et al. \(2020\)](#) show that investors who have prior trading relationships with underwriters tend to receive larger allocations. Since data on IOIs in the corporate bond market are private, it is difficult to study further determinants of primary-market bond allocations directly.⁶

The corporate bond underwriting process described here, however, is very similar to the one used for equity initial public offerings (IPOs), for which [Jenkinson et al. \(2018\)](#) are able to study the bids submitted by institutional investors during book-building. [Jenkinson et al. \(2018\)](#) find that larger IOI bid sizes are strongly associated with higher relative allocations (that is, allocations relative to the bid), even after controlling for investor size. These results indicate that investors have an incentive to place large orders on the primary market. While the evidence does not come directly from the bond market (but rather from the equity market), this highlights one reason why insurers might prefer to place bids for a few large positions rather than many smaller positions on the primary market—besides the simple fact that few large positions might be institutionally easier to manage, and that insurers might view similar bonds as largely substitutable.

As mentioned in [Section 2.4](#), the most important determinant of the volume of insurers' purchases of new bonds on the primary market in a given time period is the same-quarter net change in their policy reserves. Policy reserves correspond to the actuarial value of future expected payments to policyholders and constitute the main liabilities of insurers. A univariate regression of quarterly changes in the insurance sectors' total holdings of bonds on the sectors'

⁶The role of relationships with underwriters in the primary bond market may play a part in generating the patterns of bond-market relationship finance documented in [Section 5.3](#). For a related discussion of the role of bilateral relationships in the interbank repo market, see also [Chen et al. \(2018\)](#), who study data from Chinese banks.

growth in policy reserves, for example, yields a high R-squared value of 64 percent and shows that for each dollar of new reserves insurers invest 82 cents in new bonds.

F Heterogeneity Within the Insurance Sector

In this section I investigate heterogeneity in the impact of bond ownership by different institutions *within* the insurance sector. A first salient dimension of heterogeneity is the degree to which insurers face financial constraints. At a regulatory level, insurance firms in the United States are subject to risk-based capital (RBC) requirements and report their RBC ratios (corresponding to total capital over required risk-based capital) in their statutory filings. Besides the statutory RBC ratios, another widely followed measure of insurer capital constraints is the Best’s Capital Adequacy Ratio (BCAR) distributed by A.M. Best, a leading provider of financial strength ratings for insurance firms.

BCAR scores are an important factor in A.M. Best’s rating methodology, with each rating category having a guideline for the minimum score required to support that rating. [Koijen and Yogo \(2015\)](#) show that risk-based capital (as measured via BCAR scores) relative to the guideline for the current rating is an important determinant of insurer behavior: for instance, insurers with low risk-based capital relative to guideline were particularly likely to reduce the price of long-term policies during the financial crisis, in order to improve their capital positions and ultimately avoid a downgrade as well as possible regulatory repercussions.

One might expect this kind of heterogeneity in financial constraints to also affect the impact of insurer ownership on bond price dynamics. In crises, more financially constrained insurers might be particularly eager to shift out of corporate bond positions and into assets with lower risk-based capital requirements (such as U.S. Treasury bonds) in order to improve their capital adequacy metrics: as a result, these more financially constrained insurers may be especially prone to engaging in fire sales of their corporate bond holdings. To investigate this hypothesis, I consider the following counterpart to the baseline specification of Section 2.2:

$$\log \zeta_{i,e} = \alpha + \beta_{\text{HC}} \cdot \phi_{i,t^S-12}^{\text{High Capital}} + \beta_{\text{LC}} \cdot \phi_{i,t^S-12}^{\text{Low Capital}} + \text{Interacted Fixed Effects} + \varepsilon_{i,e}, \quad (\text{A.6})$$

where $\phi_{i,t}^{\text{High Capital}}$ is the share of bond i that is owned by insurers with above-median risk-based capital relative to guideline (measured as the difference between the insurer’s BCAR score and the guideline for its current A.M. Best rating), while $\phi_{i,t}^{\text{Low Capital}}$ is the share owned by insurers with below-median values of the same. I estimate this specification for the Great Recession, with risk-based capital relative to guideline measured as of 2007.⁷ The regression

⁷I exclude later events because of data availability and because A.M. Best modified the BCAR methodology and the corresponding rating guidelines in more recent years. I compute the median value of risk-based capital relative to guideline cross-sectionally, across all the insurance firms in the sample.

includes the fully saturated set of interacted fixed effects introduced in Section 2.2, which encompasses firm dummies as well as dummies for bond characteristics (such as credit rating, duration, issue size, and so forth).

The point estimates from this specification, displayed in Figure A.19, are $\hat{\beta}_{\text{HC}} = -.57$ and $\hat{\beta}_{\text{LC}} = -.29$, and a Wald test rejects the null hypothesis of coefficient equality at the 1 percent significance level. These estimates imply that bonds do particularly well in a crisis when they are owned by insurers that are less financially constrained, in the sense of having a robust risk-based capital position relative to rating guidelines. This finding parallels the results of Ellul et al. (2011), who show that more constrained insurers are particularly likely to sell corporate bonds that are downgraded to speculative grade. It also relates to Becker et al. (2021), who show that more financially constrained insurers are particularly likely to retain downgraded non-agency mortgage-backed securities (MBS) as compared to other downgraded assets, following a 2009 regulatory reform that nearly eliminated capital requirements associated with non-agency MBS positions.

A second notable dimension of heterogeneity within the insurance sector is in the structure of corporate ownership. While the majority of the assets of U.S. insurance firms are now held by regular stock companies, which are owned by their shareholders, a second type of ownership structure also remains prevalent in the form of *mutual* companies. In a mutual company, policyholders are also equity holders, so that the incentives of the firm's owners and those of its clients are perfectly aligned. As compared to stock companies, mutual companies on average tend to engage in less risk-taking, have better risk-based capital ratios, and have stronger balance sheet strength ratings (National Association of Mutual Insurance Companies 2019). For these reasons, it is plausible to expect mutuals to be particularly unlikely to engage in corporate bond fire sales in market downturns. I test this hypothesis with the following specification:

$$\log \zeta_{i,e} = \alpha + \beta_{\text{M}} \cdot \phi_{i,t_e}^{\text{Mutuals}} + \beta_{\text{SC}} \cdot \phi_{i,t_e}^{\text{Stock Companies}} + \text{Interacted Fixed Effects} + \varepsilon_{i,e}, \quad (\text{A.7})$$

where $\phi_{i,t}^{\text{Mutuals}}$ is the share of bond i owned by insurance mutuals, while $\phi_{i,t}^{\text{Stock Companies}}$ is the share owned by stock insurance firms.

I estimate this specification across events, as in the baseline of Section 2.2. As before, Figure A.19 displays the point estimates from this specification, which are $\hat{\beta}_{\text{M}} = -.78$ and $\hat{\beta}_{\text{LC}} = -.33$. In this case, too, a Wald test rejects the null hypothesis of coefficient equality at the 1 percent significance level. These results confirm that bonds held by mutuals have particularly mild drawdowns in crises, which is consistent with the hypothesis that mutuals have an especially low propensity to engage in fire sales.

G Shift-Share Interpretation Under Heterogeneity

For simplicity of exposition, in Section 3 I discussed the shift-share IV estimates under the null hypothesis of homogeneous treatment effects, whereby all insurers impact bond price dynamics with the same quantitative elasticities. Here I discuss the interpretation of the shift-share IV estimates if there is heterogeneity in the causal impact of ownership *across insurers*. This heterogeneity would arise, for example, if certain groups of insurers are especially unlikely to engage in fire sales, even relative to their peers within the insurance sector: as I show in Appendix Section F, these patterns can result from factors such as each insurer's degree of financial constraints. The shift-share IV approach continues to provide a valid causal estimator in the presence of heterogeneity—although it should then be interpreted as estimating a weighted average of heterogeneous treatment effects.

Without loss of generality, suppose the heterogeneity is across groups of insurers $g \in \mathcal{G}$. The share of bond i held by the group of insurers g at time t is ϕ_i^g , where for simplicity I suppress time indices. Naturally, these shares sum to the aggregate insurer share, so that $\sum_{g \in \mathcal{G}} \phi_i^g = \phi_i$. All insurers within group g are assumed to have the same causal impact β_g on bond drawdowns ζ_i :

$$\beta_g = \frac{\partial \log \zeta_i}{\partial \phi_i^g}. \quad (\text{A.8})$$

In the most extreme case, this framework can accommodate treatment effects β_g that are different for each individual insurer: in that case, one can simply take $\mathcal{G} = \{\{j\} : j \in \mathcal{J}\}$, where as in the main text $\mathcal{J}$ is the set of all insurers.

Following the notation in Goldsmith-Pinkham et al. (2020), let Z be the vector stacking the shift-share instruments Z_i defined in equation (11), X be the vector stacking the insurer shares ϕ_i , and Y be the vector stacking the outcomes ζ_i . Further, let D be the matrix containing the dummies corresponding to controls included in the regressions (that is, a constant term plus dummies for the interacted fixed effects). The shift-share estimator $\hat{\beta}_{\text{Bartik}}$ obtained from the 2SLS regressions of Section 3 is

$$\hat{\beta}_{\text{Bartik}} = (Z'X^\perp)^{-1}Z'Y^\perp, \quad (\text{A.9})$$

where $X^\perp = M_D X$ and $Y^\perp = M_D Y$, with $M_D = I - D(D'D)^{-1}D'$ the annihilator matrix for D . Next, define the following group-specific instruments, which only aggregate the extensive margin indicators $\xi_{i,j}$ for insurers in each group g :

$$Z_i^g = \sum_{j \in g} \xi_{i,j} \bar{\phi}_j. \quad (\text{A.10})$$

Each of these group-specific shift-share instruments estimates an individual heterogeneous treatment effect β_g :

$$\hat{\beta}_g = (Z_g' X^\perp)^{-1} Z_g' Y^\perp, \quad (\text{A.11})$$

where Z_g is the vector stacking the instruments Z_i^g , and I correspondingly define $\bar{\phi}_g = \sum_{j \in g} \bar{\phi}_j$.

Using the results in Proposition 3 of Goldsmith-Pinkham et al. (2020), the shift-share estimator $\hat{\beta}_{\text{Bartik}}$ can be decomposed in terms of the individual estimators $\hat{\beta}_g$:

$$\hat{\beta}_{\text{Bartik}} = \sum_g \hat{\alpha}_g \hat{\beta}_g, \quad (\text{A.12})$$

where terms $\hat{\alpha}_g$ are estimated Rotemberg (1983) weights that sum to one (so that $\sum_g \hat{\alpha}_g = 1$),

$$\hat{\alpha}_g = \frac{\bar{\phi}_g Z_g' X^\perp}{\sum_{g'} \bar{\phi}_{g'} Z_{g'}' X^\perp}. \quad (\text{A.13})$$

Goldsmith-Pinkham et al. (2020) point out that the Rotemberg weights $\hat{\alpha}_g$ can take negative values, although this does not necessarily pose a conceptual problem as long as the component of $\hat{\beta}_{\text{Bartik}}$ with negative weights $\hat{\alpha}_g$ does not have a relatively large magnitude. As a practical way to assess this, they recommend splitting the estimate into the component with positive $\hat{\alpha}_g$ (that is, $\sum_{g|\alpha_g > 0} \hat{\alpha}_g \hat{\beta}_g$) and that with negative $\hat{\alpha}_g$ (that is, $\sum_{g|\alpha_g < 0} \hat{\alpha}_g \hat{\beta}_g$), and compare their magnitudes.

To start with, I implement this check by defining the partitioning set $\mathcal{G}$ using the dimensions of heterogeneity that are motivated by prior and I explored in Appendix Section F. First I separate insurers according to their degree of financial constraints, so that $\mathcal{G}$ splits insurers into two equal-sized groups, sorting them according to their risk-based capital relative to guideline as in equation (A.6). Second, I split insurers according to their corporate ownership structure, so that $\mathcal{G}$ contains two groups—mutuals and stock companies, as in equation (A.7). In both of these cases, none of the estimated Rotemberg weights $\hat{\alpha}_g$ are negative, so that the magnitude of the negative-weighted component of $\hat{\beta}_{\text{Bartik}}$ is exactly zero. Even in the most extreme and conservative case, in which I assume that the treatment effects β_g are different for each individual insurer, so that $\mathcal{G} = \{\{j\} : j \in \mathcal{J}\}$, the negative-weighted component is more than 20 times smaller in magnitude than the positive-weighted component.

H Further Details on Bond Covenants

Section 2.3 provides an analysis of bond covenants. Here I describe in details the bond covenant data and provide statistics on covenant presence in the sample. I obtain data on bond covenants from Mergent FISD. I construct a binary variable capturing whether a bond

features each of the covenant categories listed and described in Table A.2. Overall, there are 31 covenant categories captured in the analysis.

I let $\mathcal{C}_{i,k}$ be the row vector collecting all the binary variables above associated with each bond i by issuer firm k . Note that the $\mathcal{C}_{i,k}$ vector is not time-varying, since covenants are established and observed at the time of bond issuance, and thereafter they represent unchanging characteristics of the bonds. Table A.3 displays statistics on the prevalence of each type of covenant in the sample. The most common type of covenant is consolidation and merger restrictions (present in 30.5% of the bonds in the sample). The second and third most common categories are asset sale restrictions (30.4% of the sample) and cross-acceleration covenants (22.8% of the sample).

As discussed in Section 2.3 and shown in Figure 3 (bar 6), the estimation results from the baseline specification in equation (5) are not meaningfully affected when introducing further interactions with dummies for the covenant vectors $\mathcal{C}_{i,k}$ (i.e., using one dummy for each of the possible configurations of $\mathcal{C}_{i,k}$ via an interacted fixed effects specification). Effectively, this interaction removes from the identifying sample bonds that have non-matching covenant vectors $\mathcal{C}_{i,k}$ within the fixed effect categories. This leads to similar point estimates as in the baseline exercise—naturally with wider standard errors given the smaller identifying sample.

When introducing further interactions with dummies for the entire covenant vector $\mathcal{C}_{i,k}$ in the specification in equation (5), the number of identifying observations goes down by roughly half (being reduced from 4,945 to 2,410). The reason for this relatively mild decline in the number of identifying observations is that the firm fixed effects already absorb most of the variation in covenants in the sample. This occurs, in turn, because most variation in covenants occurs *across firms* rather than within firm.

To see this formally, we can perform a variance decomposition on the collection of vectors $\mathcal{C}_{i,k}$ into within-firm and between-firm components. The variance decomposition methodology is outlined formally below, in subsection H.1. When implementing this methodology, the resulting estimate is that only 19.7% of the variation in the covenant vectors $\mathcal{C}_{i,k}$ occurs within firm, while 80.3% of the variation is between-firm. Effectively, many firms adopt a standard set of covenants that they apply to most of their bond issues. Since the baseline empirical strategy in the paper relies on within-firm estimation, there is then not much difference when additionally interacting the fixed effects with the covenant vectors.

H.1 Variance Decomposition Methodology

For each firm k , define $\bar{\mathcal{C}}_k$ as the firm’s average covenant vector. Also let $\bar{\mathcal{C}}$ be the overall mean in the entire sample. The between-firm variance is proportional to

$$\sum_k n_k \cdot (\bar{\mathcal{C}}_k - \bar{\mathcal{C}})^T (\bar{\mathcal{C}}_k - \bar{\mathcal{C}}),$$

where n_k is the number of observations for firm k . On the other hand, the within-firm variance is proportional to

$$\sum_k \sum_i (\mathcal{C}_{i,k} - \bar{\mathcal{C}}_k)^T (\mathcal{C}_{i,k} - \bar{\mathcal{C}}_k).$$

Note that these are both scalar quantities that correspond to the sums of the between-firm and within-firm variances for each individual dimension of the covenant vector.⁸ The ratio of the between-firm variance to the within-firm variance is therefore

$$\psi = \frac{\sum_k n_k \cdot (\bar{\mathcal{C}}_k - \bar{\mathcal{C}})^T (\bar{\mathcal{C}}_k - \bar{\mathcal{C}})}{\sum_k \sum_i (\mathcal{C}_{i,k} - \bar{\mathcal{C}}_k)^T (\mathcal{C}_{i,k} - \bar{\mathcal{C}}_k)}.$$

Implementing this variance decomposition results in the point estimate $\hat{\psi} = 4.07$. This estimate implies that in the sample, 19.7% (given by $\frac{1}{1+\hat{\psi}}$) of the variation is within-firm, and 80.3% of the variation is between-firm.

I A Hedonic Analysis of Primary Market Bond Prices

This section presents a hedonic analysis of primary market bond prices outside of crisis periods, as a function of bond and issuer characteristics, as well as buyer composition in the primary market. This analysis confirms and extends the results on non-crisis equilibrium secondary market prices of Section 6.

I consider the sample of US investment-grade bonds issued between 2005 and 2020 outside of the crisis period of Figure 2. I restrict the sample to start in 2005 since this is the year in which the insurance holdings data becomes available. For each bond i issued by firm k , I regress the offering yield at issuance, $\text{OY}_{i,k}$, on bond characteristics X_i including the insurer share at issuance $\phi_{i,\tau(i)}$:

$$\text{OY}_{i,k} = \alpha + \beta' X_i + \varepsilon_{i,k}. \quad (\text{A.14})$$

I also consider specifications in which I include firm fixed effects γ_k , which absorb all variation that is fixed at the issuer level:

$$\text{OY}_{i,k} = \gamma_k + \beta' X_i + \varepsilon_{i,k}. \quad (\text{A.15})$$

The resulting estimates from these hedonic specifications are reported in Table A.8. Across specifications, whether including the firm fixed effects or not, the results remain qualitatively and quantitatively consistent. As expected, the offering yield is strongly decreasing as credit

⁸This follows from the linearity of expectations. Formally, indexing the dimensions by d , we have that

$$\mathbb{E}[(\bar{\mathcal{C}}_k - \bar{\mathcal{C}})^T (\bar{\mathcal{C}}_k - \bar{\mathcal{C}})] = \sum_d \mathbb{E}[(\bar{\mathcal{C}}_{k,d} - \bar{\mathcal{C}}_d)].$$

ratings improve: controlling for characteristics other than firm effects, BBB bonds have on average offering yields 138 basis points higher than AAA bonds. The offering yields are also increasing with bond duration: the semi-elasticity of the offering yield to duration is between 112 and 122 basis points. Further, callable issuances are associated with lower offering yields.

Both in the within-firm specifications and in the specifications without firm fixed effects, higher insurance ownership at issuance is associated with slightly higher offering yields: a 10 percentage points higher insurer share associates with a 12 to 13 basis points offering yield increase. This implies that on average, across all issues, insurers tend to pay lower prices when purchasing bonds on the primary market. These primary market results confirm and extend the secondary market pricing results of Section 6, in which it was established that bonds with higher insurance ownership also tend to have slightly higher yields in the secondary market.

In the secondary market context, the higher yields can be rationalized as resulting from an illiquidity premium, since insurer-held bonds become endogenously more illiquid. Conversely, in the primary market context, the higher offering yields are evidence that firms pay a slightly higher cost of capital when their bonds are placed with insurers. In this sense, the impact of insurer ownership on firms' cost of capital is state contingent: issuers effectively pay a few basis points in higher cost of capital in bad times, but this results in a lower cost of capital during crisis times (such that placing bonds with insurers has features of an insurance contract).

J Fire Sales and Reversals

This section provides details on the return reversal analysis in the spirit of Coval and Stafford (2007) introduced in Section 2.2. I test whether the negative abnormal bond returns in the Great Recession exhibit subsequent price reversal, as would be consistent with a fire sales mechanism. To do this, I examine the cumulative abnormal returns on a suitably constructed portfolio. I construct a long-short portfolio which takes positions in investment-grade bonds that are outstanding as of 2007Q1 and have at least one counterpart bond that is issued by the same firm and has the same characteristics (duration, rating, issue size, seniority, and so forth). Within each group of matching bonds, I sort bonds according to their pre-crisis insurer share. The portfolio then takes long positions in bonds that have an insurer share below the group median, and short positions in bonds above the group median. The positions in each bond are sized so as to have equal volatility (based on the pre-crisis estimated bond volatility), such that each bond contributes an equal amount to the volatility of the overall portfolio.

Summing up, this portfolio takes matched positions in bonds within the fixed effect groups, isolating the component of the return that is coming from variation in the bond's insurer share. The level of leverage is chosen so that the ex-ante (pre-crisis) volatility of the overall long-short portfolio matches that of a value-weighted index of excess returns $R_{i,t}^e$ on US investment-grade corporate bonds. Importantly, since the overall long-short portfolio volatility depends purely

on the chosen leverage level, the analysis is intended to emphasize the dynamics of the returns rather than their absolute magnitude. The cumulative returns on this portfolio are shown in Appendix Figure A.6. The figure also shows bootstrapped 95% confidence bands. As expected, the portfolio exhibits negative abnormal returns during the crisis, capturing the fact that bonds with lower insurer ownership under-performed relative to their counterparts with higher insurer ownership. The dynamics of the portfolio return now allow us to properly assess reversal.

The portfolio return reaches a minimum towards the beginning of 2009, and then gradually reverts over time. The abnormal relative return is mostly gone by the end of 2009, and it is fully reverted by the middle of 2010. Hence, at the within-firm level, the pricing gaps are not particularly long-lived. These reversal dynamics are consistent with a fire sales theoretical mechanism (as in Williamson 1988, Shleifer and Vishny 1992).

As discussed in Section 5.2, the reversal documented in Appendix Figure A.6 is consistent with a longer-lived effect on the real side and an impact on firm investment because of the presence of different dynamics *within firm* and *across firms*. This is in turn consistent with theories that postulate different degrees and dynamics of demand-driven price impact for assets depending on the availability of close substitutes—as emphasized, for instance, in the models of Chaudhary et al. (2022), Gabaix and Koijen (2022), and Davis et al. (2023). When comparing bonds within firm that have equivalent characteristics, it is reasonable to expect faster reversal than when comparing the bonds of different firms, since mispricings in the former category are both more easily identifiable and more easily correctable. These within-firm mispricing dynamics, while helpful to achieve identification in the type of saturated specifications considered in Section 2, are not however the same channel as for real firm outcomes. Rather, fire sales at the level of the firm itself have a larger impact on firm outcomes, and these naturally have slower reversal dynamics.

The argument above explains why the reversal occurs more quickly within-firm than across firms. It is also helpful to think about the overall time scale—e.g., why the within-firm mispricing takes 4 to 6 quarters to revert during the Great Recession. This ultimately indicates the presence of some persistent limits to arbitrage in the corporate bond market, which I discuss in Section 2.2. To summarize, while a full examination of the reasons for these limits to arbitrage is beyond the scope of this paper, I highlight several institutional features that make these likely.

At a baseline level, limits to arbitrage are likely to be supported by search and matching frictions, given the over-the-counter (OTC), decentralized nature of the corporate bond market (Duffie, Gârleanu and Pedersen 2005, 2007). Moreover, regulatory frictions prevent insurers from shifting their portfolios to similar cheap bonds: as discussed by Ellul et al. (2015), insurers (particularly life insurers, which are the most important group for the corporate bond market) have flexibility to hold assets at historical cost towards regulatory capital requirements. This

means that they have incentives not to trade bonds in a downturn period—even if these have lower-priced close substitutes—since this would entail having to recognize the capital losses.

Lastly, in large crises arbitrage capital is unavailable because all investor sectors with a large presence in the corporate bond market are attempting to sell on net—as documented in Appendix Figure A.3, which uses transactions data from the U.S. Flow of Funds to demonstrate that between 2008Q1 and 2009Q1 banks, broker-dealers, government-sponsored enterprises, insurers, mutual funds and ETFs, pensions funds, and foreign investors were all net sellers of corporate bonds.

K Excluding Financials and Utilities

As outlined in Sections 2.3 and 5.2, one possible concern with the baseline empirical strategy is that financial or utility firms constitute a sizable fraction of the identifying variation, and that these might be special: for instance, bonds issued by utility firms may be more likely to have complex special covenants, and investment for these sectors might be less comparable to that of non-utilities, non-financial firms. To rule out these concerns, in this section I estimate both the security-level and firm-level specifications excluding these two sectors. Specifically, I exclude one-digit SIC codes 6 and 9, and two-digit SIC code 49.⁹

The security-level estimation results with this sample restriction are reported in panel A of Table A.9, while the firm-level estimation results are reported in panel B. In both cases, the results remain robust to excluding these industrial sectors, and enough variation is left for the effects to be estimated with sufficient statistical significance. Appendix Section H elaborates further on covenants, explaining that accounting for bond covenants does not meaningfully impact the results because variation in covenants is concentrated across firms rather than within firm—and hence a large part of it is already accounted for in the baseline empirical strategy which relies on within-firm estimation.

L The Role of ETFs: Additional Details

For simplicity of exposition, my analysis in the main text considers open-end mutual funds and exchange traded funds (ETFs) jointly. In this section, I disentangle these two types of funds and discuss the role of ETFs in greater depth, focusing particularly on differences in redemption mechanisms and their relationship to corporate bond fire sales. I investigate whether ownership by ETFs impacts corporate bond price dynamics in crises differently from

⁹This also excludes the public administration sector (one-digit SIC code 9). I refer to the sector as being about financial and utilities because there are very few public administration firms in the sample. This is because the entire analysis already focuses solely on corporate bonds, excluding central and local government bonds.

ownership by open-end mutual funds, but cannot reject the null hypothesis of no differences with a sufficiently high level of statistical confidence.

In the corporate bond market, ETFs have historically been very small, although they have been growing in size in recent year. Corporate bond holdings by ETFs account for just 1 percent of the fund sector’s total holdings in my sample in 2005, but this figure grows to 12 percent by 2020.¹⁰ Lettau and Madhavan (2018) and Ben-David et al. (2017) provide an overview of the structure of ETFs and their differences with regular open-end mutual funds, and also review the growing academic literature studying them. Most saliently, shares in ETFs are traded continuously on public exchanges, and the task of keeping an ETF’s price in line with that of the securities it holds is delegated to financial institutions known as authorized participants (APs). To perform arbitrage, APs trade in-kind with the ETF sponsor in what’s known as the *primary* ETF market. APs can carry out redemptions by returning an ETF share to the ETF provider in exchange for a basket of the underlying securities. Alternatively, they can create ETF shares via the reverse primary market process.

Importantly, trades of ETF shares among the general public—which constitute the *secondary* ETF market—do not directly involve trades in the underlying securities. This process contrasts with flows in and out of open-end mutual funds. Abstracting from cash buffers, open-end mutual funds have to physically liquidate their security holdings when faced with investor redemptions, since contractually these redemptions must be met in-kind (and conversely, they must purchase securities to accommodate inflows).

The question of whether ETFs threaten or improve financial stability, particularly in the bond market, has been contentious among researchers and policymakers. On the one hand, ETFs may cater to an investor base that is particularly prone to liquidate its investments in times of distress, thus exposing the bond markets to a source of destabilizing demand: Dannhauser and Hoseinzade (2021), for example, show that the flow-performance sensitivity of ETFs is more than double that of open-end mutual funds.

On the other hand, there are at least two reasons to believe that bond ETFs may actually be beneficial in times of crisis. First, the open-end mutual fund structure is subject to run incentives à la Diamond and Dybvig (1983), as highlighted and formally modeled by Chen et al. (2010) and Zeng (2017). The ETF structure removes these run incentives (Goldstein et al. 2017). Second, the ETF structure also removes the mechanical link between fund-investor sales and sales of the underlying securities. Large investor sales of ETF shares on the secondary ETF market do not necessarily have to lead to ETF outflows, since the latter require APs to conduct in-kind redemptions. Further, APs may decide to keep the redeemed bonds on their balance sheets (for instance, to avoid liquidation costs) and thus not go on to sell them in the secondary bond market.

¹⁰Precisely because of their small relative size, simply excluding ETFs from the sample does not meaningfully impact any of the analyses that I present throughout the paper.

To empirically assess the impact of ETF ownership on bond price dynamics during crises, I consider the following counterpart to the baseline specification of Section 2.2:

$$\log \zeta_{i,e} = \alpha + \beta_{\text{OE}} \cdot \phi_{i,t_e^S-12}^{\text{Open-End}} + \beta_{\text{ETF}} \cdot \phi_{i,t_e^S-12}^{\text{ETF}} + \text{Interacted Fixed Effects} + \varepsilon_{i,e}, \quad (\text{A.16})$$

where $\phi_{i,t}^{\text{Open-End}}$ is the share of bond i that is owned by open-end mutual funds, while $\phi_{i,t}^{\text{ETF}}$ is the share owned by ETFs. I estimate this specification across events, with fully saturated interacted fixed effects (encompassing event dummies, issuer dummies, and bond characteristics dummies) as in Section 2.2. The point estimates from this specification, displayed in Figure A.20, are $\hat{\beta}_{\text{OE}} = .39$ and $\hat{\beta}_{\text{ETF}} = .98$, but the coefficient on the ETF share is estimated imprecisely, as evidenced by the wide confidence interval. Importantly, a Wald test fails to reject the null hypothesis of coefficient equality at the 10 percent confidence level.

The low statistical power of this analysis is plausibly due to the small relative size of ETFs in the corporate bond market.¹¹ A larger point estimate for $\hat{\beta}_{\text{ETF}}$, as compared to $\hat{\beta}_{\text{OE}}$, would imply that ETF ownership is particularly destabilizing for bond prices in crises—which may result from the clientele effects discussed above. However, the high degree of statistical uncertainty around these estimates prevents me from drawing strong conclusions.

M Real Effects: A Structural Interpretation

In this section I present a simple organizing framework that yields a structural interpretation of the real-side results in the paper. More specifically, I outline how the estimated impact of firm-level insurer ownership on firm investment dynamics can be interpreted in terms of its effects on firms' marginal cost of financing and effective Tobin's q . I begin by characterizing the first-order condition for optimal firm investment as a function of marginal q and marginal costs of financing in the presence of costly external financing. Bolton, Chen and Wang (2011) derive this same first-order condition in the context of a continuous-time model of investment with endogenous cash management. To focus on the economic intuition that is relevant for my analysis, I show how the same condition can be obtained in a simpler, discrete-time model with exogenous external financing episodes. I then show how to use the first-order condition for firm investment to derive estimating equations that relate my empirical estimates to structural elasticities. I follow conventions that are common in the q -theory literature (Hayashi 1982, Kogan and Papanikolaou 2012), and my notation follows Campbell (2017).

¹¹These results are qualitatively analogous if I restrict the sample to only include observations corresponding to the COVID crash of 2020, when the relative importance of ETF corporate bond holdings was highest.

M.1 A Model of Firm Investment With Costly External Financing

A firm produces output Y_t using its capital stock K_t and labor L_t with a technology $f(K_t, L_t)$. Labor is frictionlessly adjustable, so I focus on the choice of capital investment, implicitly assuming optimal labor choice. The dynamics of the capital stock are given by

$$K_{t+1} = (1 - \delta)K_t + I_t, \quad (\text{A.17})$$

where I_t is investment. Note that K_{t+1} is known at time t . The firm's profits are Π_t . Capital investment I_t incurs physical adjustment costs that depend only on the ratio of investment to total capital. With $i_t = \frac{I_t}{K_t}$ the investment rate at time t , adjustment costs are given by

$$\Omega(i_t) = \omega(i_t) K_t, \quad (\text{A.18})$$

with $\omega(\cdot) > 0$ and convex in i_t . The firm has a cash stock W_t that evolves as

$$W_{t+1} = W_t + \Pi_t - \Omega_t - D_t, \quad (\text{A.19})$$

and pays out a share λ_t of its cash stock as dividend each period:¹²

$$D_t = \lambda_t W_t. \quad (\text{A.20})$$

For simplicity, I take the external financing policy to be exogenous. I assume that the firm raises external financing (which may be interpreted as either debt or equity) whenever its cash stocks hits a lower barrier $\underline{W}$, which brings it back to a cash level $\overline{W} > \underline{W}$. In doing this, the firm incurs a cost

$$\tau \Delta_W, \quad (\text{A.21})$$

where $\Delta_W \equiv \overline{W} - \underline{W}$, and $\tau \geq 1$ represents a wedge between the costs of internal and external financing.

The firm seeks to maximize the present discounted value V_t of present and future dividends with its investment policy. I let $M_{t,t+\tau}$ be the (possibly stochastic) time- t discount factor for cash flows happening at time $t + \tau$. The firm's value therefore satisfies

$$V_t = \max_{\{I_t\}} \mathbb{E}_t \sum_{\tau=0}^{\infty} M_{t,t+\tau} D_{t+\tau}. \quad (\text{A.22})$$

Because of the presence of costly external financing, firm value is (weakly) increasing in its

¹²As with other elements of the firm's cash management policy, I leave λ_t unmodeled. Here it is effectively taken as exogenous.

cash holdings. We can define the marginal value of cash as

$$\text{MVC}_t = \frac{\partial V_t}{\partial W_t}, \quad (\text{A.23})$$

which can be equivalently interpreted as the marginal cost of financing (MCF) that the firm faces at a given point in time:

$$\text{MCF}_t = \text{MVC}_t. \quad (\text{A.24})$$

In the frictionless benchmark, $\tau = \text{MCF}_t = \text{MVC}_t = 1$. When we have a positive financing wedge $\tau > 1$, MCF_t will be greater than one, reflecting the marginal cost of depletion of the firm's cash stock.

The solution to the firm's optimization problem specified in (A.22) is characterized recursively by the following Bellman equation:

$$V_t = \max_{I_t} \{D_t + \mathbb{E}_t[M_{t,t+1}V_{t+1}]\}. \quad (\text{A.25})$$

The first-order condition for this optimization problem equates the marginal cost of today's investment with its marginal impact on total firm value:

$$\text{MCF}_t \omega'(i_t) = q_t, \quad (\text{A.26})$$

where marginal Tobin's q is

$$q_t = \mathbb{E}_t \left[M_{t,t+1} \frac{\partial V_{t+1}}{\partial K_{t+1}} \right]. \quad (\text{A.27})$$

Importantly, the optimality condition (A.26) shows that in the presence of financial frictions the key sufficient statistic for investment is not marginal q , as in the frictionless benchmark, but rather the ratio between marginal q and marginal cost of financing, which I refer to as effective q :

$$q_t^{\text{eff}} = \frac{q_t}{\text{MCF}_t} \quad (\text{A.28})$$

For simplicity, I assume the following commonly-used quadratic form for ϕ :

$$\phi(i_t) = i_t + \frac{\theta}{2} i_t^2, \quad (\text{A.29})$$

where the scalar parameter $\theta > 0$ governs the magnitude of physical adjustment costs. With this functional specification, the firm's optimality condition takes the simple form

$$i_t = \frac{1}{\theta} \left(\frac{q_t}{\text{MCF}_t} - 1 \right). \quad (\text{A.30})$$

M.2 Relating the Empirical Estimates to the Model

The real-side estimates of the impact of *ex ante* firm-level insurer shares ϕ measure (in the absence of confounders) the cross-sectional effect $\frac{\partial i}{\partial \phi}$. Using the first-order condition for firm investment in equation (A.30), these estimated investment effects allow me to assess the cross-sectional impact of investor base composition on the marginal cost of financing and effective q for the representative firm in a crisis. The estimating equations are

$$\frac{\partial q^{\text{eff}}}{\partial \phi} = \theta \frac{\partial i}{\partial \phi}, \quad (\text{A.31})$$

and

$$\frac{\partial \text{MCF}}{\partial \phi} = -\theta \frac{q}{(1 + \theta i)^2} \frac{\partial i}{\partial \phi}. \quad (\text{A.32})$$

All terms in these estimating equations are measurable from the data, with the exception of the adjustment cost parameter θ , which needs to be calibrated. Figure A.21 plots the estimated values of $\frac{\partial q^{\text{eff}}}{\partial \phi}$ and $\frac{\partial \text{MCF}}{\partial \phi}$ as a function of the adjustment cost parameter θ . Depending on the calibrated value of θ , a 50 percentage point increase in the firm-level insurer share leads to an increase in effective Tobin's q of the order of 2 to 6 percentage points, and a decrease in the marginal cost of financing of the order of 1 to 4 percentage points.¹³

N Details on Oster Test

In this section I provide further details on the partial identification methodology of Section 3.3. I follow the notation and results in Oster (2019). I let $F_{i,e}$ be a vector including dummies for the fully-saturated set of interacted fixed effects in the baseline estimates, entering the regression model with population parameters γ . I refer to the product $\gamma F_{i,e}$ as $O_{i,e}$, which stands for *observables*. Suppose that the true model also includes some relevant omitted variables $Q_{i,e}$, which enter into the linear specification with some non-zero population parameters η :

$$\log \zeta_{i,e} = \alpha + \beta \cdot \phi_{i,t_e^S-12} + \underbrace{\gamma F_{i,e}}_{\equiv O_{i,e}} + \underbrace{\eta Q_{i,e}}_{\equiv U_{i,e}} + \varepsilon_{i,e}, \quad (\text{A.33})$$

where I refer to the product $\eta Q_{i,e}$ as $U_{i,e}$, which stands for *unobservables*. I define the following ratio, whose numerator captures the extent to which the excluded unobservables co-vary with the insurer shares ϕ_{i,t_e^S-12} , and whose denominator captures the extent to which the included

¹³Complementary work on the real effects of equity markets (as opposed to bond markets) includes Baker et al. (2003), Chen et al. (2007), Edmans et al. (2012), and Hau and Lai (2013).

observables co-vary with them:

$$\delta = \frac{\text{Cov}(U, \phi)}{\text{Var}(U)} \left[\frac{\text{Cov}(O, \phi)}{\text{Var}(O)} \right]^{-1}. \quad (\text{A.34})$$

The ratio δ therefore relays the strength of selection of unobservables relative to selection on observables in the extended model of equation (A.33). It is the object of primary interest in the analysis of Section 3.3.

The sample variances of log drawdowns and insurer shares in the regression specification of equation (A.33) are $\hat{\sigma}_y^2$ and $\hat{\sigma}_x^2$. The *short regression* estimates the model (A.33) via OLS while only including event fixed effects: this corresponds to the estimates in column 1 of Table 3. The estimated beta and R-squared values from the short regression are β° and R° . The *long regression* estimates the same model but including the fully saturated set of interacted fixed effects: this corresponds to the estimates in column 5 of Table 3. The estimated beta and R-squared values from the long regression are $\tilde{\beta}$ and $\tilde{R}$. Let $R_{\max}$ be the R-squared from a hypothetical regression that also includes all relevant unobservables. As in Oster (2019), I let $R_{\max} = \min(1.3 \tilde{R}, 1)$.

We are interested in estimating the value of δ consistent with a true treatment effect $\hat{\beta} = 0$. Also define $\hat{\tau}_x$ as the variance of the residuals from a regression of the insurer shares ϕ_{i,t_e^S-12} in equation (A.33) on the interacted fixed effects. Then the threshold value $\bar{\delta}$ is computed as

$$\bar{\delta} = \frac{\nu_1}{\nu_2}, \quad (\text{A.35})$$

where

$$\begin{aligned} \nu_1 = & (\tilde{\beta} - \hat{\beta})(\tilde{R} - R^\circ)\hat{\sigma}_y^2\hat{\tau}_x + (\tilde{\beta} - \hat{\beta})\hat{\sigma}_x^2\hat{\tau}_x(\beta^\circ - \tilde{\beta})^2 \\ & + 2((\tilde{\beta} - \hat{\beta}))^2 \left(\hat{\tau}_x(\beta^\circ - \tilde{\beta})\hat{\sigma}_x^2 \right) + (\tilde{\beta} - \hat{\beta})^3 (\hat{\tau}_x\hat{\sigma}_x^2 - \hat{\tau}_x^2) \end{aligned} \quad (\text{A.36})$$

and

$$\begin{aligned} \nu_2 = & \left(R_{\max} - \tilde{R} \right) \hat{\sigma}_y^2(\beta^\circ - \tilde{\beta})\hat{\sigma}_x^2 + (\tilde{\beta} - \hat{\beta}) \left(R_{\max} - \tilde{R} \right) \hat{\sigma}_y^2 (\hat{\sigma}_x^2 - \hat{\tau}_x) \\ & + (\tilde{\beta} - \hat{\beta})^2 \left(\hat{\tau}_x \left((\beta^\circ - \tilde{\beta})\hat{\sigma}_x^2 \right) + (\tilde{\beta} - \hat{\beta})^3 (\hat{\tau}_x\hat{\sigma}_x^2 - \hat{\tau}_x^2) \right). \end{aligned} \quad (\text{A.37})$$

As mentioned in the main text, both Altonji et al. (2005) and Oster (2019) argue that results with a threshold value $\bar{\delta}$ greater than or equal to 1 should be viewed as robust to the potential presence of selection on unobservables. A first argument for the $\bar{\delta} \geq 1$ bound is empirical. Oster (2019), for example, studies a sample of all articles published in the *American Economic Review*, the *Quarterly Journal of Economics*, the *Journal of Political Economy*, and *Econometrica* between 2008 and 2010, and finds that only 30 percent of published empirical results are robust to a $\bar{\delta} \geq 1$ bound. A second argument is theoretical: since the observables included in a researcher's baseline specification are normally the ones that she *a priori* reputes to be the most important, the $\bar{\delta} \geq 1$ bound follows from a straightforward ordering argument.

O Investor Composition and Bond Turnover Volatility

In this section I provide additional results showing that higher fund ownership forecasts a larger increase in secondary-market bond turnover during downturns, which is consistent with stronger redemptions-induced selling by funds during crises. I define monthly *turnover* for each bond as the ratio of total trading volume to the value outstanding of the bond. I then consider the following counterpart of equations (A.1) and (22), which regresses turnover on quarter-lagged insurer shares:

$$\text{Turnover}_{i,t} = \alpha + \bar{\Lambda} \phi_{i,t-12} + \tilde{\Lambda} \phi_{i,t-12} \text{GZ}_t + \text{Interacted Fixed Effects} + \varepsilon_{i,t}. \quad (\text{A.38})$$

As in the previous specifications, I include an interaction with the [Gilchrist and Zakrajšek \(2012\)](#) index (GZ_t) to capture variation in the estimated effects over the credit cycle. I also continue to include issuer dummies, bond characteristics dummies, and time dummies in the fixed effects interaction.

Figure [A.22](#) plots the estimated total effect $\Lambda_t = \bar{\Lambda} + \tilde{\Lambda} \text{GZ}_t$ over time. Two features are apparent from these estimates. First, insurer ownership substantially reduces bond turnover in normal times, when credit spreads are low. For example, in quiet periods such as 2005-07 and 2017-19, the estimated total effect is $\Lambda_t = -.06$, which is economically large given that the unconditional average of monthly turnover in the sample is about 10 percent. Second, investor base composition is a strong predictor of spikes in bond turnover in crisis times, as evident from the dynamics of the estimated Λ_t . Lower insurer ownership (and therefore higher fund ownership) forecasts a larger increase in secondary-market bond turnover during downturns.

References for Online Appendix

- Altonji, Joseph G, Todd E Elder, and Christopher R Taber**, “An evaluation of instrumental variable strategies for estimating the effects of Catholic schooling,” *Journal of Human Resources*, 2005, 40 (4), 791–821.
- Baker, Malcolm, Jeremy C Stein, and Jeffrey Wurgler**, “When does the market matter? Stock prices and the investment of equity-dependent firms,” *The Quarterly Journal of Economics*, 2003, 118 (3), 969–1005.
- Becker, Bo, Marcus M Opp, and Farzad Saidi**, “Regulatory forbearance in the US insurance industry: The effects of removing capital requirements for an asset class,” *Forthcoming at the Review of Financial Studies*, 2021.
- Ben-David, Itzhak, Francesco Franzoni, and Rabih Moussawi**, “Exchange-traded funds,” *Annual Review of Financial Economics*, 2017, 9, 169–189.
- Bolton, Patrick, Hui Chen, and Neng Wang**, “A unified theory of Tobin’s q, corporate investment, financing, and risk management,” *The Journal of Finance*, 2011, 66 (5).
- Campbell, John Y**, *Financial decisions and markets: A course in asset pricing*, Princeton University Press, 2017.

- Chaudhary, Manav, Zhiyu Fu, and Jian Li**, “Corporate bond multipliers: Substitutes matter,” *Working Paper*, 2022.
- Chen, Qi, Itay Goldstein, and Wei Jiang**, “Price informativeness and investment sensitivity to stock price,” *The Review of Financial Studies*, 2007, *20* (3), 619–650.
- Chen, Qi, Itay Goldstein, and Wei Jiang**, “Payoff complementarities and financial fragility: Evidence from mutual fund outflows,” *Journal of Financial Economics*, 2010, *97* (2), 239–262.
- Chen, Yin, Zhong Xu, and Chuanwei Zou**, “Trading relationships in the Chinese repo market,” *Working Paper*, 2018.
- Coppola, Antonio, Matteo Maggiori, Brent Neiman, and Jesse Schreger**, “Redrawing the map of global capital flows: The role of cross-border financing and tax havens,” *The Quarterly Journal of Economics*, 2021, *136* (3), 1499–1556.
- Coval, Joshua and Erik Stafford**, “Asset fire sales (and purchases) in equity markets,” *Journal of Financial Economics*, 2007, *86* (2), 479–512.
- Dannhauser, Caitlin D and Saeid Hoseinzade**, “The unintended consequences of corporate bond ETFs: Evidence from the Taper Tantrum,” *The Review of Financial Studies*, 2021.
- Davis, Carter, Mahyar Kargar, and Jiacui Li**, “Why is asset demand inelastic?,” *Working Paper*, 2023.
- Diamond, Douglas W and Philip H Dybvig**, “Bank runs, deposit insurance, and liquidity,” *Journal of Political Economy*, 1983, *91* (3), 401–419.
- Duffie, Darrell, Nicolae Gârleanu, and Lasse H Pedersen**, “Over-the-counter markets,” *Econometrica*, 2005, *73* (6), 1815–1847.
- Duffie, Darrell, Nicolae Gârleanu, and Lasse H Pedersen**, “Valuation in over-the-counter markets,” *The Review of Financial Studies*, 2007, *20* (6), 1865–1900.
- Edmans, Alex, Itay Goldstein, and Wei Jiang**, “The real effects of financial markets: The impact of prices on takeovers,” *The Journal of Finance*, 2012, *67* (3), 933–971.
- Ellul, Andrew, Chotibhak Jotikasthira, and Christian T Lundblad**, “Regulatory pressure and fire sales in the corporate bond market,” *Journal of Financial Economics*, 2011, *101* (3), 596–620.
- Ellul, Andrew, Chotibhak Jotikasthira, Christian T Lundblad, and Yihui Wang**, “Is historical cost accounting a panacea? Market stress, incentive distortions, and gains trading,” *The Journal of Finance*, 2015, *70* (6), 2489–2538.
- Fama, Eugene F and Kenneth R French**, “Luck versus skill in the cross-section of mutual fund returns,” *The Journal of Finance*, 2010, *65* (5), 1915–1947.
- Gabaix, Xavier and Ralph SJ Koijen**, “In search of the origins of financial fluctuations: The inelastic markets hypothesis,” *Working Paper*, 2022.
- Gilchrist, Simon and Egon Zakrajšek**, “Credit spreads and business cycle fluctuations,” *American Economic Review*, 2012, *102* (4), 1692–1720.
- Goldsmith-Pinkham, Paul, Isaac Sorkin, and Henry Swift**, “Bartik instruments: What, when, why, and how,” *American Economic Review*, 2020, *110* (8), 2586–2624.
- Goldstein, Itay, Hao Jiang, and David T Ng**, “Investor flows and fragility in corporate bond funds,” *Journal of Financial Economics*, 2017, *126* (3), 592–613.
- Hau, Harald and Sandy Lai**, “Real effects of stock underpricing,” *Journal of Financial Economics*,

- 2013, *108* (2), 392–408.
- Hayashi, Fumio**, “Tobin’s marginal q and average q: A neoclassical interpretation,” *Econometrica*, 1982, pp. 213–224.
- Jenkinson, Tim, Howard Jones, and Felix Suntheim**, “Quid pro quo? What factors influence IPO allocations to investors?,” *The Journal of Finance*, 2018, *73* (5), 2303–2341.
- Kogan, Leonid and Dimitris Papanikolaou**, “Economic activity of firms and asset prices,” *Annual Review of Financial Economics*, 2012, *4* (1), 361–384.
- Koijen, Ralph SJ and Motohiro Yogo**, “The cost of financial frictions for life insurers,” *American Economic Review*, 2015, *105* (1), 445–75.
- Lettau, Martin and Ananth Madhavan**, “Exchange-traded funds 101 for economists,” *Journal of Economic Perspectives*, 2018, *32* (1), 135–54.
- Maggiori, Matteo, Brent Neiman, and Jesse Schreger**, “International currencies and capital allocation,” *Journal of Political Economy*, 2020, *128* (6), 2019–2066.
- National Association of Mutual Insurance Companies**, “The mutual factor,” *Technical Report*, 2019.
- Nikolova, Stanislava, Liying Wang, and Juan Julie Wu**, “Institutional allocations in the primary market for corporate bonds,” *Journal of Financial Economics*, 2020, *137* (2), 470–490.
- Oster, Emily**, “Unobservable selection and coefficient stability: Theory and evidence,” *Journal of Business & Economic Statistics*, 2019, *37* (2), 187–204.
- Rotemberg, Julio**, “Instrument variable estimation of misspecified models,” *MIT Sloan Working Paper*, 1983.
- Sen, Ishita and Varun Sharma**, “Internal models, make believe prices, and bond market cornering,” *Working Paper*, 2020.
- Shleifer, Andrei and Robert W Vishny**, “Liquidation values and debt capacity: A market equilibrium approach,” *The Journal of Finance*, 1992, *47* (4), 1343–1366.
- The Bond Market Association**, “Corporate operational underwriting process: Business practices in “plain English,”” *Technical Report, Jointly with the Depository Trust and Clearing Corporation*, 2004.
- Williamson, Oliver E**, “Corporate finance and corporate governance,” *The Journal of Finance*, 1988, *43* (3), 567–591.
- Zeng, Yao**, “A dynamic theory of mutual fund runs and liquidity management,” *Working Paper*, 2017.

Variable	Definition
Insurer share ($\phi_{i,t}$)	The share of bond i which is held by insurers at time t . Defined as $\phi_{i,t} = \frac{X_{i,t}^I}{V_{i,t}}$, where $V_{i,t}$ is the outstanding market value of bond i and $X_{i,t}^I$ is the market value of all insurers' holdings in bond i .
Bond return ($R_{i,t}$)	The raw weekly net return on bond i , defined as $R_{i,t} = \frac{P_{i,t} + C_{i,t}}{P_{i,t-1}} - 1$, where $P_{i,t}$ be the traded price of bond i in week t , which is constructed as the median price in the last day of week t in which bond i trades plus accrued coupon interest, and $C_{i,t}$ is the coupon payment in the relevant time period, if any.
Bond duration ($D_{i,t}$)	The duration of bond i at time t .
Excess return ($R_{i,t}^e$)	The excess return on bond i , defined as $R_{i,t}^e = R_{i,t} - R_t^f(D_{i,t})$, where $R_t^f(D)$ is the weekly return on a basket of Treasury bonds of duration D .
Drawdown ($\zeta_{i,e}$)	A summary measure capturing the total loss in value of a bond i during a market downturn episode e . Calculated as $\zeta_{i,e} = \max_{\tau \in \mathcal{T}_e} \left(1 - \frac{\text{CR}_{i,\tau}}{\text{CR}_{i,t_e^S}} \right)$, where $\mathcal{T}_e$ is the event window, τ indexes time, and $\text{CR}_{i,t} = \prod_{\tau=t_i^0}^t (1 + R_{i,\tau}^e)$ is the cumulative excess return on bond i since its first trading date t_i^0 .
Trading activity ($\mathcal{T}_{i,t}$)	A measure of quarterly trading activity in bond i , either the quarterly number of trades or quarterly dollar trading volume.
Instrument shifter ($\bar{\phi}_j$)	The average size of insurer j 's primary market bond purchases relative to those of the aggregate insurance sector (constructed so that $\sum_{j \in \mathcal{J}} \bar{\phi}_j = 1$).
Purchase indicators ($\xi_{i,j}$)	Dummy variable indicating whether insurer j acquired bond i at issuance.
Shift-share instrument (Z_i)	Shift-share instrument capturing the collective size of the set of insurers that acquire bond i at issuance, defined as $Z_i = \sum_{j \in \mathcal{J}} \xi_{i,j} \bar{\phi}_j$. The purchase indicators $\xi_{i,j}$ function as the shares, while the size coefficients $\bar{\phi}_j$ function as the shifters.
Bond yield spread ($s_{i,t}$)	Yield spread to Treasury of bond i at time t .

(Table continues on next page)

Table A.1: **List of variables and corresponding definitions.** This table lists key variables used throughout the paper and provides their corresponding definitions, for ease of reference.

(Table continues from previous page)

Variable	Definition
Gilchrist-Zakrajšek index (GZ_t)	An aggregate measure of credit spreads on U.S. corporate bonds, constructed and published by Gilchrist and Zakrajšek (2012).
Proxy safe hands share ($\tilde{\phi}$)	Defined as $\tilde{\phi}_{i,t} = 1 - \frac{X_{i,t}^F}{V_{i,t}}$, where $X_{i,t}^F$ is the market value of total mutual fund and ETF holdings in bond i , and $V_{i,t}$ is the outstanding market value of bond i .
Firm-level insurer share (Φ_k)	Share of a firm's bonds held by insurers in 2007Q1, defined as $\Phi_k = \frac{\sum_{i \in \mathcal{I}_k} X_{i,2007Q1}^I}{\sum_{i \in \mathcal{I}_k} V_{i,2007Q1}}$, where $V_{i,t}$ is the outstanding market value of bond i , $X_{i,t}^I$ is the market value of insurers' holdings in bond i , and $\mathcal{I}_k$ is the set of bonds by firm k outstanding in 2007Q1.
Bond size weights ($\rho_{i,k}$)	Weights proportional to bond size within firm, defined as $\rho_{i,k} = \frac{V_{i,2007Q1}}{\sum_{i' \in \mathcal{I}_k} V_{i',2007Q1}}$. These satisfy $\sum_i \rho_{i,k} = 1$.
Firm-level instrument (Z_k)	Version of the shift-share instrument aggregated to the firm level, using bond size weights. Defined as $Z_k = \sum_{i \in \mathcal{I}_k} \rho_{i,k} \cdot Z_i$.
Investment rate ($\text{Investment}_{k,t}$)	Investment rate of firm k in year t , measured either as yearly capital expenditures (CAPX) over lagged total assets or as yearly CAPX plus acquisitions over lagged total assets.
Firm issuance outcomes ($Y_{k,t}$)	One of two outcomes: either an indicator for whether firm k issued a new bond in year t , or the average offering yield on new bonds offered by firm k in year t , conditionally on issuing a new bond.
Purchase indicators: extended notation ($\xi_{i,j,k,t}$)	Dummy variable indicating whether investor j purchased bond i , issued by firm k , at the time of its issuance t .
Yield spread decile dummies ($D_{s,k,t}$)	Dummy indicating whether the average secondary-market yield spread to Treasury of firm k at time t falls within the decile s of firm-level yield spreads.

Table A.1: **List of variables and corresponding definitions (continued).** This table lists key variables used throughout the paper and provides their corresponding definitions, for ease of reference.

Covenant type	Covenant description
Negative pledge covenant	Restricts the issuer from issuing secured debt, unless the current issue is also secured on a pari passu basis.
Cross-default covenant	The bond automatically enters default if the issuer defaults on any other debt.
Cross-acceleration covenant	Allows the bondholder to accelerate the bond, if any other bond by the issuer has been accelerated because of default.
Put trigger on change of control (“poison put”)	The bondholder can exercise a put option on the bond (sell the bond back to the issuer) in the event of a change of corporate control.
Put trigger on rating decline	The bondholder can exercise a put option on the bond in the event of a credit rate decline.
Declining net worth trigger	The bondholder can exercise protection clauses if the issuer’s net worth falls below a minimum threshold.
After-acquired property clause	Any assets acquired by the issue at a future date will be added as collateral for the current debt (most common for utility firms).
Asset sale clause	Requires that proceeds from the sales of certain assets be used to redeem the bonds, but does not restrict the issuer’s ability to engage in asset sales.
Consolidation and merger restriction	Restricts the issuer’s ability to consolidate or merge with other entities.
Asset sale restriction	Limits the issuer’s ability to sell assets.
Investment restriction	Imposes restrictions on the issuer’s investment policy.
Net worth maintenance restriction	Requires the issuer to maintain a net worth above a minimum threshold.
Indebtedness restriction	Restricts the issuer from issuing additional debt, with limits on the absolute dollar amount of debt outstanding or percentage of total capital.
Funded debt restriction	Restricts the issuer’s ability to issue additional long-term debt (with an initial maturity of one year or longer).
Dividends-related payments restriction	Restricts the ability of the issuer to make dividend payments to shareholders.
Other payments restriction	Restricts the ability of the issuer to make payments (other than dividend payments) to shareholders or other parties.

(Table continues on next page)

Table A.2: **List of covenants.** This table lists the bond covenant categories used in the analysis.

(Table continues from previous page)

Covenant type	Covenant description
Liens	Establishes bondholders' right to sell assets used as collateral to recover unpaid debt.
Sales leaseback restriction	Restricts the ability of the issuer to engage in sales leaseback transactions, and/or restricts the use of proceeds from such transactions.
Senior debt issuance restriction	Restricts the issuance of additional senior debt by the issuer.
Subordinated debt issuance restriction	Restricts the issuance of additional subordinated (junior) debt by the issuer.
Stock issuance restriction	Restricts the issuance of additional common equity by the issuer.
Stock transfer, sale, or disposition restriction	Restricts the issuer's ability to transfer, sell, or dispose of its common equity or the common equity of a subsidiary.
Affiliate transactions restriction	Restricts the issuer's ability to engage in intra-group transactions with its affiliates and/or subsidiaries.
Net earnings test for issuance	Sets criteria based on the issuer's profitability as a precondition for the issuance of additional debt.
Fixed charge coverage restriction	Requires the issuer to maintain a minimum ratio of earnings to fixed charges.
Leverage test	Sets limits on the issuer's total leverage.
Subsidiary borrowing restriction	Restricts the ability of the issuers' subsidiaries to raise debt financing.
Covenant defeasance allowance with no tax consequences	Allows the issuer to be released from certain covenants under specific conditions without triggering tax liabilities.
Covenant defeasance allowance with tax consequences	Allows the issuer to be released from certain covenants under specific conditions, but doing this may trigger tax liabilities.
Monetary defeasance allowance without tax consequences	Allows the issuer to be released from the monetary obligation of the debt security and remove it from its balance sheet by pre-funding all payments via an escrow account, without tax consequences to the bondholders.
Legal defeasance allowance	Allows the issuer to be released from the monetary obligation of the debt security and remove it from its balance sheet by pre-funding all payments via an escrow account, with possible tax consequences to the bondholders.

Table A.2: **List of covenants (continued).** This table lists the bond covenant categories used in the analysis.

Covenant type	Prevalence
Negative pledge covenant	21.2%
Cross-default covenant	8.7%
Cross-acceleration covenant	22.8%
Put trigger on change of control (“poison put”)	11.1%
Put trigger on rating decline	0.4%
Declining net worth trigger	0.0%
After-acquired property clause	0.2%
Asset sale clause	0.5%
Consolidation and merger restriction	30.5%
Asset sale restriction	30.4%
Investment restriction	0.1%
Net worth maintenance restriction	0.9%
Indebtedness restriction	3.7%
Funded debt restriction	0.4%
Dividends-related payments restriction	1.1%
Other payments restriction	0.8%
Liens	1.8%
Sales leaseback restriction	14.9%
Senior debt issuance restriction	0.0%
Subordinated debt issuance restriction	0.4%
Stock issuance restriction	0.3%
Stock transfer, sale, or disposition restriction	0.3%
Affiliate transactions restriction	0.7%
Net earnings test for issuance	0.2%
Fixed charge coverage restriction	6.8%
Leverage test	0.0%
Subsidiary borrowing restriction	0.0%
Covenant defeasance allowance, no tax consequences	22.4%
Covenant defeasance allowance, tax consequences	4.0%
Monetary defeasance allowance, no tax consequences	21.9%
Legal defeasance allowance	5.4%

Table A.3: **Covenant prevalence in the sample.** This table shows the share of the bonds in the sample that contain covenants in each of the categories listed in Table A.2.

	US Insurers				US Funds*				Non-US Funds*			
	2005	2010	2015	2020	2005	2010	2015	2020	2005	2010	2015	2020
Asset Allocation (%)												
Bonds: Corporate, US	49	51	54	57	18	25	31	29	6	14	23	23
Bonds: Corporate, Non-US	5	4	3	3	2	3	2	2	33	40	37	33
Bonds: Sovereign, US	23	23	21	17	16	21	21	21	3	3	5	6
Bonds: Sovereign, Non-US	2	2	2	1	4	6	7	6	51	39	32	31
Bonds: Other, Incl. ABS	6	6	5	5	59	44	39	41	7	3	3	7
Equities	14	14	15	17	1	1	1	1	0	0	0	0
<i>*Fixed income specialists only</i>												

Table A.4: **Overview of overall portfolio holdings of insurers and fixed income specialist funds.** This table shows the composition of the portfolio of invested assets of insurance companies and fixed income specialist funds. I show the share of the aggregate portfolio of these institutions that is invested in US corporate bonds (defined as TRACE-reportable bonds), non-US corporate bonds, US sovereign bonds, non-US sovereign bonds, other bonds including asset-backed securities, and equities.

	US Investment Grade Corporate Bonds			
	2007	2010	2015	2020
Number of Securities	7,951	7,835	9,561	11,293
Value Outstanding (\$M)				
Mean	407	501	645	698
Median	250	300	474	500
95 th Percentile	1,500	1,750	2,000	2,000
Total Value (\$B)	3,236	3,925	6,165	7,880
Number of Issuers	1,255	1,411	1,492	1,868
Other Characteristics				
Average Duration (Yrs)	5.4	5.6	6.0	7.4
Share Junior (%)	7	6	5	7
Share Floating (%)	13	12	7	7
Share Callable (%)	48	57	66	76

Table A.5: **U.S. investment-grade corporate bonds: additional summary statistics.** This table reports summary statistics for the U.S. investment grade corporate bonds in the sample, including number of securities, statistics on their face value outstanding, the total size of the market, the number of issuers, average bond duration, and the share of bonds that is junior, floating-rate, or has callable features. Data is shown for the years 2007, 2010, 2015, and 2020. Bonds with a reported face value of less than \$25 million are excluded.

	Bond Excess Return	
	(1)	(2)
Insurer Share	.08 (.02)	.08 (.02)
<i>Fixed Effects Interaction:</i>		
Event Dummy	Yes	Yes
Bond Size	Yes	Yes
Bond Duration	Yes	Yes
Bond Rating	Yes	Yes
Issuer Dummy	Yes	Yes
Callable Dummy	—	Yes
Senior Dummy	—	Yes
Coupon Type	—	Yes
Identifying Observations	5,328	4,945
Identifying Issuers	452	437
R^2	.86	.88

Table A.6: **Estimates using arithmetic excess returns.** This table reports estimates from a version of the regression specification in equation (5) in which the left-hand side variable is defined simply as the net arithmetic excess return on each bond during each downturn event, $\bar{R}_{i,e} = \prod_{\tau=t_e}^{t_e^E} (1 + R_{i,\tau}^e) - 1$:

$$\bar{R}_{i,e} = \alpha + \beta \cdot \phi_{i,t_e^S-12} + \text{Interacted Fixed Effects} + \varepsilon_{i,e}.$$

The estimates are from weighted least squares (WLS) regressions, with bond sizes as weights. Standard errors are clustered at the issuer by event level and are reported in parentheses. The estimates are fully consistent with the baseline analysis: since the average drawdown in the sample, across events, is 19 percentage points, the baseline estimate $\hat{\beta} = -.41$ of Table 3 implies that an increase in the insurer share from 0 to 1 should lead on average to a roughly 8 percentage points higher excess return, which is what this alternative specification finds.

	OLS		2SLS	
	(1)	(2)	(3)	(4)
Insurer Share	-.40 (.10)	-.41 (.10)	-.56 (.20)	-.48 (.20)
<i>Fixed Effects Interaction:</i>				
Bond Size	Yes	Yes	Yes	Yes
Event Dummy	Yes	Yes	Yes	Yes
Bond Duration	Yes	Yes	Yes	Yes
Bond Rating	Yes	Yes	Yes	Yes
Issuer Dummy	Yes	Yes	Yes	Yes
Callable Dummy	—	Yes	—	Yes
Senior Dummy	—	Yes	—	Yes
Coupon Type	—	Yes	—	Yes
Identifying Observations	5,328	4,945	2,381	2,304
Identifying Issuers	452	437	373	363
R^2	.84	.85	.88	.89
First Stage F	—	—	56.7	54.3

Table A.7: **Shift-share IV estimates, excluding bonds issued prior to 2005.** This table repeats the estimates in Table 4, excluding bonds issued prior to 2005 from the shift-share instrumental variable (IV) regressions. Since the insurance holdings data start in 2005, in constructing the shift-share instrument for bonds issued prior to 2005 the baseline analysis uses insurers' holdings in 2005 to proxy for holdings at issuance. The estimates in this table confirm that excluding these bonds does not materially impact the results. The estimates are from weighted least squares (WLS) regressions, with bond sizes as weights. Standard errors are clustered at the issuer by event level and are reported in parentheses.

	Offering Yield (%)			
	(1)	(2)	(3)	(4)
Log Duration	1.22 (.06)	1.18 (.07)	1.16 (.06)	1.12 (.07)
Log Size	-.11 (.06)	.13 (.08)	-.01 (.06)	.17 (.08)
Senior Dummy	-.02 (.25)	-.06 (.15)	-.02 (.25)	-.05 (.16)
Callable Dummy	-.84 (.12)	-.64 (.23)	-.89 (.12)	-.65 (.23)
<i>Coarse Rating Dummy:[†]</i>				
AA	.61 (.19)	—	.60 (.19)	—
A	.90 (.15)	—	.89 (.14)	—
BBB	1.38 (.16)	—	1.38 (.16)	—
Insurer Share	—	—	1.17 (.21)	1.34 (.25)
Issuer FE	No	Yes	No	Yes
Identifying Observations	6,604	6,318	6,604	6,318
R^2	.25	.48	.26	.49

[†]Excluded category: AAA

Table A.8: **Hedonic analysis of primary market bond prices.** This table reports the estimates of the hedonic regression specification in equations (A.14) and (A.15) in the sample of investment-grade U.S. corporate bonds issued from 2005 onwards. The dependent variable is the bond's offering yield in percentage points. The independent variables are a bond's log duration, log size, dummies for the bonds being senior and callable, dummies for coarse bond rating categories (AAA, AA, A, and BBB, with AAA being the excluded category), and the insurer share at issuance. Columns 2 and 4 include issuer fixed effects. The estimates are from weighted least squares (WLS) regressions, with bond sizes as weights. Standard errors are clustered at the issuer level and are reported in parentheses.

(A) Security-Level Analysis

	Log Bond Drawdown	
	(1)	(2)
Insurer Share	-.52 (.14)	-.45 (.14)
<i>Fixed Effects Interaction:</i>		
Event Dummy	Yes	Yes
Bond Size	Yes	Yes
Bond Duration	Yes	Yes
Bond Rating	Yes	Yes
Issuer Dummy	Yes	Yes
Callable Dummy	—	Yes
Senior Dummy	—	Yes
Coupon Type	—	Yes
Identifying Observations	2,764	2,573
Identifying Issuers	223	220
R^2	.84	.86

(B) Firm-Level Analysis

	Investment Rate (%)	
	(1)	(2)
Insurer Share ($\hat{\beta}$)	3.06 (1.49)	6.10 (2.43)
Firm FE	Yes	Yes
Time FE	Yes	Yes
Controls	Yes	Yes
Outcome	CAPX	CAPX and Acquisitions
Observations	4,461	4,461
Firms	567	567
R^2	.76	.49
First-Stage F	47.4	51.6

Table A.9: **Estimates excluding utilities and financials.** This table reports results excluding utilities and financial firms, as well as public administration, as outlined in Appendix Section K. Panel A reports estimates of the security-level specification in equation (5), with standard errors clustered at the issuer by event level. Panel B reports estimates of the firm-level specification in equation (19), using two-stage least squares, with standard errors clustered at the firm level.

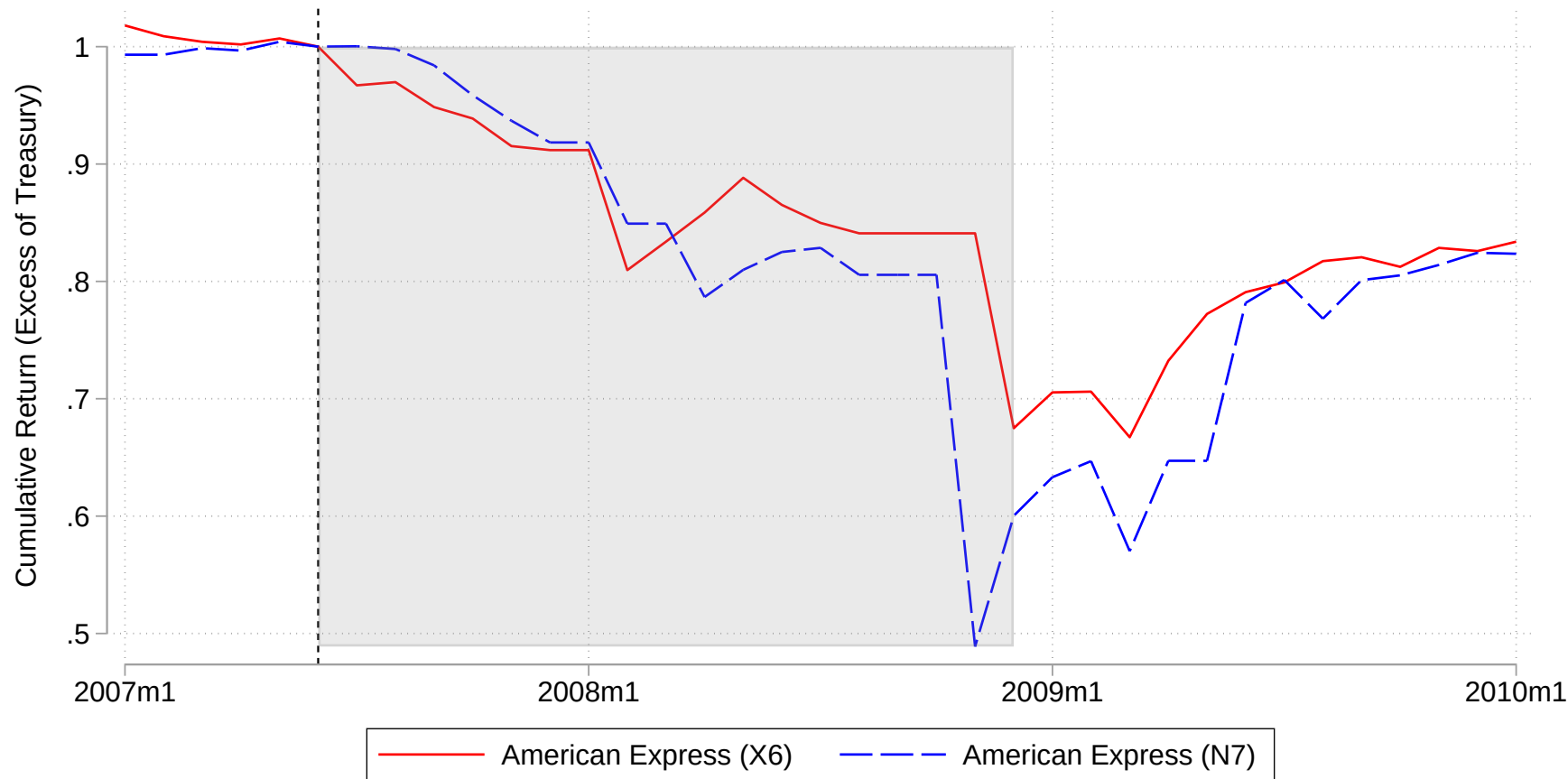


Figure A.1: **An example of the identifying variation.** This figure provides an example of the identifying variation. The *dashed blue line* plots the cumulative returns in excess of Treasury, $CR_{i,t}$, on American Express Credit Corporation's bond with CUSIP code 0258M0CN7, normalized to one at the beginning of the Great Recession event window (*gray shaded area*). The *solid red line* plots the cumulative returns in excess of Treasury on a second bond issued by American Express Credit Corporation, with CUSIP code 0258M0BX6, with matching characteristics and also normalized to one on the same date. These bonds, issued in December 2005 and June 2006, were both senior, five-year, A+ rated, \$500 million notes, and matched on all other characteristics included in the fixed effect set.

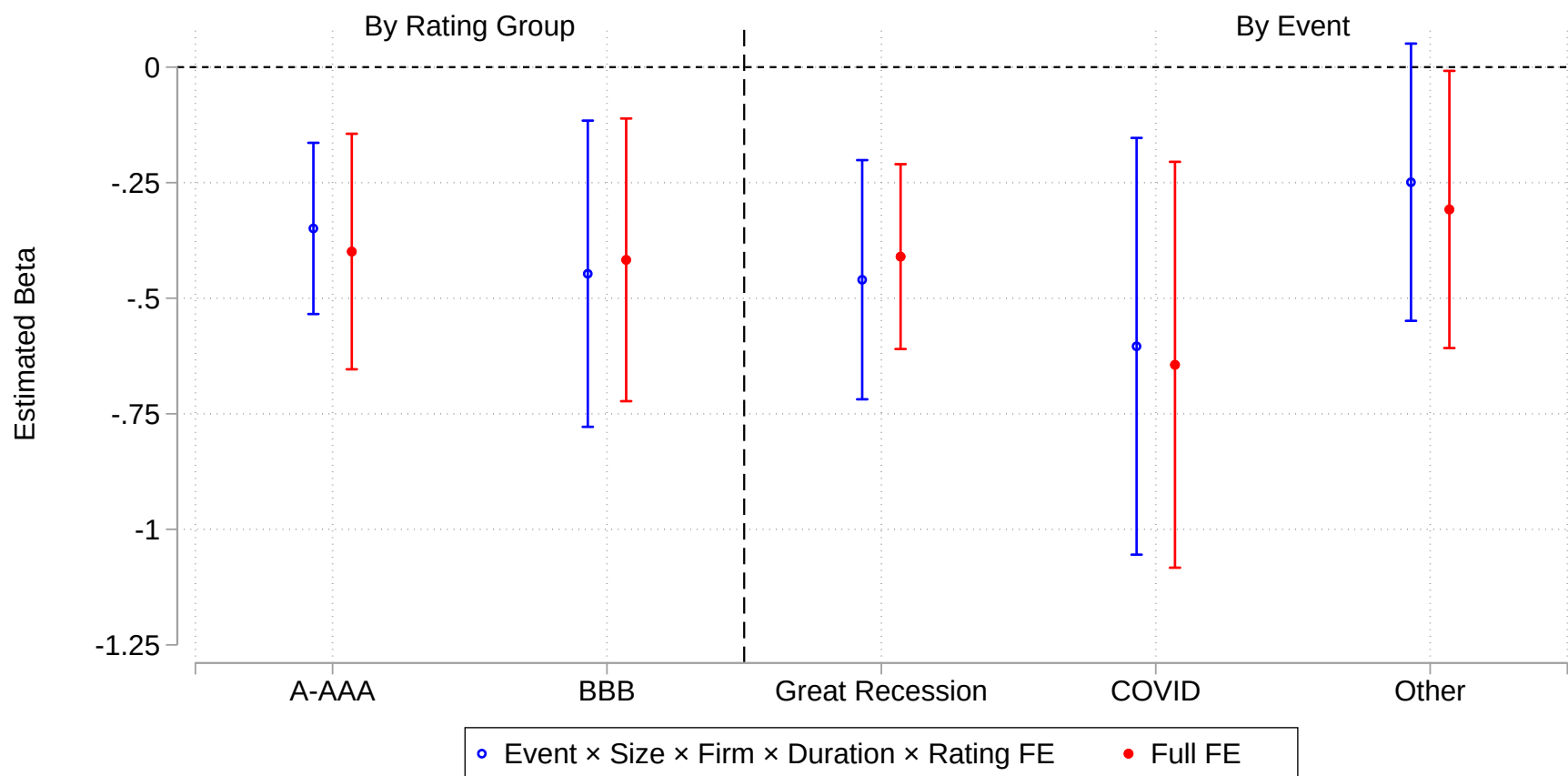
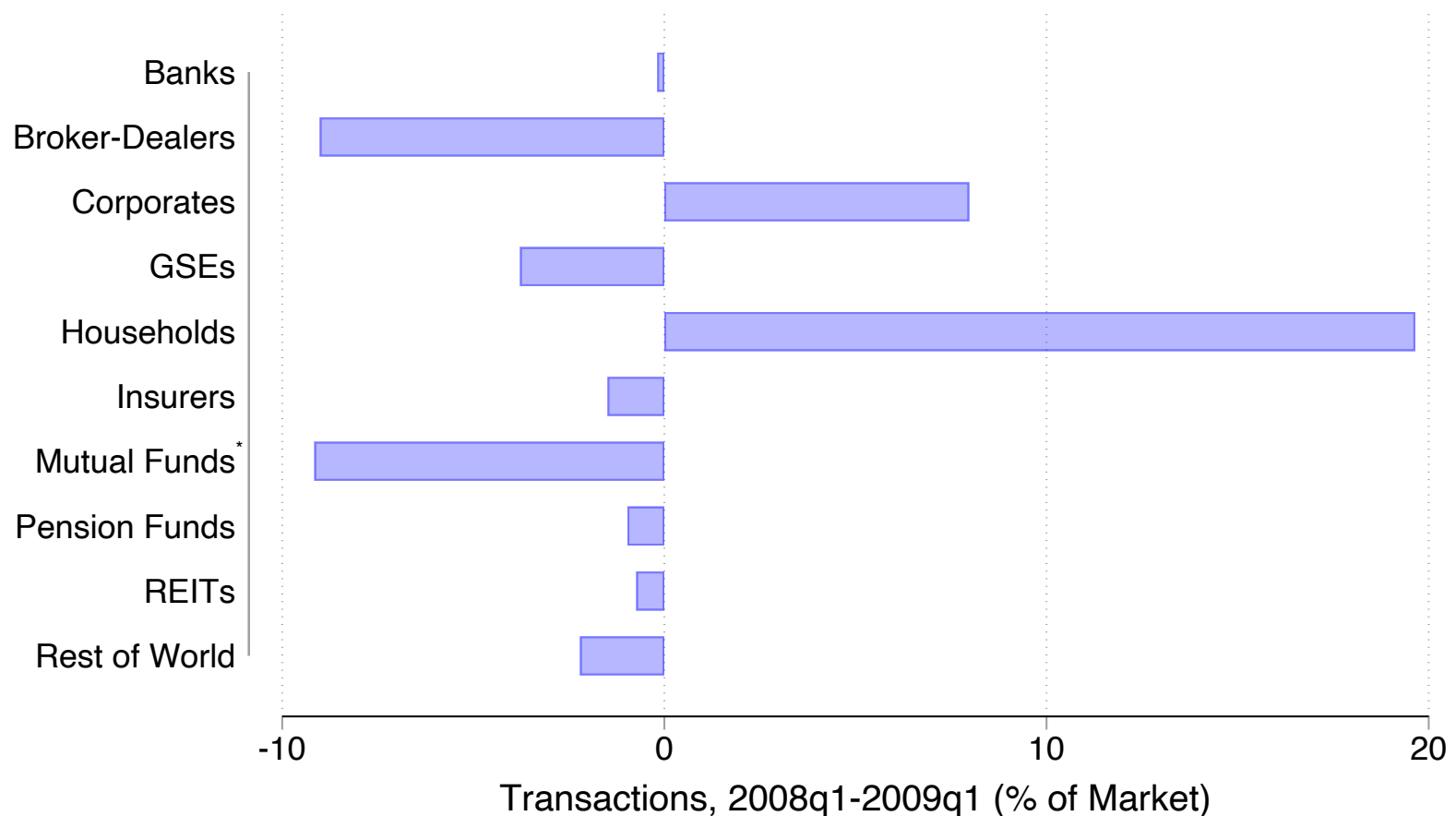


Figure A.2: **Heterogeneity in the estimated effects across rating groups and events.** This figure shows estimates from the regression specification in equation (5) on various subsamples for U.S. investment grade bonds. The first two groups of bars show estimates in the subsample of bonds rated AAA through A- and in the subsample of bonds rated BBB+ through BBB-, respectively. The remaining bars display separate estimates for different events. The estimates are from weighted least squares (WLS) regressions, with bond sizes as weights. The displayed confidence intervals are at the 95 percent confidence level and are clustered at the issuer by event level.



*Includes ETFs and money market funds

Figure A.3: Net transactions in corporate and foreign bonds by institutional sector, from the U.S. Flow of Funds. This figure plots the net transactions in corporate and foreign bonds by institutional sector between 2008Q1 and 2009Q1 from the Flow of Funds (FoF). The net transactions are scaled by the lagged total amount of corporate and foreign bonds reported in the FoF. By construction, the bars sum to zero, so that the figure shows the reallocation of ownership across sectors over the course of the financial crisis. The household sector in the FoF is a residual category that includes entities such as nonprofits and hedge funds. Transactions by the corporate sector include net bond issuance.

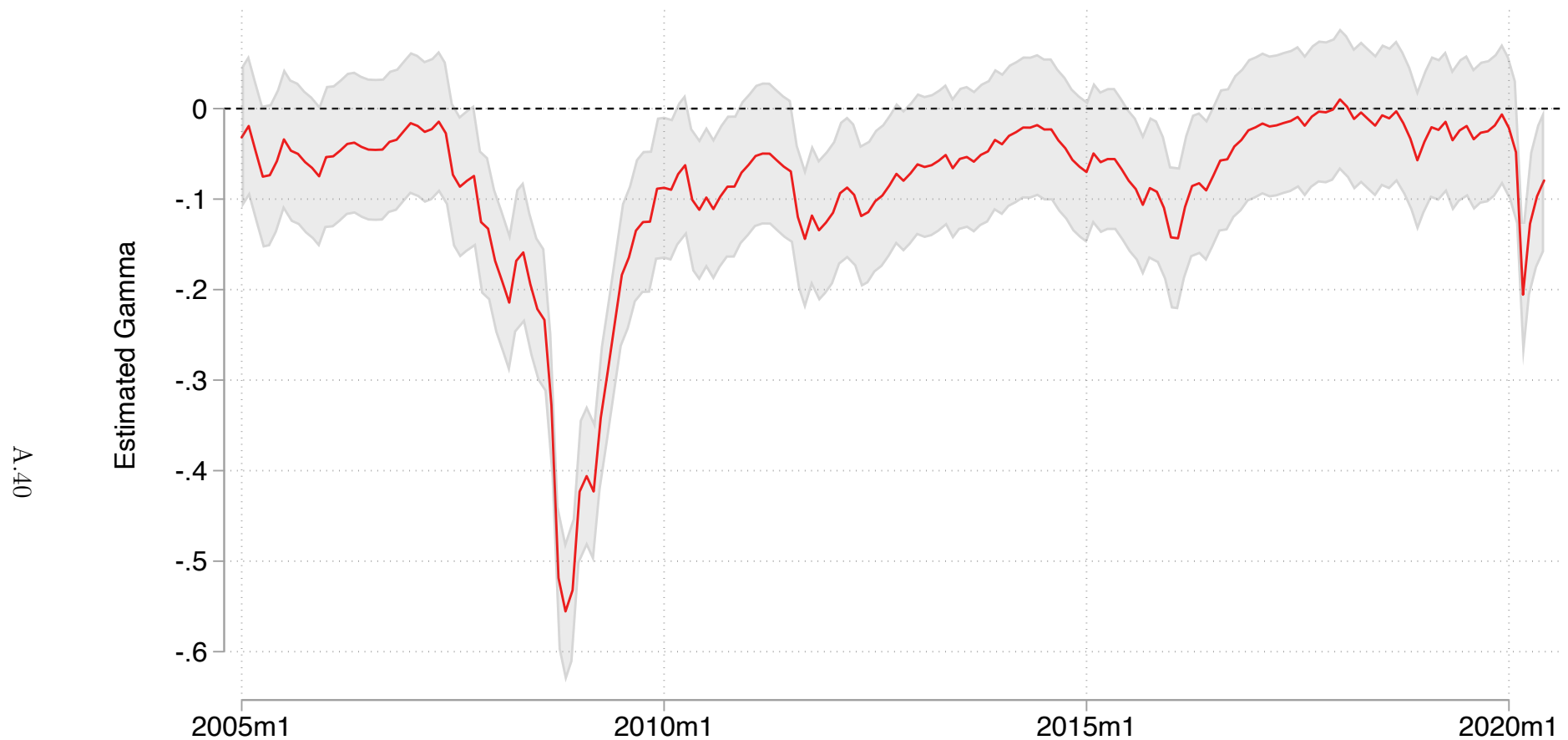


Figure A.4: **Insurer ownership and bond betas over the credit cycle.** This figure shows the estimates from the regression specification in equation (A.1). The *red line* plots the total estimated effect $\Gamma_t = \bar{\Gamma} + \bar{\Gamma} GZ_t$ of security-level insurer shares on bond betas against time. The *blue shaded area* corresponds to a 95 percent confidence band estimated using Newey and West (1987) HAC standard errors, with lag selection as in Newey and West (1994).

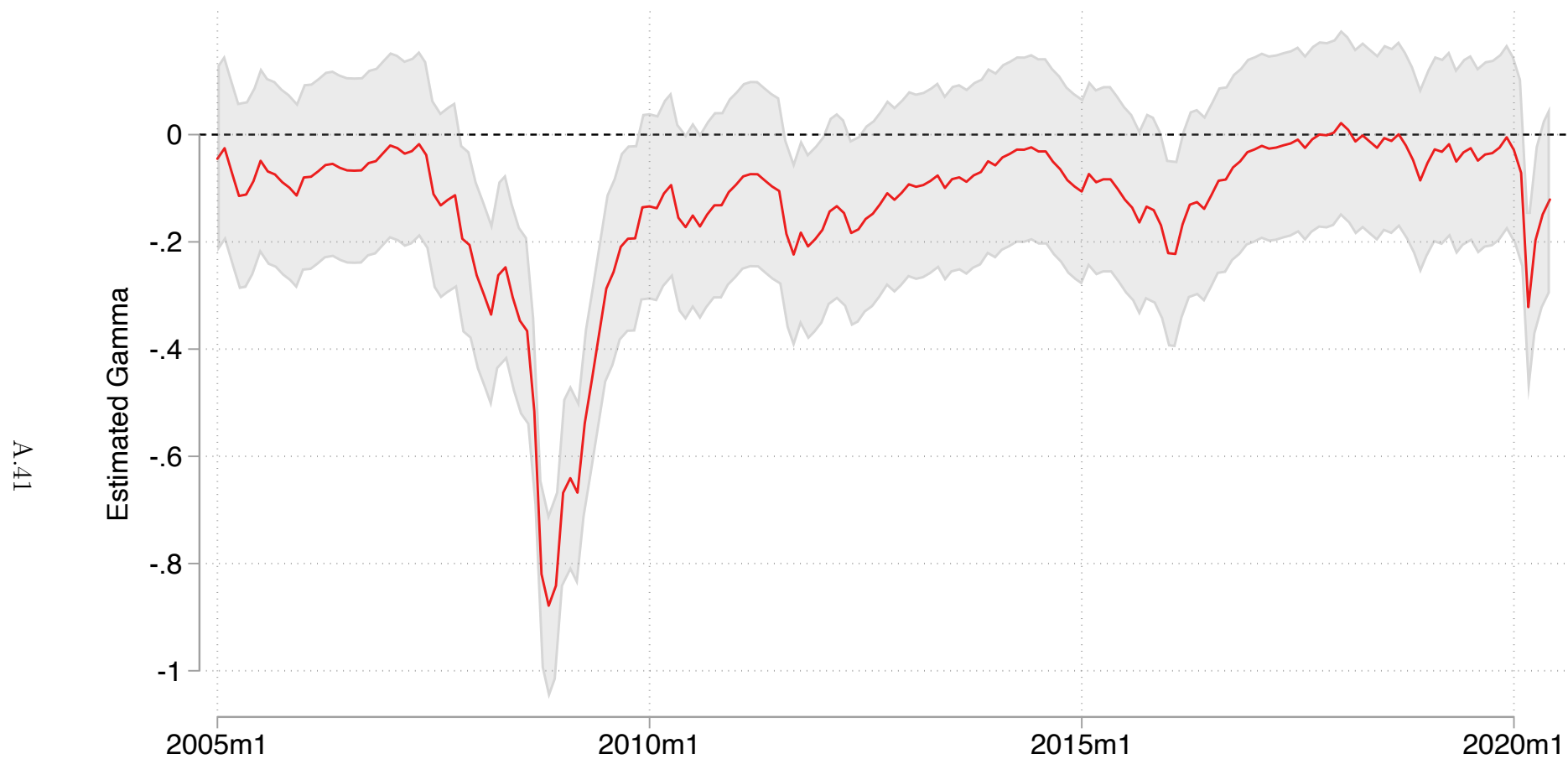


Figure A.5: **Insurer shares and bond betas: specification with one-quarter betas.** This figure replicates the results in Figure A.4 using betas that are estimated on a one-quarter rolling window.

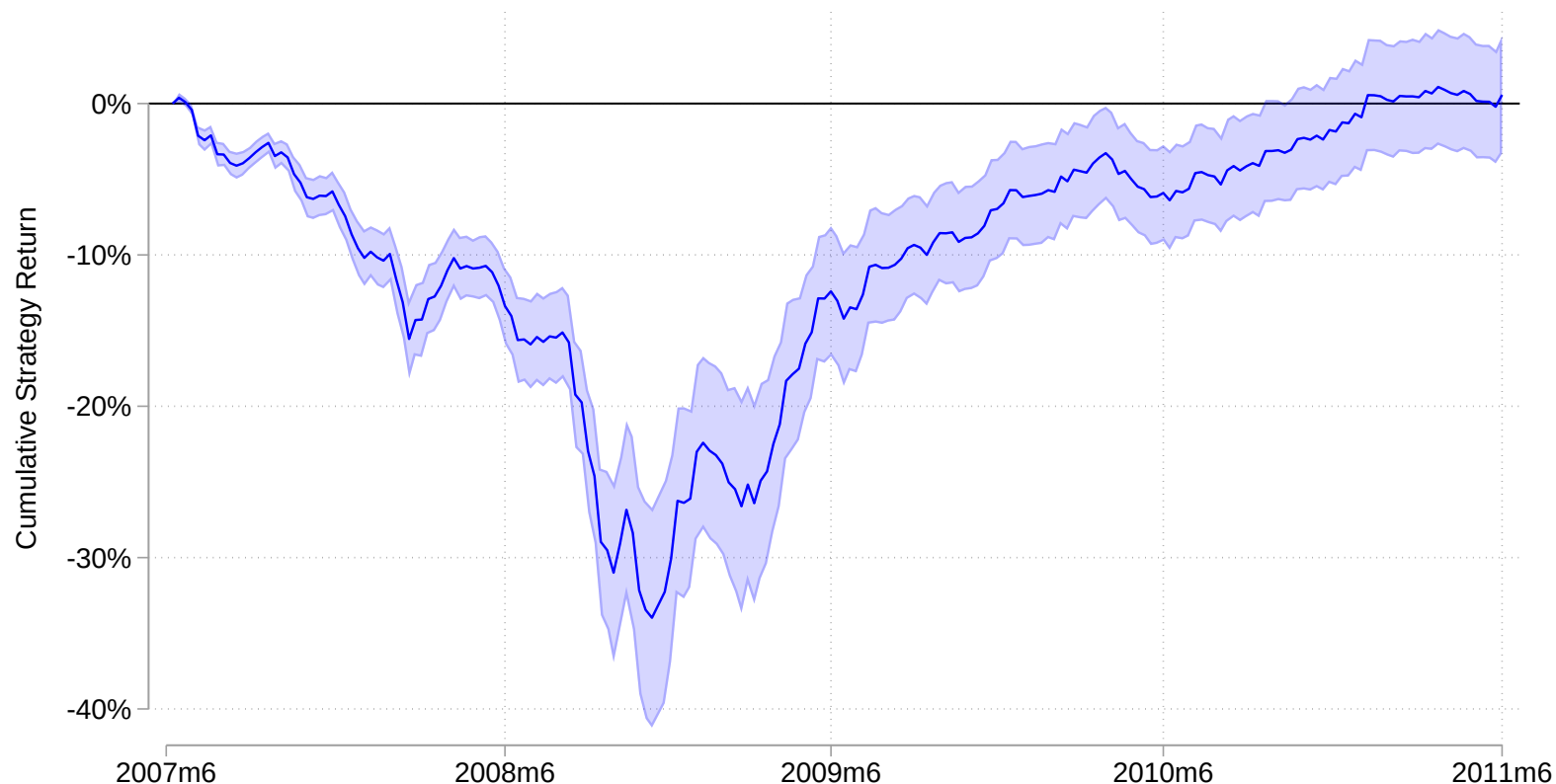


Figure A.6: **Price reversal in within-firm dynamics.** This figure shows reversion in the abnormal returns due to fire sales in the Great Recession. I construct a portfolio with positions in investment-grade bonds that are outstanding as of 2007Q1 and have at least one counterpart bond that is issued by the same firm and has the same characteristics (duration, rating, issue size, seniority, and so forth). Within each group of matching bonds, I sort bonds according to their pre-crisis insurer share. The portfolio takes long positions in bonds that have an insurer share above the group median, and short positions in bonds below the group median. The positions in each bond are equal-sized, in dollar terms. The level of leverage is chosen so that the ex-ante (pre-crisis) volatility of the overall long-short portfolio matches that of a value-weighted index of excess returns $R_{i,t}^e$ on US investment-grade corporate bonds. The graph shows the cumulative returns on this portfolio in excess of a basket of Treasuries of matched duration, abstracting from transaction costs.

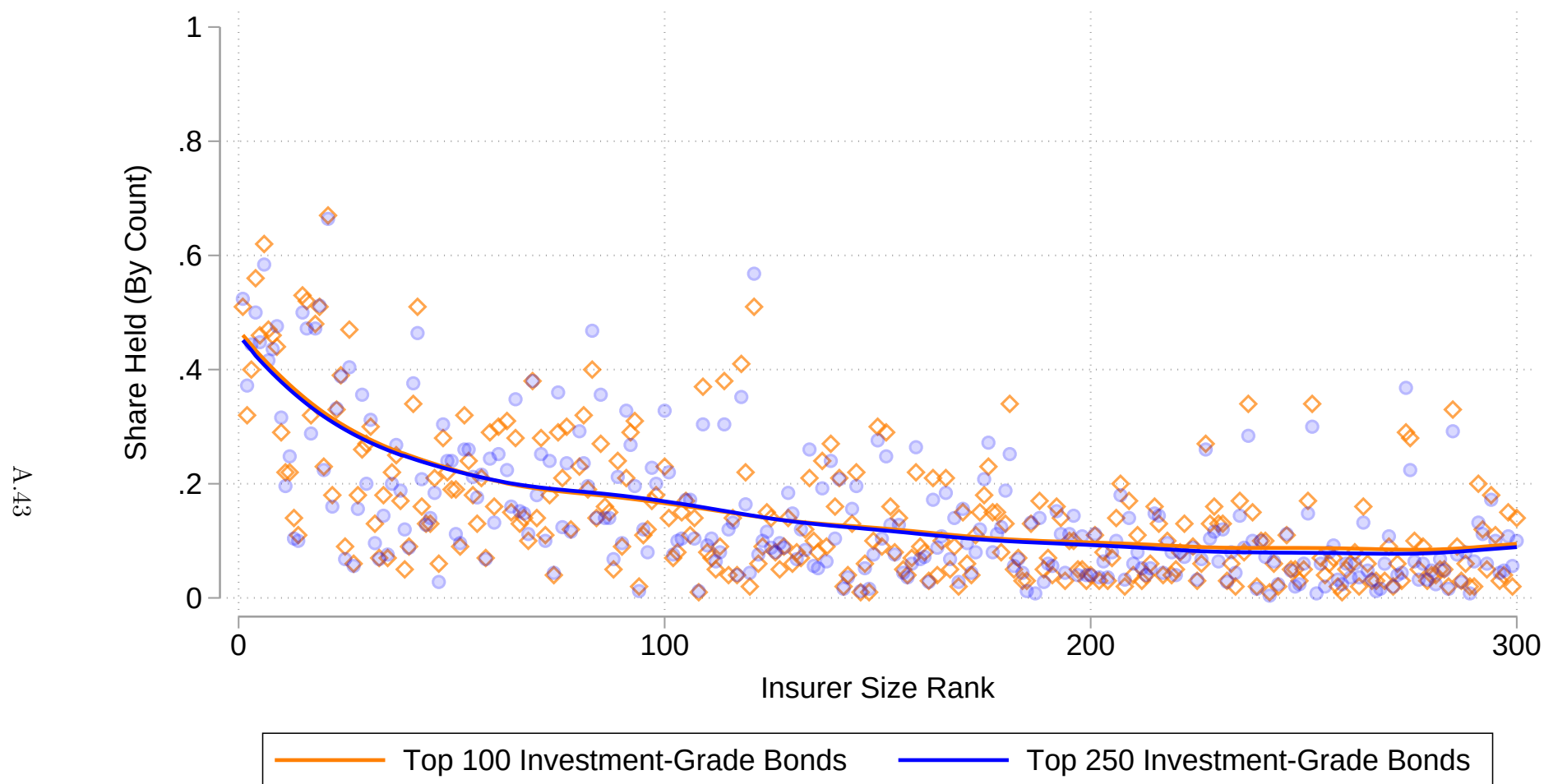


Figure A.7: **Holdings idiosyncrasy: robustness.** This figure replicates the analysis in Figure 7 with smaller sets of bonds. The figure ranks insurers by the overall size of their portfolio investment, and shows the share of the 100 largest (*orange diamonds*) and 250 largest (*blue dots*) actively traded investment-grade U.S. corporate bonds that they hold an open position in, all measured cross-sectionally as of 2017Q4. The thick lines show the fit from lowess regressions of the investment shares on the fund rank.

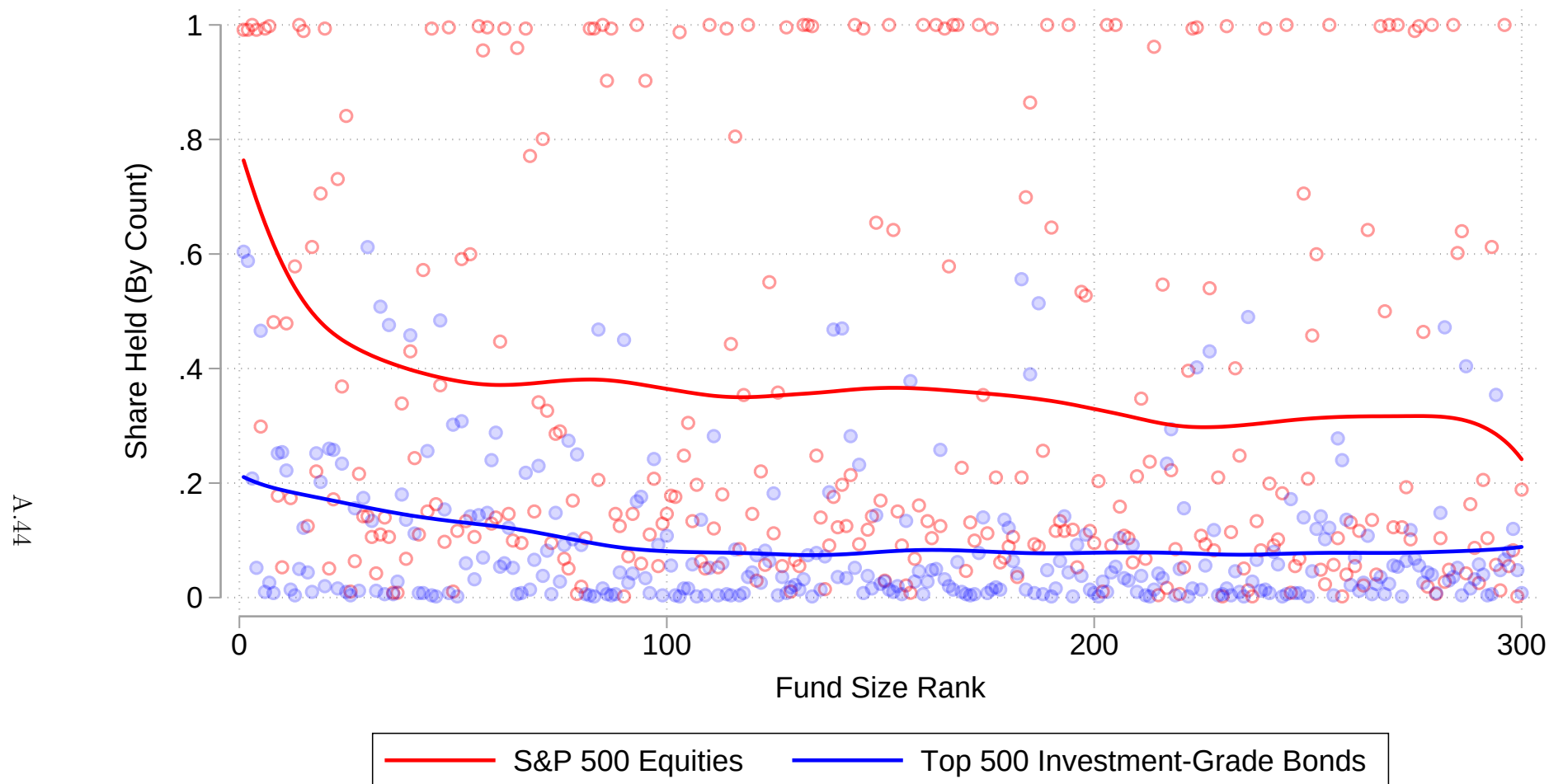


Figure A.8: **Holdings idiosyncrasy: mutual funds and ETFs.** This figure replicates the analysis in Figure 7 for mutual funds and ETFs. The figure ranks funds by the overall size of their portfolio investment, separately for equity specialists (*red dots*) and fixed income specialists (*blue dots*). The *blue dots* show the share of the 500 largest actively traded investment-grade U.S. corporate bonds that each fund holds an open position in, while the *red dots* show the share of S&P 500 equities that it holds an open position in, all measured cross-sectionally as of 2017Q4. The thick lines display the fit from lowess regressions of the investment shares (shown by the dots) on the fund rank.

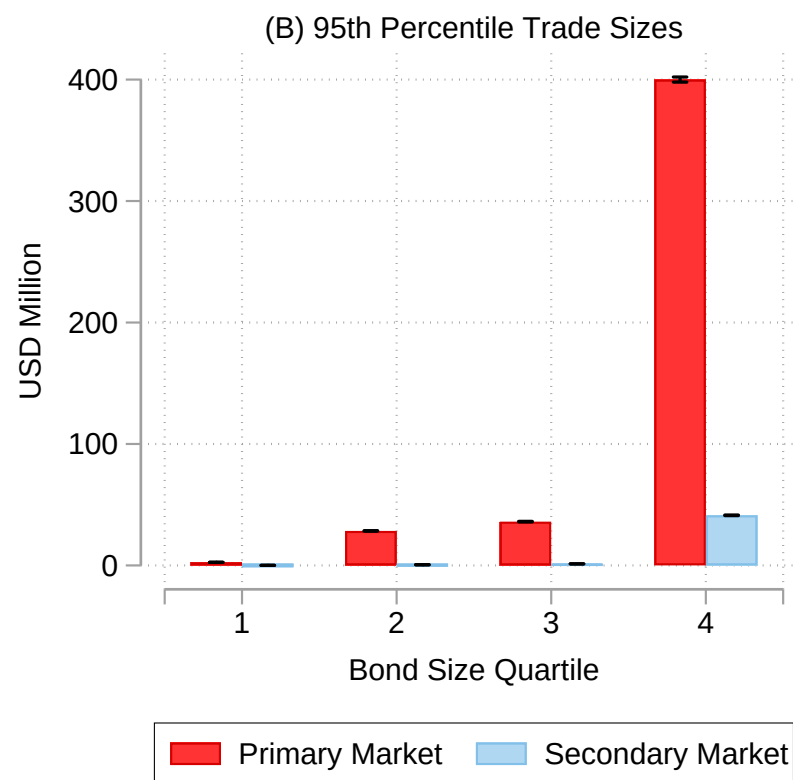
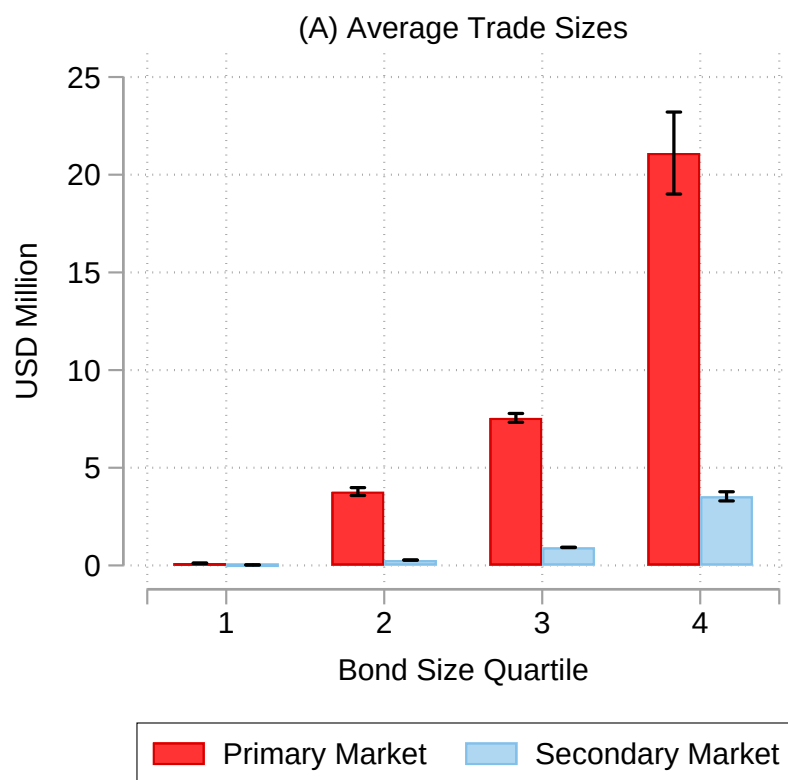


Figure A.9: **Trade sizes in the primary and secondary U.S. investment-grade corporate bond market.** Panel A plots the average trade size on the primary market (*red bars*) and on the secondary market (*blue bars*), by quartile of bond trade size, in the full sample of transactions in U.S. investment-grade corporate bonds from TRACE Academic. Panel B plots the size of trades at the 95th percentile of the trade size distribution.

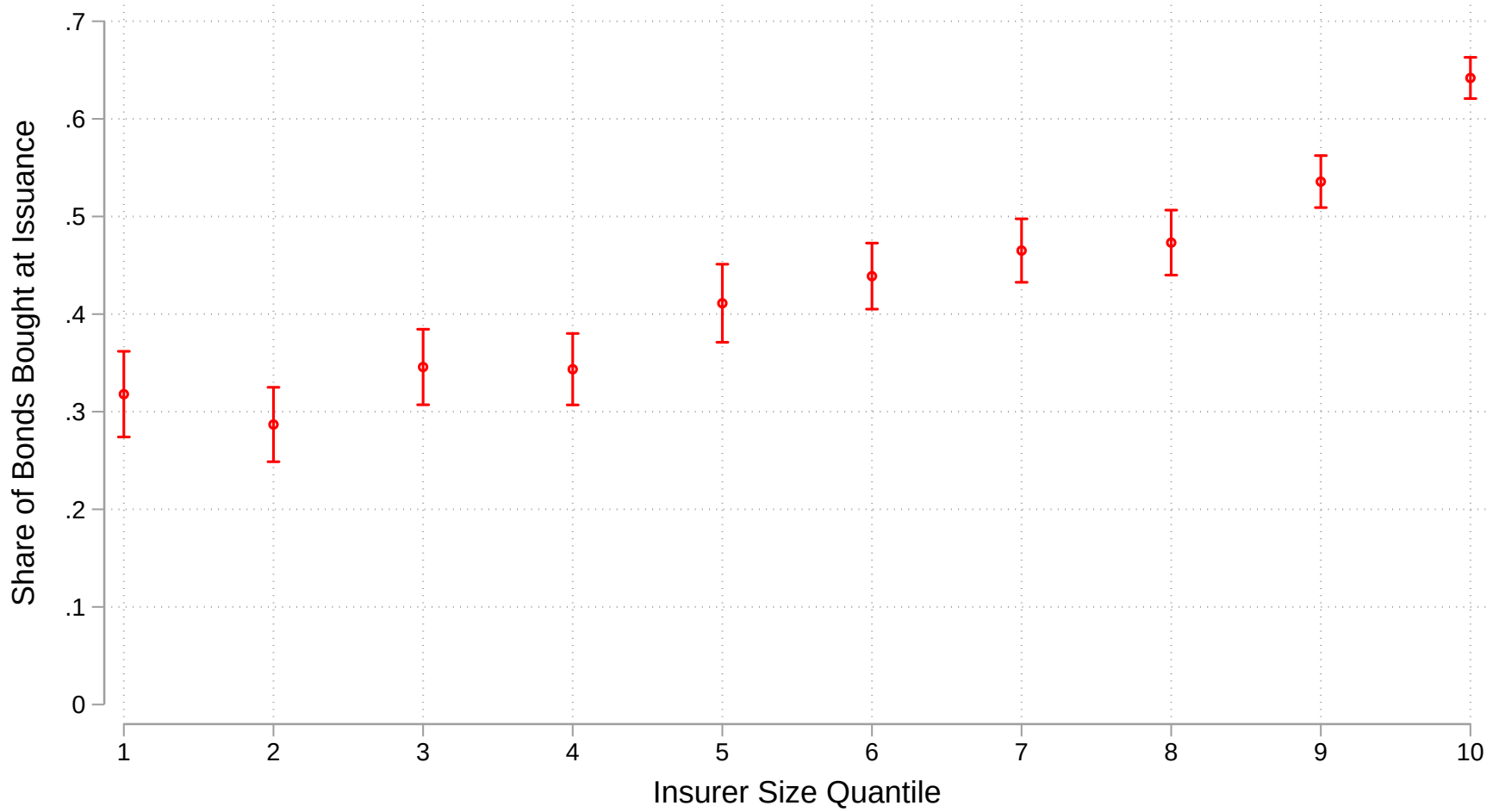


Figure A.10: **Share of bonds purchased on the primary market across the insurer size distribution.** This figure plots the share of all insurer bond purchases that are made on the primary market by insurer size decile. The bars correspond to 95 percent confidence intervals.

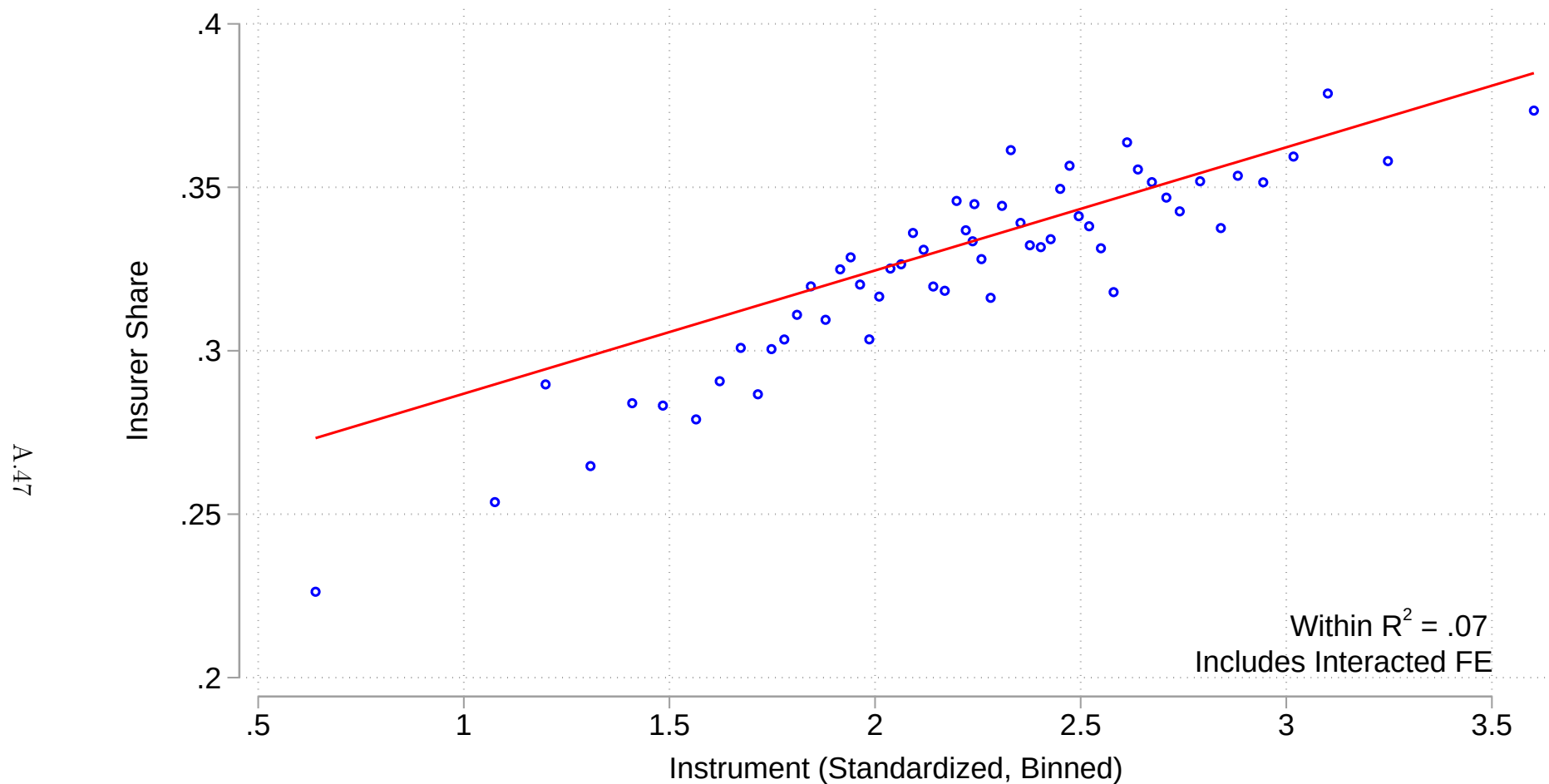


Figure A.11: **Shift-share IV: first stage relationship.** This plot shows a binned scatter plot of insurer shares against the instrument Z_i from the first stage of the 2SLS regression with fully saturated interacted fixed effects in Section 3 and Table 4. For expositional purpose, the instrument is standardized so that its standard deviation is equal to one.

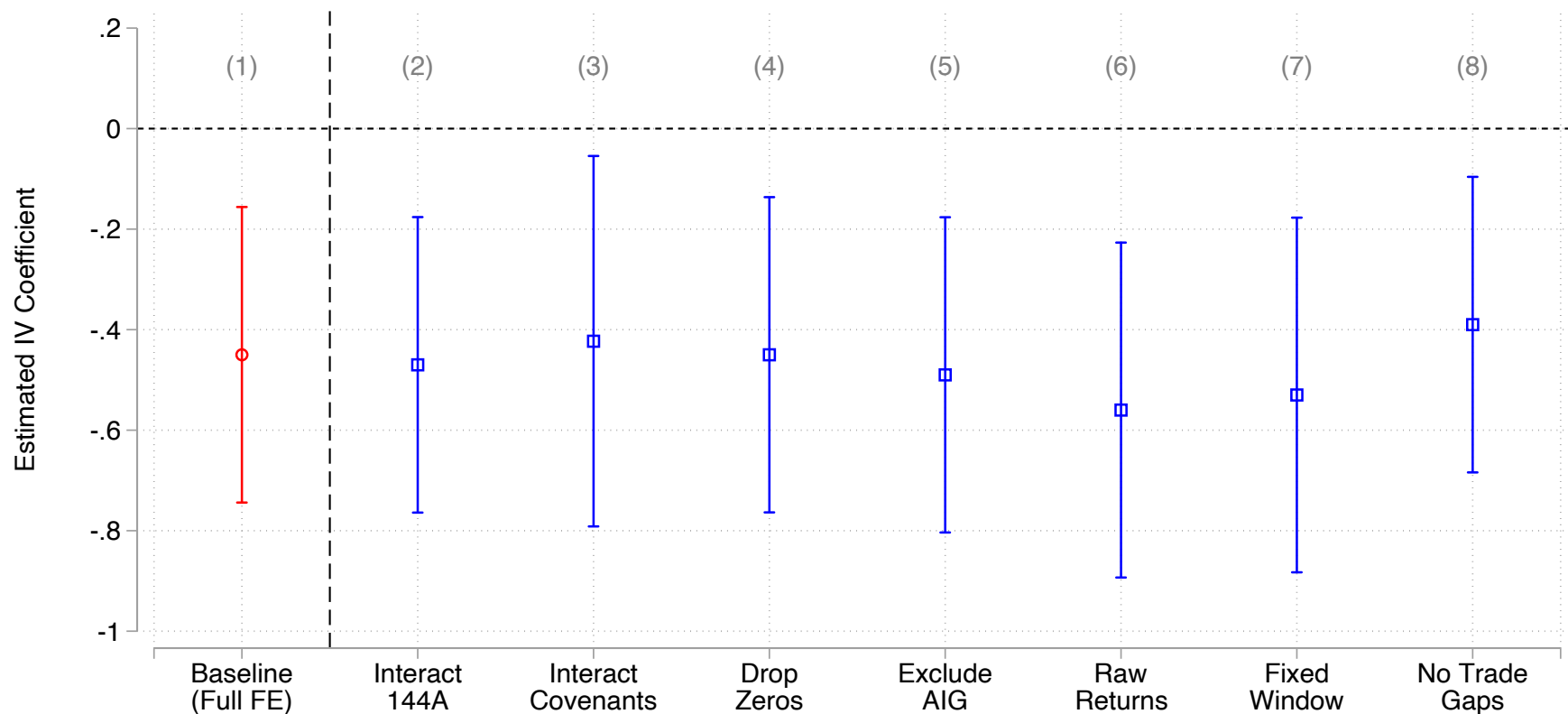


Figure A.12: **Shift-share IV: robustness to alternative specifications.** This figure demonstrates the robustness of the baseline 2SLS estimate (*red bar*) from column 8 of Table 4 to further variations in the empirical specification. Bar 2 introduces an additional interaction with a dummy indicating whether a bond is issued under SEC rule 144A. Bar 3 adds an interaction with dummies for the set of a bond's covenants, hence removing bonds with non-matching covenants from the estimation sample. Bar 4 excludes bonds with a zero insurer share from the sample. Bar 5 excludes the positions of AIG from the computation of the insurer positions $X_{i,t}^I$. Bar 6 uses raw returns $R_{i,t}$ in place of the excess returns $R_{i,t}^e$. Bar 7 uses an alternative measure of drawdowns which applies the same window to all bonds for a given event. Bar 8 drops any bonds that trade less frequently than once a week in a given event period. The displayed confidence intervals are at the 95 percent confidence level and are clustered at the issuer by event level.

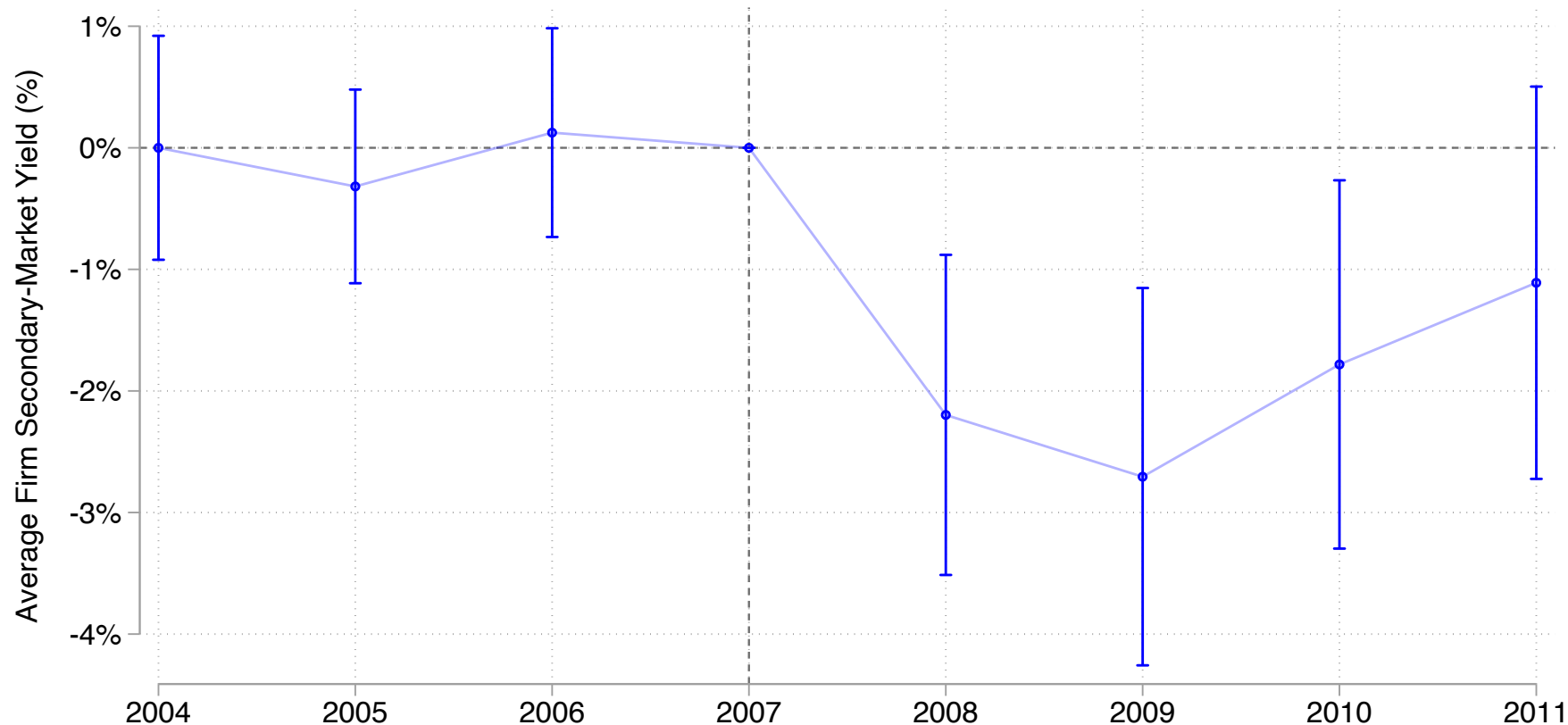


Figure A.13: **The association between insurer bond ownership and firm-level secondary-market yield dynamics in the Great Recession.** This figure shows the estimated coefficients from the dynamic difference-in-differences specification in equation (20), when using average firm secondary-market bond yields rather than primary-market offering yields. The specification is estimated on the sample of U.S. firms with outstanding bonds in 2007Q1, excluding speculative-grade issuers. The estimates are from just-identified two-stage least-squares, using the firm-level instrument in equation (17), interacted with year dummies. The coefficients quantify the association of average firm secondary-market yields (*blue line*) with the share of a firm's bonds held by domestic insurers in 2007Q1. The controls consist of firm-level covariates, measured in 2007 and interacted with time dummies: these include fixed effects for issuer rating, average duration of the firm's outstanding bonds, two-digit industry, and deciles of firm size and firm leverage. Standard errors are clustered at the firm level. Bars correspond to 95 percent confidence intervals.

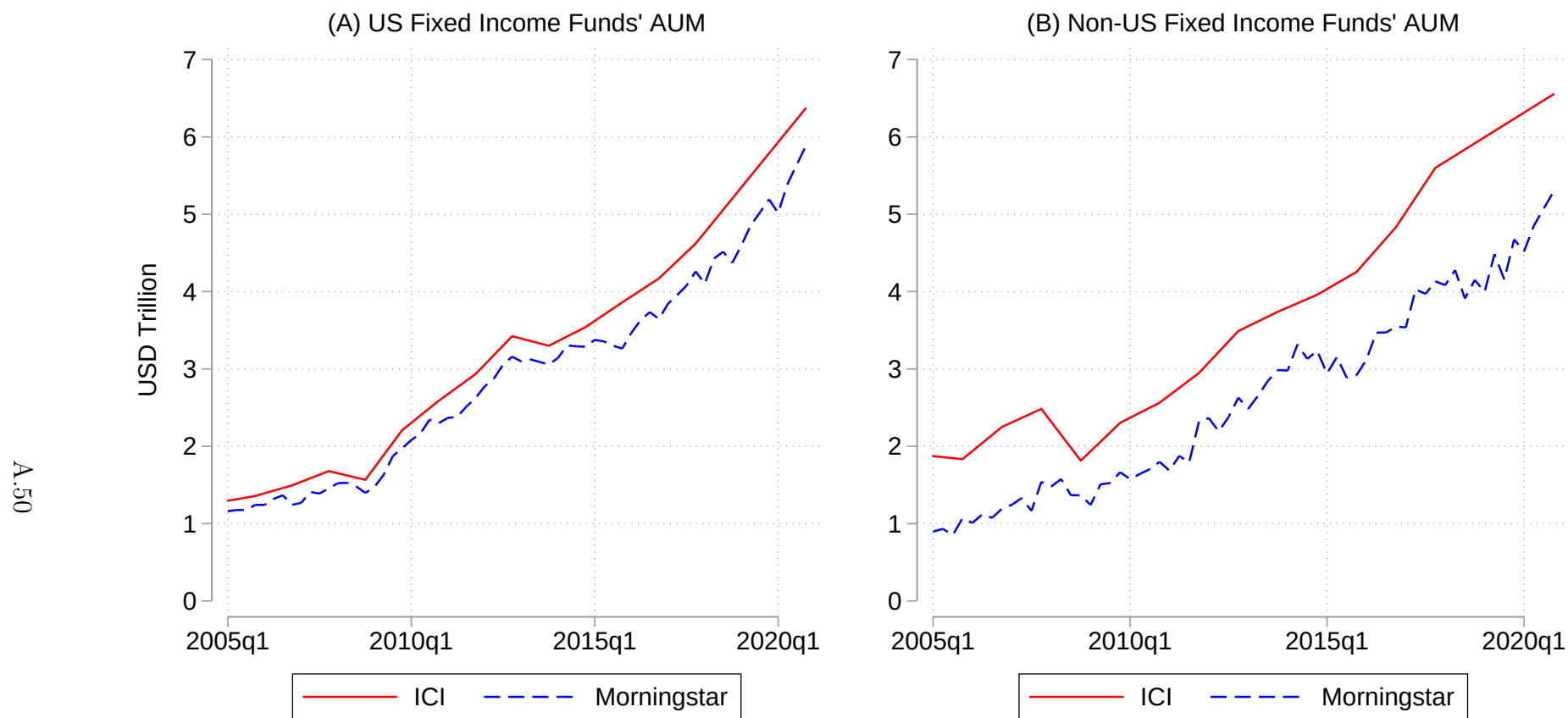


Figure A.14: **Aggregate value of fixed income funds' positions in the Morningstar data, compared to ICI benchmarks.** Panel A shows the aggregate market value of all positions held by U.S. domiciled fixed income investment funds in the Morningstar data (*dashed blue line*), alongside the aggregate value of the positions of U.S. fixed income mutual funds and ETFs reported by the Investment Company Institute (ICI, *solid red line*). Panel B shows the aggregate market value of all positions held by fixed income investment funds domiciled outside of the United States in the Morningstar data (*dashed blue line*), alongside the corresponding ICI benchmark (*solid red line*).

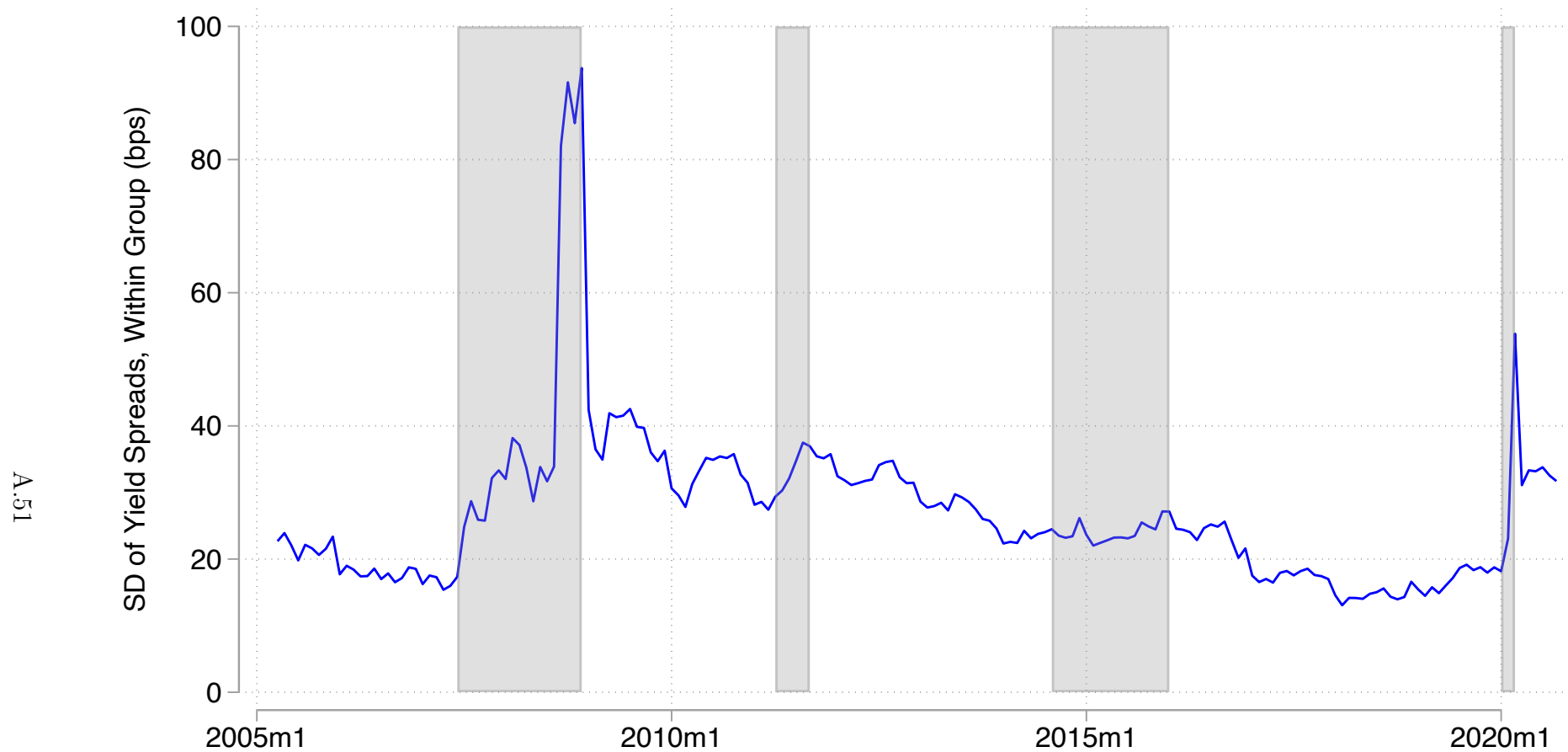


Figure A.15: **Variation of yield spreads to Treasury within fixed effect categories, U.S. investment-grade corporate bonds.** This figure shows the standard deviation of yield spreads to Treasury among corporate bonds that match along all dimensions included in the fully-saturated baseline interacted fixed effects specifications: that is, standard deviations are computed within groups of bonds from the same issuers and with the same characteristics. The series is computed at a monthly frequency. The *shaded gray areas* show the event windows used to estimate drawdowns in the baseline analysis.

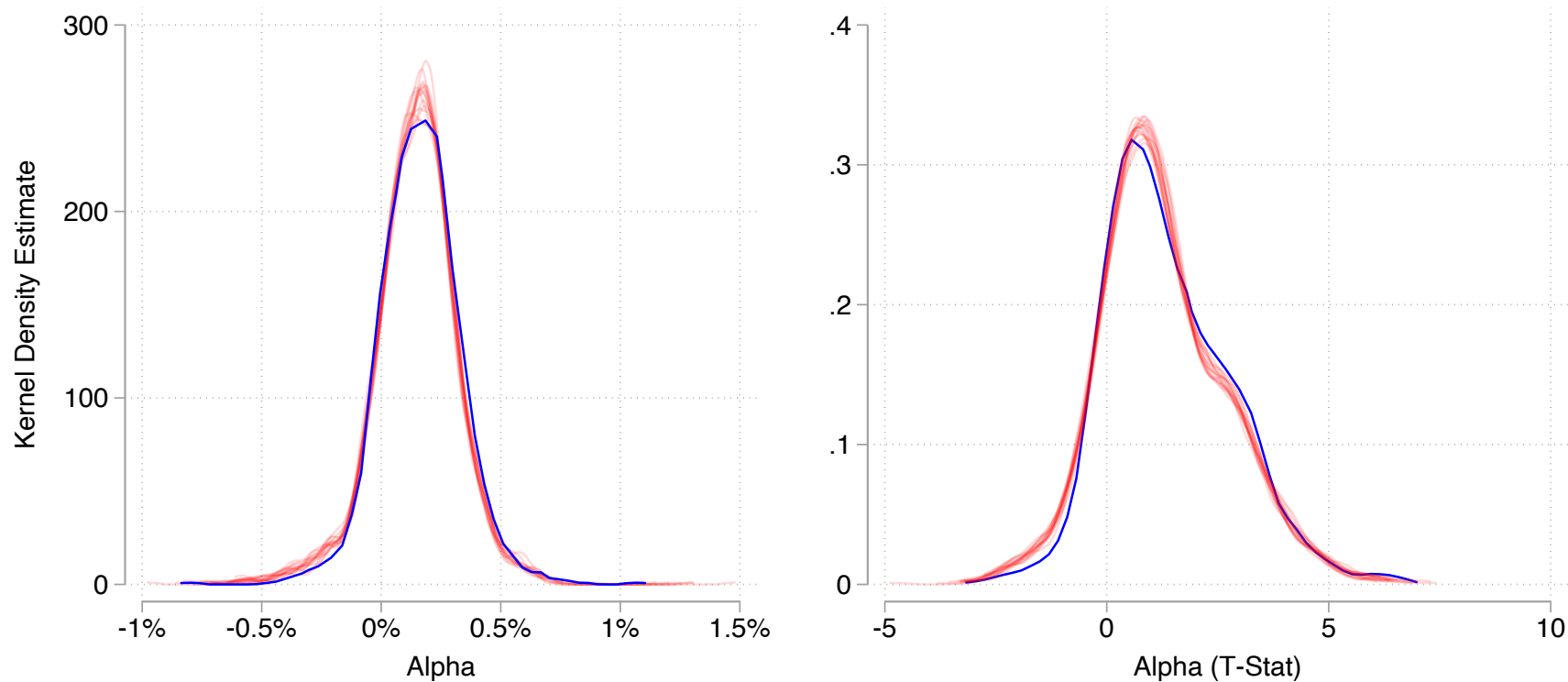


Figure A.16: **Testing for the presence of skill in the cross-section of insurers: Fama-French bootstrap approach.** This figure plots kernel density estimates of the empirical distribution of residual alphas and their t statistics in the cross-section of insurance firms (*blue curves*), estimated as described in Appendix Section D, alongside bootstrap realizations of the same (*red curves*). The bootstrap simulations impose zero residual alphas by construction. Each red curve averages over 1000 independent bootstrap simulations.

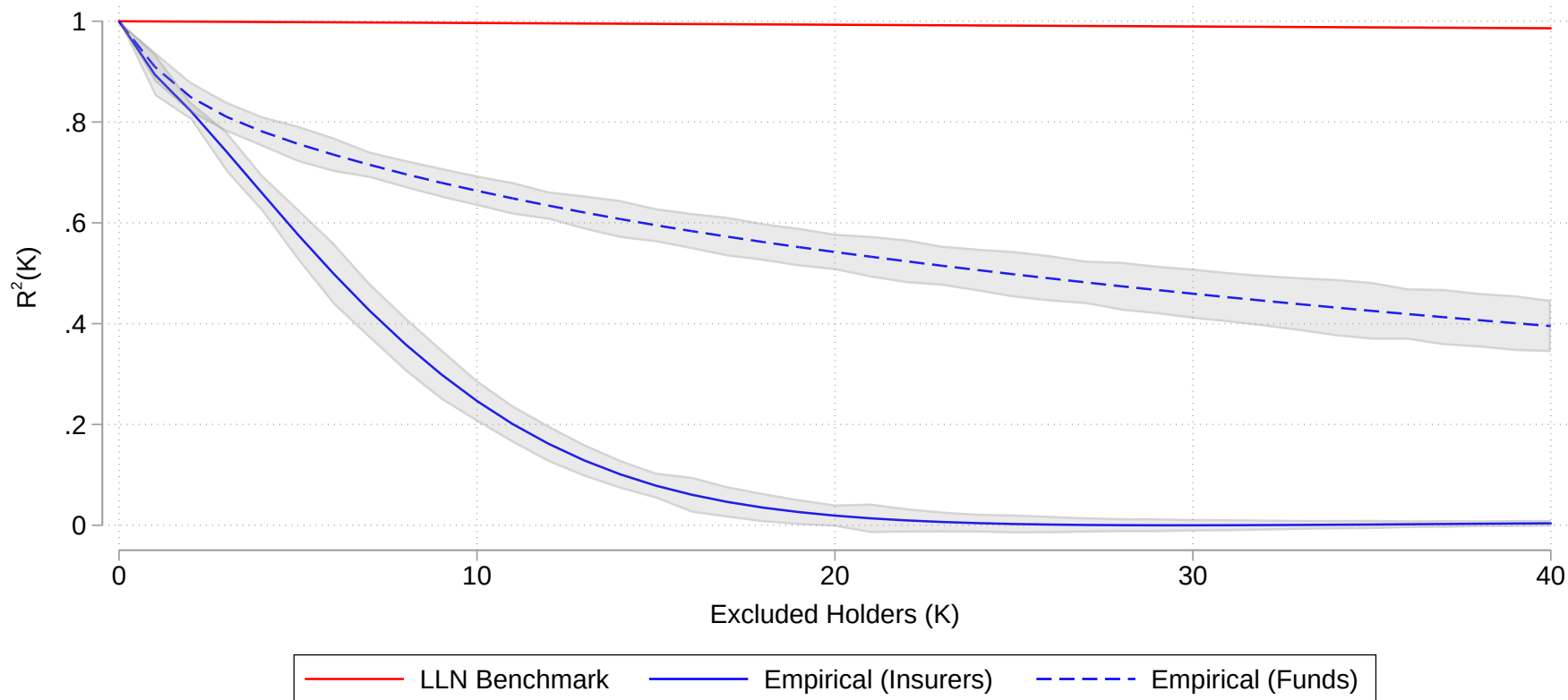


Figure A.17: **Deviations from the law of large numbers in insurer holdings.** The *solid blue line* plots the empirical R-squared values obtained from estimating equation (A.3). These estimates regress insurer shares $\phi_{i,t}$ against truncated insurer shares $\phi_{i,t}^{(K)}$, which exclude the holdings of the largest K holders underlying each position. The *solid red line* plots the theoretical R-squared values as we vary K from the benchmark model in equation (A.3), in which the law of large numbers applies. For comparison, the *dashed blue line* also plots the same estimates for mutual funds and ETFs.

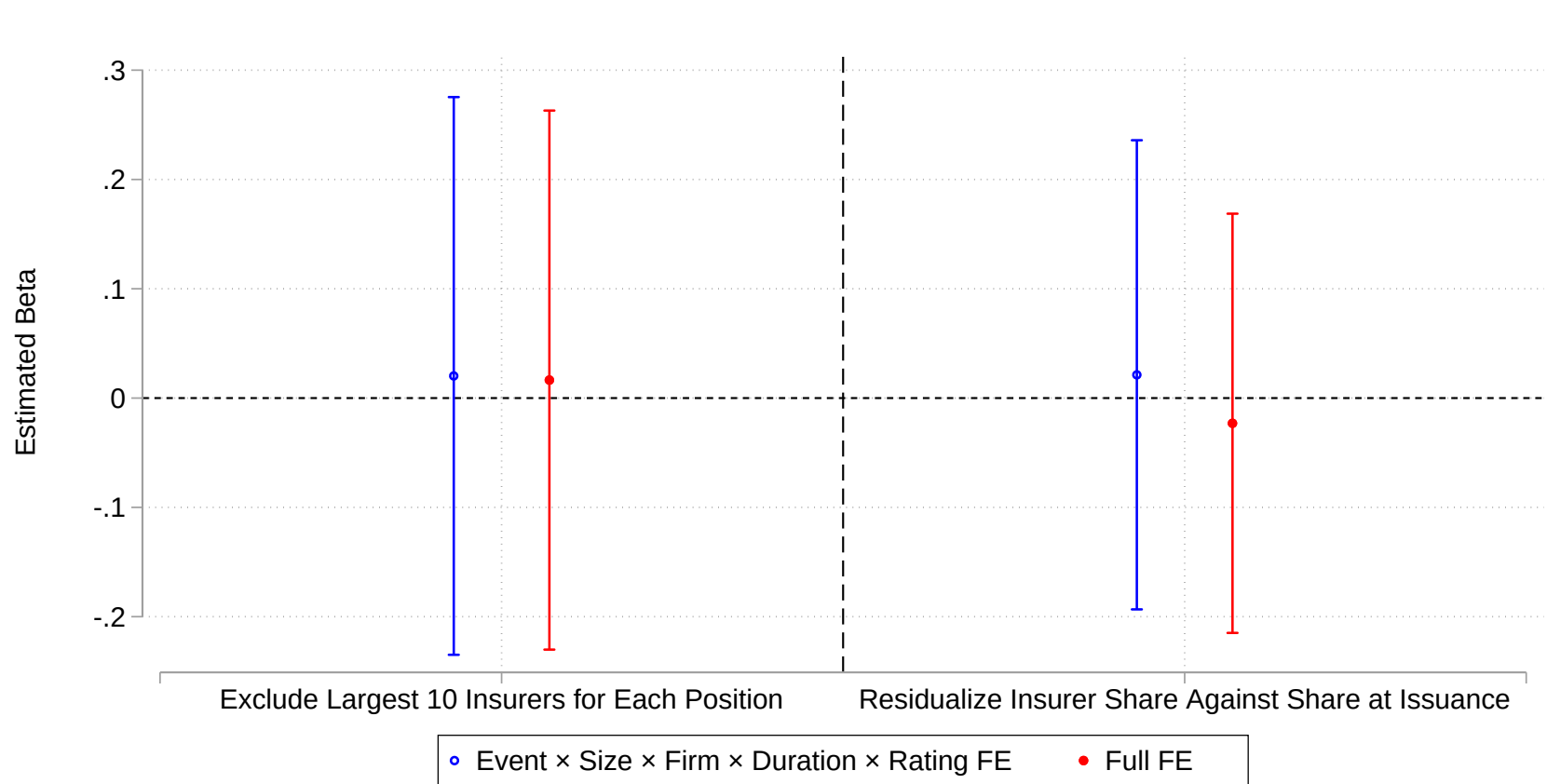


Figure A.18: **Further placebo exercises for baseline OLS regressions.** This figure shows placebo estimates obtained from estimating two variants of equation (5) for U.S. investment-grade corporate bonds. The first placebo exercise replaces the true insurer shares ϕ_{i,t_e^S-12} with the truncated shares $\phi_{i,t_e^S-12}^{(10)}$, which exclude the largest underlying 10 holders for each bond. The second placebo exercise regresses the true insurer shares ϕ_{i,t_e^S-12} against the insurer shares at issuance $\phi_{i,\tau(i)}$, and then uses the residuals from this regression as the right-hand side argument in equation (5). The displayed confidence intervals are at the 95 percent confidence level and are clustered at the issuer by event level.

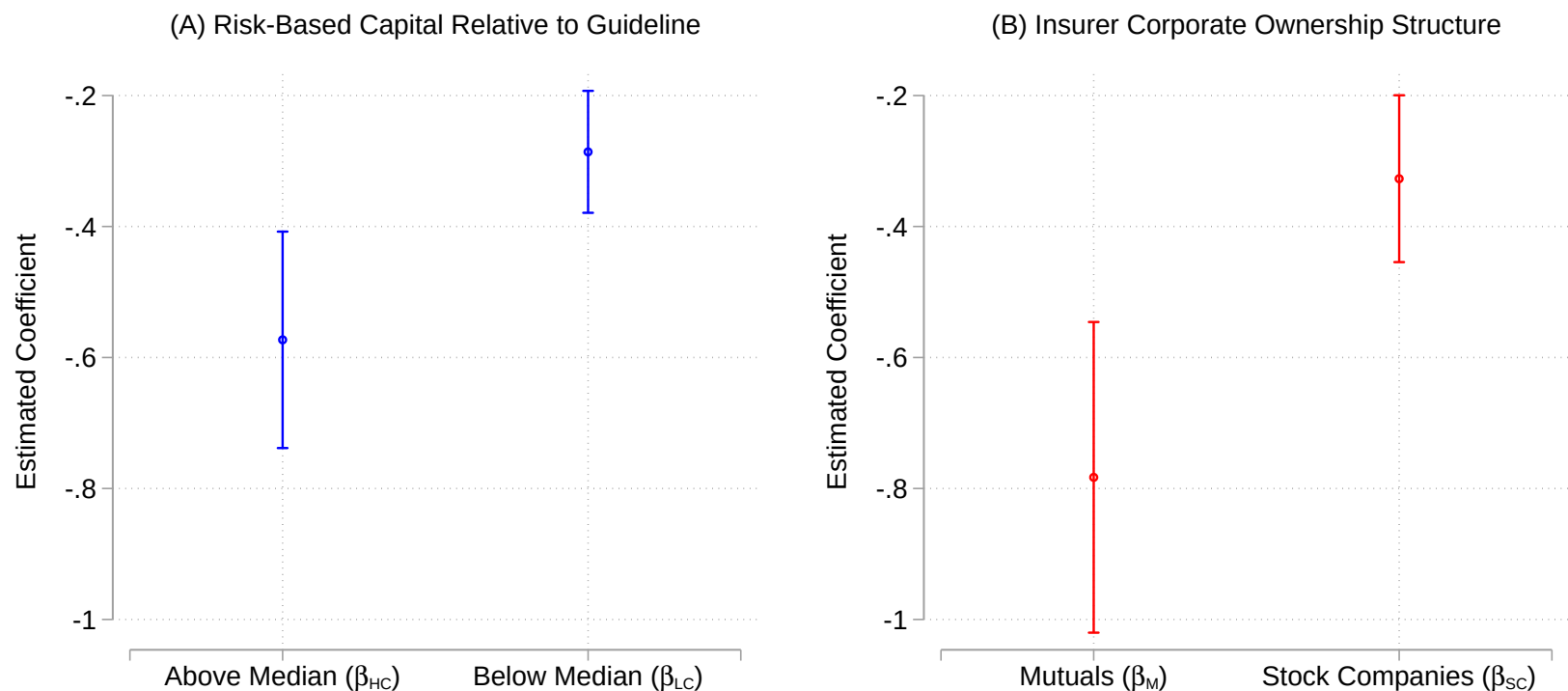


Figure A.19: **Heterogeneity within the insurance sector.** This figure displays the results concerning heterogeneity within the insurance sector introduced in Appendix Section F. Panel A shows the estimated coefficients $\hat{\beta}_{HC}$ and $\hat{\beta}_{LC}$ for the specification in equation (A.6). These coefficients respectively quantify the impact on bond drawdowns of ownership by insurers with above-median and below-median risk-based capital relative to guideline (as measured by the difference between each insurer's A.M. Best Capital Adequacy Ratio and the guideline for the insurer's current rating). Panel B shows the estimated coefficients $\hat{\beta}_M$ and $\hat{\beta}_{SC}$ for the specification in equation (A.7), which measure the impact on bond drawdowns of ownership by mutuals and stocks companies. Standard errors are clustered at the issuer by event level, and bars correspond to 95 percent confidence intervals. For both pairs of estimates, a Wald test rejects the null hypothesis of coefficient equality with $p < .01$.

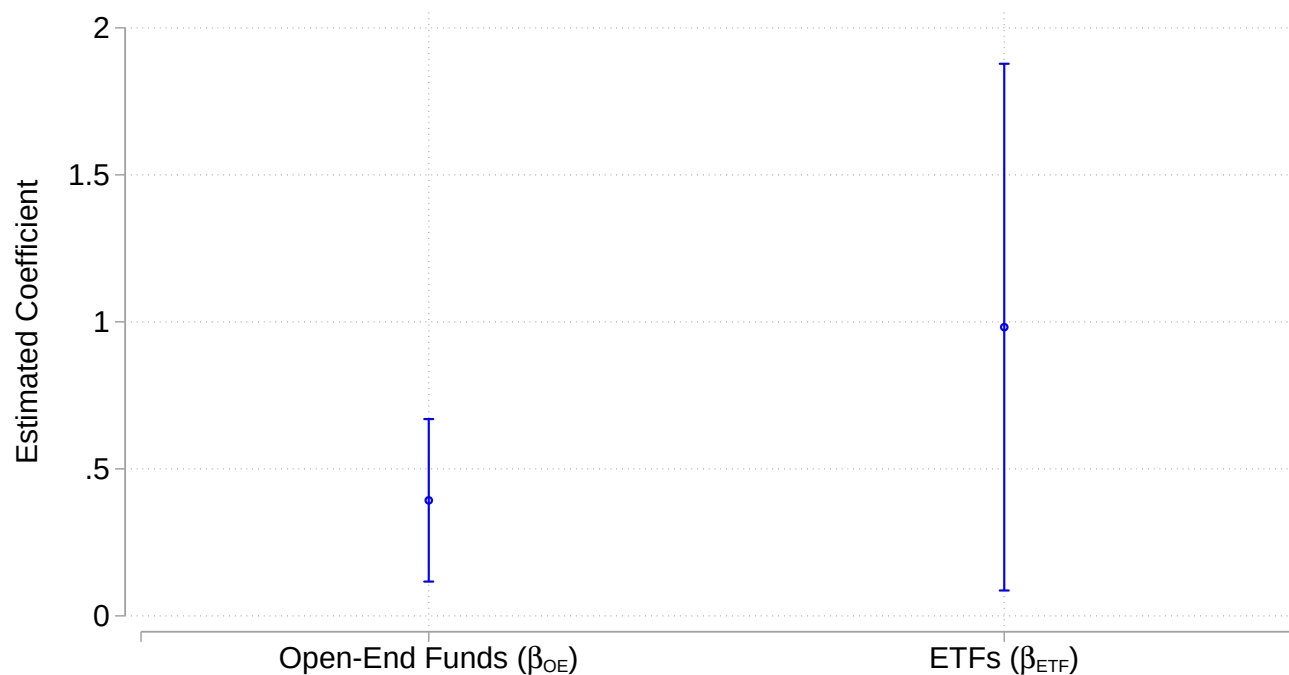


Figure A.20: **Heterogeneity between open-end mutual funds and ETFs.** This figure displays the results concerning heterogeneity between open-end mutual funds and exchange traded funds (ETFs) introduced in Appendix Section L. The dots show the estimated coefficients $\hat{\beta}_{OE}$ and $\hat{\beta}_{ETF}$ for the specification in equation (A.16). These coefficients respectively quantify the impact on bond drawdowns of ownership by open-end mutual funds and ETFs. Standard errors are clustered at the issuer by event level, and bars correspond to 95 percent confidence intervals. A Wald test fails to reject the null hypothesis of coefficient equality at the 10 percent confidence level.

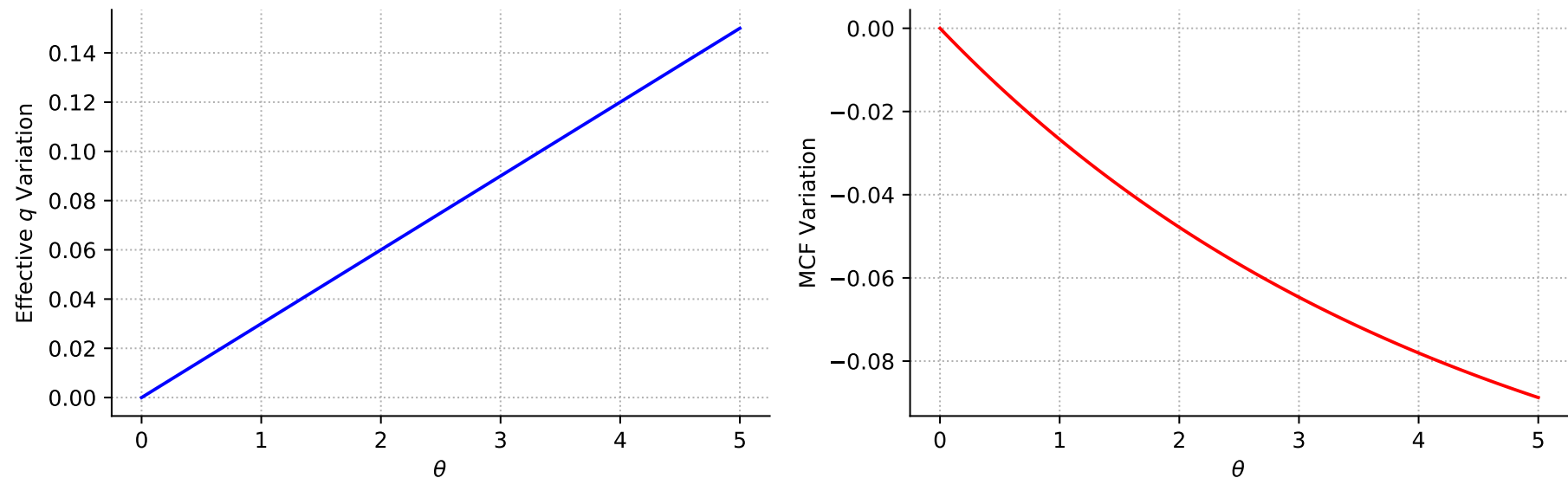


Figure A.21: **Structural interpretation under q -theory with costly external financing: calibrated results.** This figure plots the calibrated values of the impact of ex-ante firm-level insurer ownership on effective marginal Tobin's q , $\frac{\partial q^{\text{eff}}}{\partial \phi}$, and on a firm's marginal cost of financing, $\frac{\partial \text{MCF}}{\partial \phi}$, as derived in Appendix Section M. These are plotted against the adjustment cost parameter θ in the *blue curve* and *red curve*, respectively.

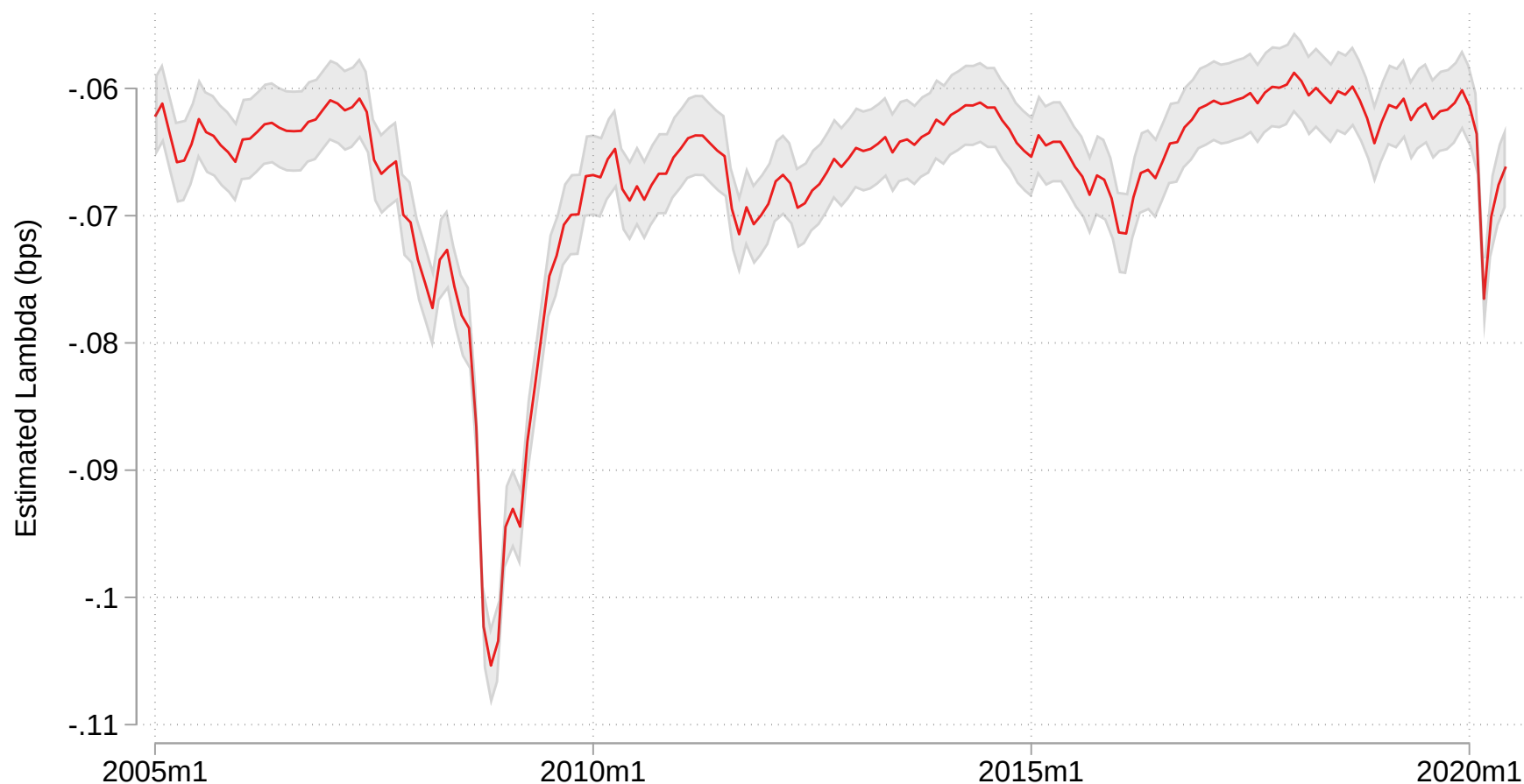


Figure A.22: **Insurer ownership and secondary-market bond turnover.** This figure shows the estimates from the regression specification in equation (A.38). The *red line* plots the total estimated effect $\Lambda_t = \bar{\Lambda} + \tilde{\Lambda}GZ_t$ of security-level insurer shares on monthly bond turnover. The *blue shaded area* corresponds to a 95 percent confidence band estimated using Newey and West (1987) HAC standard errors, with lag selection as in Newey and West (1994).